

**Investigating the role of socioeconomic status
in determining urban habitat quality for the
house sparrow, *Passer domesticus***

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EXETER AS A THESIS FOR THE DEGREE OF DOCTOR OF PHILOSOPHY
IN BIOLOGY; 29TH SEPTEMBER 2009.**

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ABSTRACT

Urban areas are increasingly recognised as an important resource for wildlife, as studies have shown that gardens, parks and brownfield sites can contain high insect and plant diversity. Urban centres can also provide resources for species of conservation concern, and it is therefore important to monitor urban habitat quality and ensure the maintenance of urban biodiversity. However urban habitats are often difficult to monitor effectively due to access and sight restrictions in built up areas. This thesis investigates urban habitat quality in relation to an urban specialist species, the House Sparrow *Passer domesticus*. After considering the importance of urban habitats for biodiversity in general, I review the current status and distribution of the house sparrow in urban areas, with particular reference to the possibility that human socioeconomic status has influenced the decline of the species in some urban areas. I then consider which features of urban houses and gardens may provide a potential explanation for inter-city variation in habitat quality for urban birds. I present evidence that the age of houses; the prevalence of roof repairs; and the presence of extensive paved areas such as driveways are linked to areas with low levels of socioeconomic deprivation. I then use nationwide data to establish that house sparrows in English cities are more likely to occur in areas that are relatively deprived. Furthermore, analysis of land use data confirms that house sparrow occurrence decreases with increasing levels of building and paving, and increases with the area of green space available. However, house sparrow occurrence also appears to decrease with increasing garden area, a surprising finding given that gardens are important foraging habitats for urban birds. By radio tracking house sparrows in urban Bristol, I show that gardens are heavily utilised by house sparrows, but that those with a high proportion of paving are avoided. It appears that changes to areas with low levels of socioeconomic deprivation, notably an increase in paved areas, may have contributed to the urban decline of house sparrows in less deprived parts of English urban areas. These findings are discussed in relation to future urban planning requirements, and the need to mitigate for the detrimental effects of urban development on species of conservation concern. The contribution of large, nationwide datasets to the monitoring of urban habitats, and the implications of these findings for other urban species, including humans, are also highlighted.

ACKNOWLEDGEMENTS

This PhD was funded by an NERC “blue skies” CASE studentship, in partnership with the British Trust for Ornithology (BTO). The study was supervised by Matthew Evans (University of Exeter) and Dan Chamberlain (BTO), and I thank them for the essential advice and assistance given to me during my research. I would like particularly to thank Dan and the many other staff at the BTO for their support and encouragement on the many visits I paid to the BTO headquarters at Thetford – your suggestions were invaluable. In particular I thank Mike Toms for providing me with BTO GardenBirdWatch data on house sparrows; Steve Freeman and Simon Gillings for statistical advice. Thank you also to the BTO GardenBirdWatch volunteers, who enabled the collection of the nationwide data on sparrows, without which this PhD would not have been possible.

Data from a number of sources was used during this PhD. As well as the BTO GardenBirdWatch, National Statistics data on deprivation and land use was used from www.statistics.gov.uk. Ordinance survey maps were obtained from DIGIMAP (<http://edina.ac.uk/digimap>).

I would also like to thank the staff at the Centre for Ecology and Conservation at Exeter University, many of whom have given up their time to help me at some stage or other. In particular I would like to thank Andy MacGowan for taking time out to travel to my field sites and help me catch and radio tag birds, and for providing feedback on my work. Thanks also are due to David Carslake for spending a substantial amount of time helping me get to grips with GLMMs and R software. I am also grateful to Thais Martins for help with data sorting; and to members of the Exeter and Plymouth Bird Biology group for advice and feedback on my work, in addition to the many other people who have answered my queries from time to time.

I would like to acknowledge in particular the many people who helped me to carry out the local scale work, in particular John Tully and Richard Bland, whose detailed studies of bird populations in Bristol first provided the inspiration for this PhD. Thanks also go to the Bristol residents who took part in the questionnaire surveys and radio-tracking study, in particular to Rodney Holbrook, Nicola Ramsden, Mitch Crossingham, Susan

Slatter, Mr and Mrs Brookman, Pru and Brian Comben, and Mrs Wilmer for allowing me to catch birds in their gardens. Radio tags were provided by Biotrack, thanks to Brian Cresswell and staff for advice and assistance. Thank you also to Bristol City Council, Horfield housing association and the Friends of Troopers Hill, for assistance in distributing surveys.

I would also like to thank Denis Summers-Smith, who has been researching sparrows for over 50 years and whose advice, information and support have been invaluable. I hope that the results of this study will be a valuable addition to the growing literature on the species, and help further the study of this familiar yet often overlooked bird. In addition to Denis, there are many other house sparrow researchers who have provided helpful ideas and advice, notably the members of the house sparrow discussion forum.

Finally, thank you to my family and friends, especially those in the office who have been in close proximity on days when I was losing the battle to get sense out of R. All those who have aided the de-stressing process by sharing walks, chocolate breaks and the occasional pint are greatly appreciated! Special thanks go to James Goodey, for not only managing to be pleased by the news that my PhD would take me 400 miles away for a few years, but for putting up with all the subsequent PhD-related stress. You deserve a medal!

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CHAPTER 1

INTRODUCTION: URBAN AREAS AS A HABITAT FOR WILDLIFE

Humans are changing the structure and composition of habitats on a global scale, and with human population density showing no sign of decreasing, the importance of incorporating the impact of human activities into the study of ecology is becoming more widely recognised (Palmer *et al*; 2004). One of the most widespread changes in habitat composition is the increase in urbanisation, defined by Marzluff (2001) as: “the process by which human settlement increases in: 1) population density and 2) intensity of land use in an area”. The global urban population is projected to increase from 3.1bn people in 2005 to 6.4bn by 2050, by which time nearly 70% of people globally are expected to live in an urban area (United Nations, 2007). This will have a consequent effect on global patterns of biodiversity and raises important questions regarding the composition of urban habitats and their suitability for wildlife. The implications of this change on planning policy and habitat management will need careful consideration in order to maintain urban biodiversity in the future.

IMPACTS OF URBANISATION ON BIODIVERSITY:

As human population density increases and settlements become more urbanised, the environment undergoes a number of physical changes. These typically include an increase in the ambient temperature (the “heat island” effect); increasing levels of air and soil pollution; and changes in land use composition, notably an increase in paved areas at the expense of vegetation (McKinney; 2002). These changes will undoubtedly have an effect on the habitat available to urban wildlife, and therefore the species composition and diversity that can be supported by the changing landscape. There is evidence that ecosystem performance declines as urban housing density increases (Tratolos *et al*; 2007), but to understand why this occurs it is necessary to look more closely at the effects that these physical changes have on the urban landscape.

Natural or semi-natural habitats undergoing urbanisation tend to become more fragmented as development encroaches. This can affect species assemblages, particularly among higher trophic groups as the resulting mosaic of small habitat patches will support fewer species than equivalent areas of continuous habitat (Gibb & Hochuli; 2002). Habitat fragmentation also increases the risk of populations becoming vulnerable to the effects of inbreeding depression, especially in species with limited dispersal abilities, as lack of connectivity between fragments of suitable habitat hinders dispersal between subpopulations and reduces gene flow. This can have detrimental fitness consequences for small populations in isolated areas, and can lead to localised extinctions (Dickman; 1987). However, the effects of reduced patch size appear to be greater than the effects of isolation in urban habitats (Evans *et al*; 2009a), probably due to the relatively small scale over which the habitat changes in urban areas.

The physical changes common to most urban habitats are usually maintained by the importation of vast quantities of resources, which serve to artificially maintain the habitat in a state of disequilibrium from the surrounding environment (McKinney; 2006). In this way, urban areas become more similar to each other and less like their local environment, a circumstance leading to the decline of native species and an increase in those better adapted to urban habitats in general. As a result, the species composition of urban habitats worldwide also tends to be relatively similar, regardless of local patterns in biodiversity. For example, the bird species richness of similar urban habitats within France, Finland and Canada were found to be more alike than the species richness of different habitats within the same city (Clergeau *et al*; 2001). Most studies of urban biodiversity report an increase in population density, but a decrease in the species diversity of wildlife as urbanisation increases (Beissinger & Osbourne,

1982; Marzluff, 2001). This is particularly true for core areas within cities, that typically contain very few species at high density.

The species favoured by urbanisation tend to be generalists that are adaptable enough to alter their nesting and foraging habits to take advantage of man-made structures and resources (DeVitor *et al*; 2007). These tend to be resident or partially migrant omnivores and granivores (Bezzel, 1985) and include species such as gulls, pigeons and corvids (McKinney, 2002). These generalist species have adapted well to life in urban centres, often reaching nuisance proportions (Savard, *et al*; 2000).

In addition, a number of avian predators have adapted to life in urban and suburban areas, following the increased densities of their prey species. These include the sparrowhawk, *Accipiter nisus* and peregrine falcon *Falco peregrinus* (e.g. Wieserbs & Jacob, 2005; Chamberlain *et al*; 2009b). Conversely, ground-nesters, long-distance migrants and forest species, especially foliage-gleaning insectivores, are typically under-represented in the urban avifauna (Bezzel, 1985). These groups are particularly vulnerable to the effects of urbanisation as they are affected to a greater extent by disturbance than other species, especially in the case of ground nesting birds, as the presence of humans increases the perceived predation risk (Beale & Monaghan, 2004). In addition, valuable food resources for insectivores are likely to be depleted by the increase in non-native plant species and reduction in mature vegetation that is associated with increased levels of development (White *et al*; 2005). Furthermore, it appears that individuals within a species also show variation along the urbanisation gradient, both behaviourally and physiologically. For example, urban house sparrows *Passer domesticus* in Hungary were consistently smaller than their rural counterparts, even when given access to ad libitum food (Liker *et al*; 2008). Similarly, a study of urban

and rural blackbirds *Turdus merula* found evidence for earlier timing of reproduction in urban birds, which appeared to be caused by a combination of both genetic effects and phenotypic flexibility (Partecke *et al*; 2004).

Urban areas also contain species that can be considered urban specialists, as they are found at higher densities in urban areas than rural habitats. These include the house sparrow, the Starling *Sturnus vulgaris*, and the feral pigeon *Columbia livia* (Rolando *et al*; 1997, Siriwardena *et al*; 2002, Robinson, *et al*; 2005). In addition human activities often introduce non-native species into urban areas, either deliberately or accidentally. Domestic pets, particularly cats may well have a detrimental effect on wild bird and mammal populations, particularly as their density is not limited by prey abundance (Woods *et al*; 2003, Sims *et al*; 2008). The planting of ornamental shrubs is also likely to have an influence on biodiversity. A survey of gardens in Sheffield, UK found that on average over half of the plants they contained were non-native species (Smith *et al*; 2006). Introduced species usually harbour fewer invertebrate species than native flora, and increases in these species have been associated with the progressive loss of insectivorous and nectarivorous bird species from urban areas (White *et al*; 2005). Retaining native plants in urban areas therefore has important implications for the maintenance of biodiversity in general.

The mosaic of habitats available in urban areas can provide many resources for adaptable species, both as a result of the new habitats created, and through direct resource provisioning by householders. Urban green spaces, private gardens, allotments and “brownfield sites” – land that has been previously developed – all provide substantial resources for biodiversity (MoErtberg, & Wallentius; 2000, Small *et al*; 2002, Eyre *et al*; 2003, Chamberlain *et al*; 2007a, Loram *et al*; 2007). Similarly, private

gardens in general contain a high diversity of both plant and insect life, particularly those that do contain a high proportion of native plants (Thompson *et al*; 2004, 2007, Wilkinson, 2006).

There is an increasing trend towards supplementary feeding by many urban households, which further increases the value of urban gardens as a resource for wildlife (Cannon *et al*; 2005). It is estimated that nearly half of all households provide some form of supplementary food for birds, and many also provide other resources such as nestboxes (Davies *et al*; 2009). The prevalence of bird feeders is positively associated with avian species richness and abundance in urban areas (Fuller, 2008; Evans *et al*; 2009a), suggesting that supplementary food is a valuable resource for urban passerines. In addition, many urban areas provide food resources unintentionally, through refuse and the provision of food for pets. Supplementary feeding has the potential to influence every aspect of avian biology, including the timing and success of reproduction and the distribution of populations. Beneficial effects reported as a result of supplementary feeding include earlier laying dates, the increased probability of second broods, and increased availability of essential nutrients such as Vitamin E (Robb *et al*; 2008a). Furthermore, recent evidence has shown that the beneficial effects of supplementary feeding can be long-lasting, for example winter feeding can lead to earlier laying dates and increased fledging success in subsequent breeding seasons (Robb *et al*; 2008b). However, supplementary feeding could also be associated with negative effects, for example an increased risk of disease transmission at feeding stations (Robb *et al*; 2008a).

For those species that can exploit human resources, food availability is often much higher in urban areas than in natural habitats. In addition, close proximity to humans

can also have the effect of reducing the number of natural predators that pose a threat to urban wildlife populations (Gering & Blair, 1999). Both of these factors are thought to contribute towards the artificially high population densities of urban specialists in city centres. However, it appears that whilst artificially provided resources are more plentiful, they may not be of as high quality as naturally available foodstuffs. This may have a consequent effect on juvenile survival rates and the breeding success of adults. A comparison of the productivity of urban and non-urban passerines has suggested that adult birds in urban areas are able to lay earlier, implying they survive the winter in better condition. In spite of this, urban birds subsequently had smaller clutches, lower productivity per nesting attempt and produced nestlings of lower weight than non-urban birds, suggesting that whilst food availability during the winter may be relatively good in urban areas, this is not the case during the breeding season (Chamberlain *et al*; 2009a). This is supported by evidence from a number of studies suggesting that the nutritional quality of food in urban areas is often low, particularly with regard to the protein content, an essential requirement for growing chicks. Juveniles fed on poor quality food are often of relatively low fledging weight, a factor known to reduce their survival chances (Mennechez & Clergeau, 2001; Pierotti & Annett, 2001; Peach *et al*; 2008). There may therefore be a trade-off between the benefits to biodiversity gained from reduced predation rates and increased winter food availability, and the costs of reduced breeding success for many species.

Large aggregations of birds at artificial feeders may also facilitate disease transmission between individuals (Bradley & Altizer, 2006), which may help to explain apparent reductions in the fitness of urban populations in relation to those in rural areas.

However, the prevalence of disease in urban areas is likely to depend to a large extent on the type of pathogen. For example, the prevalence of tick infestations appears

relatively low in urban areas, whereas the prevalence of other diseases such as avian malaria may show variation according to city (Evans *et al*; 2009b). There may also be an enhanced risk of disease for some species, particularly those that become stressed by the effects of urbanisation, or those that are native and come into contact with multi-host pathogens introduced to their local habitat (Bradley & Altizer, 2006). For example disease has been implicated as a contributory factor in the decline of the native red squirrel in the UK, even though the pathogen concerned only circulates at low levels in the population (Tomkins *et al*; 2003). Similarly, antibody prevalence for West Nile Virus was found to be significantly higher amongst Northern Cardinals (*Cardinalis cardinalis*) than in other songbirds along an urbanisation gradient, suggesting that they are more tolerant of infection than other, less common species (Bradley *et al*; 2008). Some of these diseases may also pose a threat to the human population, and the progress of wildlife diseases in urban habitats therefore warrants careful monitoring.

Although core urban areas such as city centres usually have lower species diversity throughout than the surrounding rural habitat, suburban areas can on occasion have relatively high biodiversity, particularly if the local rural habitat is heavily cultivated (Mason, 2000). This has been observed for birds; plants; mammals; butterflies; and insects in general (Racey & Euler, 1982; Blair & Launer, 1997; Marzluff, 2001; McKinney, 2002). High species diversity at intermediate levels of urbanisation is often attributed to the “intermediate disturbance hypothesis”, whereby urban sprawl has the effect of creating a mosaic of different habitats in close proximity, thereby supporting more species than would be sustained by one habitat alone (Connell, 1978). For this reason, suburban areas often support many “edge species” such as blackbirds (*Turdus merula*) and song thrushes (*Turdus philomelos*), some of which have declined in other habitats (Mason, 2000). Although urbanisation has potentially many detrimental effects

on biodiversity, it may therefore also have beneficial effects on certain species, some of which are of conservation concern.

CONSERVATION IMPORTANCE OF URBAN AREAS:

Urban areas are capable of supporting large species assemblages, and as a consequence can provide important habitats for species of conservation concern (Cannon *et al*; 2005, Smith *et al*; 2005). In the past, urban habitats were generally neglected in biodiversity research, with a few exceptions, most notably Eastern Europe where much ornithological research was carried out in the 1970s and 1980s (e.g. Bezzel, 1985). As urbanisation has increased on a global scale, urban areas have become more widely recognised as an important conservation habitat in their own right.

Brownfield sites support a high diversity of invertebrates, particularly Coleoptera, many of which are scarce nationally (Small *et al*; 2002). Many avian species in particular are strongly associated with human habitation, and urban areas provide refuges for many species that have declined in other areas (Gregory & Baillie, 1998; Marzluff *et al*; 2001). This is the case for the song thrush and black redstart *Phoenicurus ochruros* in eastern England, species that have declined in their traditional habitats due to the intensification of land use in more rural habitats (Mason, 2000; www.blackredstarts.org.uk).

The provision of supplementary food by urban landowners may also be a valuable resource for species that predominantly use other habitats such as farmland, when food is scarce in adjacent rural areas (Fuller *et al*; 2004). This is particularly true during winter, as modern farming methods leave fewer stubble fields available as a source of food for many farmland species. Bird feeders also provide important extra resources in those years when naturally occurring food is scarce (Chamberlain *et al*; 2007b).

Gardens also provide important nesting habitats for many species, such as the house sparrow and European starling, *Sturnus vulgaris*, which are red-listed in the UK and of conservation concern across Europe (Crick *et al*; 2002, Bland *et al*; 2004).

The importance of urban habitats are becoming more widely recognised, both as resources for wildlife in their own right, and for their beneficial effects on human health and wellbeing. Evidence suggests that the presence of urban green space can enhance recovery from surgery, and that access to green spaces such as urban parks can have both physical and psychological benefits to humans (Sutherland *et al*; 2006, Ulrich, 1984). These benefits appear to increase with increasing biodiversity, suggesting that loss of green space may have detrimental effects on human health as well as on species abundance if active management strategies are not employed (Fuller *et al*; 2007). Conservation of biodiversity in urban areas should therefore be considered important not only for maintaining wildlife, but also for promoting human health and wellbeing.

Given that increasing urbanisation is inevitable as human population density continues to increase, it is important to consider the effects of new developments on species diversity in cities. Prior knowledge of what constitutes good habitat in terms of conserving biodiversity is crucial to maximising the potential of any given habitat to attract a diverse range of species. Similarly, only obtaining sufficient knowledge of habitat requirements for urban wildlife can prevent biodiversity loss in areas where redevelopment takes place. It is important for ecologists to work with urban planners and local government in order to see that recommendations are implemented to maintain existing biodiversity in the long term.

BIODIVERSITY MONITORING IN URBAN AREAS:

There is a practical difficulty in quantifying the extent of the resources available in urban areas, as most land is privately owned. Sight and access restrictions are especially problematic in urban areas, due to the high building density and high number of landowners within a given area compared with other habitat types (Cannon, 1999). However, it is important to base conservation decisions in urban areas on sound evidence, given that the result of such decisions will affect a large number of landowners. In addition, the variation between different urban areas suggests that management policies suitable for one area will not necessarily apply to another. Even so, establishing a database of broad trends in biodiversity and habitat use will prove a valuable aid to conservation, management and development policies (Sutherland *et al*; 2004).

The quality of individual urban habitat patches is likely to be extremely variable, and this will have an impact on potential biodiversity. The level of urbanisation, type and density of buildings and the structure of urban gardens will all affect the resources available to both birds and other urban wildlife. Therefore finding ways to assess and estimate urban habitat quality for wildlife will become an increasingly important part of future ecological studies.

One potential solution to access and site restrictions due to the number of different landowners in urban habitats is to utilise the knowledge of those who are themselves interested in wildlife. So called “citizen science” has been used for many years in ornithology, for example to track bird migration routes, and is likely to play a major role in ecology and conservation in the future, particularly with the development of internet and mobile phone based applications that can be utilised to engage with the public and facilitate data collection (Greenwood, 2007; Silvertown, 2009). It is successfully

employed by a number of organisations in the UK, such as the British Trust for Ornithology (BTO), the Royal Society for the Protection of Birds (RSPB), and the Mammals Trust. These organisations ask interested parties to collect information about the species they see in their locality, and submit data either on paper or online. This data has limitations; although those involved are interested enough to submit data, they may not be experts, and therefore the data collected is less detailed than that collected by more thorough surveying techniques, and inaccuracies may arise. In addition, as sites are self selecting, there may be gaps in data coverage, or biases in the areas covered by this means. However, these surveys allow data to be collected on a scale that would be impossible to cover using other means, and because they use standardised methodologies, they allow comparisons to be drawn between different geographical regions, and long term trends to be recorded over a variety of habitats. Encouraging urban landowners themselves to obtain data also allows for better representation of urban areas as a habitat type, as they are underrepresented in early nationwide surveys such as the BTO Common Bird Census (Marchant *et al*; 1990).

Collecting data to monitor the changes that are taking place in urban areas will enable better management of these areas as habitats for both humans and wildlife in the future. This is particularly important given the potential for conflict between those who enjoy the benefits wildlife can bring, and those who wish to develop urban landscapes. Many wild animals in the UK, particularly birds, are protected by law from disturbance. The Wildlife and Countryside Act of 1981 prevents the damage or destruction of wild bird nests or eggs, and places restrictions on disturbance to many other species (DEFRA, 2009). In addition, many species that use urban areas are afforded extra protection by schemes such as the UK Biodiversity Action Plan (BAP, 2009).

These restrictions have important implications for urban developments, as those wishing to disturb sites containing species of conservation concern may be required to mitigate for the effects of the disturbance on vulnerable wildlife populations. In the case of small mammals or protected amphibians this may involve translocating populations to a different site to make way for development projects. However, when dealing with bats or birds the situation is more complex, and in many cases populations or individuals must not be disturbed at all, particularly in the breeding season. For some species mitigation measures to protect them only at certain times of year may not be enough to prevent population decline, and alternative habitat may need to be created in order to replace that which is lost due to development. This is particularly true for species that need specific habitat requirements such as suitable nest sites, or a specific food. Where this is the case, only having a thorough understanding of how the species interacts with the urban habitat will enable appropriate mitigation measures to be adopted.

It has been suggested that studies should focus on gaining a better understanding of how wildlife interacts with the urban landscape by investigating a combination of biotic and non-biotic factors, rather than focusing solely on environmental features (Loss *et al*; 2009). This approach avoids many of the problems associated with collecting data in urban areas, and allows larger scale studies to be undertaken than would be possible by more conventional means. In addition, nationwide surveys of both biotic and abiotic factors are increasingly available, and the standardised methodology they employ allows comparisons to be made between urban areas in different regions. Although obtaining direct measures of urban development is problematic, we argue that other indices provide a useful indication of the effects of urban habitat change.

One source of such data in the UK is data derived from census statistics, which have been used in a number of studies of urban land use change (E.g. Pauleit *et al*; 2005, Tratolos *et al*; 2007). Measures of urban development encompass the relationships between a number of habitat and environmental variables. These would be difficult to collect directly, and the complex interactions between different groups of environmental variables would be extremely difficult to quantify by other means, given that their effects are likely to vary both within and between cities.

THE HOUSE SPARROW, *PASSER DOMESTICUS*: AN URBAN SPECIALIST:

This thesis examines the potential role of urban habitat change in the decline of an urban specialist, the house sparrow (*Passer domesticus*, Linnaeus). House sparrows have been associated with human habitation since the Bronze Age (Ericson *et al*; 1997). The species is common and widespread, and is traditionally associated with human habitation, both in agricultural and urban areas. In the UK two thirds of the house sparrow population is associated with urban areas (Siriwardena *et al*; 2002). However, the species has declined across its core range in North Western Europe since the 1970s, and in recent years urban populations have declined particularly strongly.

The house sparrow is typically 14-15cm in length, with a wingspan of 21-25.5cm (Snow, Perrins *et al*; 1998). The male is brown, with a grey crown and rump, and a black bib and eye stripe, contrasting with greyish-white underparts. The female is pale brown, and lacks the distinguishing characteristics of the male, with the exception of two pale wing bars that are found in both sexes. Seeds form the bulk of the diet, but vegetable food, human refuse and insects are also eaten, with the latter being especially important as a source of food for chicks during the breeding season (Summers-Smith, 1988; Snow, Perrins *et al*; 1998).

House sparrows are a gregarious, colonial species, but are also highly sedentary, rarely moving more than 1km from their colony site (Crick, 2002). Pairs remain together from year to year, and are generally monogamous, although polygamy does occur. Nests are often sited in cavities found under the eaves of buildings, but are also constructed in trees and hedges. However, the adaptable nature of the house sparrow enables it to utilize a wide variety of nest sites and nesting materials (Indykiewicz, 1991). Egg-laying begins in April in Europe, and pairs can produce up to four broods a year although two or three is more usual. Clutch size is generally between three and five, and both sexes take part in nest building, incubating and subsequent feeding of the young. Incubation lasts between 11-14 days, and nestlings fledge between 11 and 19 days after hatching, after which they continue to receive food from the parent birds for up to two weeks (Snow *et al*; 1998).

The house sparrow has been chosen as the focus for this thesis because it is an urban specialist with a wide distribution covering almost every continent. It is very successful as an introduced species, and was previously considered a pest (Crick, 2002). However, the species has declined in urban areas by over 50% in recent decades, and is currently a species of conservation concern in the UK and Western Europe. Chapter Two of this thesis discusses the decline of the house sparrow in more detail, and reviews the current distribution of the species within urban areas with a view to identifying potential mechanisms for the decline. In particular, the possibility of a link between urban deprivation and the occurrence of house sparrows in cities is discussed in relation to anecdotal evidence obtained from across Europe. Chapter Three uses data on house and garden features in a UK city, Bristol, to look for a link between the perception of house sparrow occurrence in the city and urban habitat features that may explain the apparent

link with deprivation. Chapter Four uses data from cities across England to investigate whether a nationwide measure of deprivation status is linked to house sparrow occurrence in urban areas. Chapter Five explores the possibility that differences in land use may be linked to deprivation and house sparrow occurrence in cities. Chapter Six reports the findings of a radio-tagging study that sheds light on the type of habitat in which sparrows spend the majority of their time, and the likely home range of adult birds on a day-to-day basis. Chapter Seven discusses the observed localised and nationwide habitat changes in urban house sparrow occurrence in relation to socioeconomic deprivation. The implications of these changes on future development policies are also considered.

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OVERALL AIM OF THESIS:

To use both national and local scale data in order to investigate the potential causes of decline of the house sparrow, *Passer domesticus* in urban areas. In particular, to look for large-scale trends in house sparrow occurrence in relation to human socioeconomic conditions with a view to identifying factors which may be causing the decline; and areas where further declines may be expected in the future. In addition, to use data collected at a more localised scale to determine whether particular human socioeconomic factors are likely to affect habitat quality for house sparrows in an urban environment.

OBJECTIVES:

1. To review the changes to urban house sparrow populations since the 1970s at a local, national, and where appropriate an international scale. In addition, to determine whether the current evidence supports a potential link between large-scale trends in house sparrow occurrence and human socioeconomic conditions in urban areas.
2. To collect data on house sparrow occurrence on a local (i.e. within city) scale, in order to determine whether particular aspects of urban houses and gardens were, firstly, associated with house sparrow occurrence at a local scale, and secondly, associated with urban deprivation.
3. To use large scale data from the British Trust for Ornithology GardenBirdWatch, and datasets from the Office for National Statistics to determine whether there is a link between the current distribution of house sparrows in urban areas and variation in the level of human socioeconomic deprivation at a nationwide scale.
4. To use nationwide data from the GardenBirdWatch survey, and the Office for National Statistics to look for other potential factors that may be linked to house sparrow occurrence on a large scale. To use both these data and data on socioeconomic deprivation to define areas where further declines are likely to occur in house sparrow populations at the city-wide and nationwide level.
5. To collect information on the occurrence of house sparrows at a localised scale by determining the home ranges of urban house sparrows by means of a small radio tracking study. In addition, to look for habitat preferences of house sparrows for particular land use and housing types within their actual and potential home range.

AUTHOR'S DECLARATION:

CONTRIBUTION OF THIS WORK TO THE FIELD OF STUDY:

1. The publication of an updated review of current work investigating the causes of the decline of the urban house sparrow, drawing attention to the link that human socioeconomic deprivation has with the current distribution of house sparrows in urban areas. (Citation: Shaw, L., Chamberlain, D. & Evans, M. (2008). The house sparrow *Passer domesticus* in urban areas: reviewing a possible link between post-decline distribution and human socio-economic status. *Journal of Ornithology* **149** 293–299. DOI: 10.1007/s10336-008-0285-y).
2. Completion of the first statistical analysis to confirm anecdotal evidence that house sparrow occurrence in urban areas is linked to human socioeconomic deprivation at a nationwide scale. In addition, this work presents evidence that areas which show increases in their socioeconomic status over time are likely to report a consequent reduction in sightings of house sparrows via the BTO GardenBirdWatch survey.
3. The completion of the first statistical analysis to link the presence of house sparrows in urban areas to land use factors at a nationwide scale.
4. The completion of an analysis of house and garden features that may affect house sparrow populations in relation to socioeconomic deprivation.
5. The completion of a novel radio tracking study to determine the home range and habitat use of house sparrows in an urban setting.

STATEMENT OF CONTRIBUTION MADE:

CHAPTER 1: The introduction was researched and written by myself with assistance from Dr Dan Chamberlain and Professor Matthew Evans.

CHAPTER 2: This review is my own interpretation of the current literature, and was written and submitted for publication by myself, with guidance and feedback from Professor Matthew Evans, and Dr Dan Chamberlain. I am lead author on the published manuscript.

CHAPTER 3: I collected data with assistance from John Tully, Richard Bland and local groups in Bristol. I analysed all data and wrote the manuscript with guidance from Professor Matthew Evans and Dr Dan Chamberlain.

CHAPTER 4: The analysis of house sparrow occurrence in relation to deprivation was carried out by myself with the assistance of Dr David Carslake (University of Exeter), and advice from Dan Chamberlain, Mike Toms and Steve Freeman of the British Trust for Ornithology. I am lead author on the manuscript, and have prepared it for publication with the assistance of Professor Matthew Evans and Dr Dan Chamberlain.

CHAPTER 5: Analysis of house sparrow occurrence in relation to land use was conducted by myself with guidance from Professor Matthew Evans and Dr Dan Chamberlain. I am lead author on the manuscript.

CHAPTER 6: I carried out all fieldwork, with assistance from Dr Andy McGowan in catching and tagging birds. Analysis of the resulting data was carried out by myself with assistance from Dr Matthew Witt (University of Exeter), Dr Dan Chamberlain and staff at the British Trust for Ornithology.

CHAPTER 7: The discussion is my interpretation of the findings of this thesis, and of the relevant literature. Professor Matthew Evans and Dr Dan Chamberlain provided advice and assistance.

CHAPTER 2

THE HOUSE SPARROW, *PASSER DOMESTICUS* IN URBAN AREAS: REVIEWING A POSSIBLE LINK BETWEEN POST- DECLINE DISTRIBUTION AND HUMAN SOCIOECONOMIC STATUS

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Published in the *Journal of Ornithology* (2008) **149** 293-299.

DOI 10.1007/s10336-008-0285-y

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Summary:

The house sparrow *Passer domesticus* is traditionally associated with human habitation. However, the species has undergone dramatic declines in many urban areas in north-western Europe. There are many theories as to why this decline has occurred, but the lack of data on house sparrow numbers prior to their decline has hampered efforts to investigate these theories in detail. This review summarises the demographic changes in urban house sparrow populations since the 1970s, and considers evidence that the current distribution of house sparrows may reflect changes in urban habitats caused by socioeconomic change. Evidence is mounting that within urban landscapes, house sparrows appear to be more prevalent in areas with a relatively low human socioeconomic status. Here we present evidence to suggest that house sparrows may have disappeared predominantly from more affluent areas, and that these areas are more likely to have undergone changes to habitat structure. We also show how these changes in habitat could influence house sparrow populations via impacts upon nesting success, foraging and predation risk.

Keywords: brownfield sites, deprivation, habitat, regeneration

As a landscape becomes more urbanised, avian population density often increases. However, species diversity tends to decrease in comparison to other habitat types (Beissinger & Osbourne 1982; Blair 1996; Marzluff *et al*; 2001). The house sparrow (*Passer domesticus*, Linnaeus) is traditionally associated with human habitation, and is one of the few species that generally persists as urbanisation increases. It is a widespread species, but in recent years there has been an abrupt decline in house sparrow numbers in urban areas throughout north-western Europe, the cause of which is not known at present, although several hypotheses have been suggested (Summers-Smith 2003).

House sparrows were so common they were regarded as a pest species during the 18th and 19th centuries (Crick 2002), particularly in towns and cities. However, populations have been declining in many urban areas since the mid 1980's (Crick 2002). In the UK this trend is particularly apparent in the south and east, which hold just over 50% of the sparrow population (Siriwardena *et al*; 2002). The overall population of UK house sparrows is estimated to have declined from approximately 13 million pairs in the early 1970's to around 6 million pairs by the late 1990's (Crick *et al*; 2002; Robinson *et al*; 2005). This decline is so widespread that the house sparrow is currently on the red list reserved for species of special conservation concern in the UK (Gregory *et al*; 2003). Similar declines have been reported in urban areas throughout north-western Europe (e.g. Prowse 2002, Mulsow 2005; 2006, Stastny *et al*; 2005; Heij 2006). As a result, the house sparrow is now listed as a species of European conservation concern (SPEC category 3, Birdlife International, 2006). Although little historical data is available, population monitoring of house sparrows has increased since the 1970s, and a more detailed picture of the state of the house sparrow in Europe is beginning to emerge.

Here we review the demographic changes to house sparrow populations in the UK, and summarise the changes to populations in European cities where data is available. We discuss recent evidence suggesting that the current distribution of house sparrows in many urban areas is linked to human socioeconomic status, and consider potential mechanisms to explain a possible link between house sparrow populations and social deprivation. We focus on three ways in which socioeconomic status could influence house sparrow populations: by creating differences in habitat structure which may impact upon foraging success; by causing indirect effects such as increased predation risk; and potential effects on nest site availability. Finally, we consider how urban regeneration and redevelopment may affect house sparrow populations on a city wide scale.

DEMOGRAPHIC CHANGES

Population monitoring in the UK has provided a relatively detailed picture of house sparrow numbers. A number of surveys have shown declines both in the wider countryside (Marchant *et al*; 1990; Robinson *et al*; 2005, Raven & Noble 2006) and especially in private gardens (Cannon *et al*; 2005, Chamberlain *et al*; 2005, Toms 2006) since the mid to late 1970s. There is evidence that declines vary both geographically and according to habitat. Raven & Noble (2006) showed that declines are occurring in most urbanized parts of England, whereas in Scotland and Wales, populations are increasing. Furthermore, Chamberlain *et al*; (2005) suggest that declines started earlier in suburban gardens than rural gardens. The timing of the decline in urban areas and the sedentary nature of house sparrows would suggest that urban and rural populations are relatively isolated from each other, and that the urban population decline is not a result of decreased immigration from farmland areas as a result of agricultural intensification (Siriwardena *et al*; 2002). House sparrow populations within a city are likely to overlap

more than those in rural areas, as they exist in closer proximity to each other. However, adult house sparrows rarely disperse over distances of more than 2km even outside the breeding season (Snow *et al*; 1998). A genetic study of four rural house sparrow populations showed very restricted movement between populations (Hole *et al*; 2002), which suggests that rural house sparrows are unlikely to be a source of recruitment to the urban breeding population.

Data from long-term UK-wide nest monitoring under the Nest Record Scheme (NRS) show that nest failure rates in house sparrows were higher in urban/suburban than rural habitats (Crick 2002). However, a caveat on the interpretation of NRS data is that measures are only per nesting attempt, rather than over a whole season. If, for example, a lack of food led to fewer nesting attempts and lower overall productivity per breeding season, this would not be detected. This, and differences in population growth rates in relation to breeding performance suggest that the mechanisms driving population trends in rural and urban/suburban habitats differ, and it is therefore appropriate to consider the two habitats separately.

Detailed data on house sparrow populations in other European countries is not as readily available as for the UK; however, population monitoring in individual cities has also provided evidence for a broad scale population decline. This appears to be the case for many cities across north-western Europe, although in others no evidence of a decline has been noted. Table 1 provides a summary of cities for which trends in the house sparrow population are available.

A LINK TO SOCIOECONOMIC STATUS

The distribution of sparrows within urban areas is often very patchy, and recent evidence suggests that human socioeconomic status may provide a partial explanation for this. It appears that the remaining house sparrow colonies are biased towards relatively deprived areas in many cities. For example in Bristol, UK, sparrows are still common in the most deprived areas of the city as measured by the English Indices of Deprivation (Noble *et al*; 2004), but they are almost completely absent from areas which have a higher socioeconomic status (Bland 1998). Similarly, in Norwich, most remaining sparrow colonies are to be found on council estates in the north of the city (Paston 2000). Evidence from Paris and Berlin also links house sparrows with deprived areas (Malher, unpublished data; Witt 2000; 2005). If this link is confirmed, it is likely that changes to areas with a high socioeconomic status have contributed to the decline of sparrow populations in those areas, whilst lower status areas have been relatively unaffected. Recent evidence suggests that areas of varying socioeconomic status do experience different levels of habitat change over time (Pauleit *et al*; 2005). Possible changes in habitat, and their effects on biodiversity are considered below, with particular reference to the potential impacts on house sparrow populations in areas of differing socioeconomic status.

GARDEN MANAGEMENT

Private gardens provide substantial resources for urban birds. However in recent years an increase in home improvements and demand for off-road parking has led to changes in habitat structure within gardens (Pauleit *et al*; 2005). The presence of rough grass, nettles and weedy patches is linked to increased bird diversity in urban areas (Chamberlain *et al*; 2007a), which suggests that tidy gardens with high proportion of paving are less suitable habitat for foraging birds. This may cause problems for adult

sparrows in the breeding season, when insect-based food is required for nestlings.

Whilst there have been few studies of urban arthropods (McIntyre 2000), Vincent (2005) found that sparrow fledging success was higher in areas where insect abundance was high, and chicks were less likely to starve when fed a high proportion of insectivorous as opposed to vegetable food. This study also showed that urban house sparrows have more success in fledging young in areas containing a high proportion of deciduous shrubs, and relatively little concrete.

It is likely that gardens in more deprived areas will consist of more suitable habitat for house sparrows than those in areas with a higher socioeconomic status. Relatively deprived areas are likely to contain a high proportion of native shrubs, and traditional council estates, provide large blocks of garden habitat (especially back-to-back terraced housing) which can provide cover and insects for foraging birds. Evidence from areas such as Norwich, UK appears to support this theory (Paston 2000). However, in more affluent areas, the planting of predominantly ornamental shrubs and increased demand for off-street parking is likely to reduce the amount of habitat available to house sparrows. In London for example, it is estimated that around two thirds of front gardens are now at least partially covered by paving as opposed to vegetation (London Assembly, 2005a). A recent study of land cover change in Merseyside, UK, noted that loss of front gardens in order to provide paved parking areas was of particular concern in more affluent areas (Pauleit *et al*; 2005).

INDIRECT EFFECTS OF HABITAT STRUCTURE

Reducing the amount of foraging habitat available to house sparrows is also likely to have indirect effects on survival and productivity. Factors such as predation and pollution are likely to affect urban birds at a city wide scale, but may have a greater

impact in areas of high socioeconomic status, where loss of greenery is already a problem. The interaction between predation risk and habitat structure has been documented by Whittingham and Evans (2004) in farmland habitats. This paper suggests that a mosaic of short and long grass may provide the optimal habitat to maximise foraging efficiency. Urban gardens also have a mosaic-like structure. However, the trend towards replacing native shrubs with paving or concrete (London Assembly 2005a) could cause the habitat structure of some areas to become more homogeneous, thereby increasing predation risk for foraging birds.

Sparrowhawks *Accipiter nisus* are common predators in urban habitats (Chamberlain *et al*; 2007b). House sparrows appear to be especially vulnerable to predation by sparrowhawks (Götmark and Post, 1996), however, there is no evidence that declines in house sparrow abundance in urban gardens are linked to the recent population increase of sparrowhawks in the UK (Chamberlain *et al*; 2009). Domestic cats *Felis catus* are also capable of having a substantial negative impact on house sparrow populations, and this has been identified as a possible factor in their decline (Churcher & Lawton 1987; Woods *et al*; 2003). The UK population of cats was estimated to be capable of predating 27 million birds in a 5 month period during 1997 (Woods *et al*; 2003), and a recent study in Bristol by Baker *et al* (2005) found that house sparrows were among the few species where predation rates by cats were high enough to have a detrimental effect on the local population. In areas where individuals are under increased foraging pressure due to suboptimal habitat, the risk of predation by cats or other birds may increase enough to affect population levels. Even if survival of adult birds is not immediately affected by increased predation risk, the survival of juvenile birds may be affected if adults spend more time avoiding predation.

Evidence from previous studies suggests that nest failure rates are high in some urban areas due to increased foraging or predation pressure. Although house sparrow nest failure rates declined between 1975 and 2000 (Crick 2002), overall failure rates were significantly higher in urban/suburban habitats. This could indicate greater predation pressure in these habitats, but information on the cause of failures (e.g. predation, starvation or other factors) is not available. In addition, evidence from a study of sparrows in Leicester, UK showed that the weight of chicks just prior to fledging was relatively low (Vincent 2005), a factor which is known to be linked to decreased post-fledging survival in a number of passerines (e.g. Garnet 1981, Magrath 1991). This, combined with a lack of food in late summer, could mean that recruitment of young birds to the breeding population is very low in some urban areas, particularly those with a low proportion of greenery.

It should also be noted that there may be more subtle effects of increased predation pressure that may alter birds' behaviour, including habitat selection (Suhonen 1993), flocking tendency in winter (Székely *et al*; 1989), and fat loading (Gosler *et al*; 1995). A recent study by MacLeod *et al* (2006) found that adult sparrows in the UK appear to limit their uptake of fat in winter, suggesting they may be subject to mass-dependent predation effects. Increased predation may therefore have subtle and indirect effects that are not detectable through simple correlative studies, but may be enough to affect the survival and future reproductive success of individuals, particularly in suboptimal habitat. In areas with a low level of social deprivation, it is therefore possible that loss of vegetation in urban gardens may increase predation risk to a level that the population cannot sustain.

NEST SITE AVAILABILITY

House sparrows often nest in the roof cavity of buildings, either under the tiles themselves or in and around the soffits and fascias. There is evidence to suggest that certain tile types and building methods may provide more suitable nesting sites for sparrows than others, and that these more suitable nesting spaces may be more abundant in areas with a low socioeconomic status. A recent small-scale survey of sparrow nests in Bristol, UK, found that house sparrows were strongly associated with roof tile types that are curved, as these provide large holes which are more accessible to sparrows (J. Tully, pers. comm.). In addition, plastic is replacing wood and cast iron as the material commonly used for soffits and fascias of houses. As wooden fascia boards rot, they create holes that can be used by nesting birds. Replacing rotten wooden boards with hard wearing plastic ones may lead to a substantial decrease in the number of nest holes available on a city-wide scale. Therefore older houses, or those in a worse state of repair, are likely to be more suitable as nesting sites.

Evidence linking breeding house sparrows to houses in poor condition exists in a number of urban areas. A nationwide survey of UK houses found that older (pre-1919) houses were important for hole-nesting birds such as sparrows, and that houses which had recently had repairs to the roofs were less likely to be associated with nesting birds (Wotton *et al*; 2002). House sparrows appeared to avoid newer (post-1985) buildings, and were also much less likely to occur in houses that were built post-1945, and those that had undergone roof repairs in the last 10 years. In Harwich, UK, house sparrows are also associated with post-war housing, but not with newer developments (Mason 2006). Less information is available for European cities, but in Vienna and Valencia, house sparrows are also associated with damaged and neglected buildings (Murgui, 2005; Sziemer & Holzer, 2005). House sparrows in Vienna are seen in old and new

developments, but breeding sparrows are commonly associated with the older buildings (Sziemer & Holzer, 2005). As neglected buildings are associated predominantly with areas of low socioeconomic status, it appears that these areas may provide more opportunities for nesting sparrows than areas which have newer buildings or those which have undergone repairs or regeneration. The loss of sparrow colonies from areas following regeneration has been recorded in Berlin (Witt 2000; 2005), which appears to support a possible link between changing socioeconomic status and house sparrow population decline.

THE WIDER URBAN ENVIRONMENT

The loss of brownfield sites and green space from urban areas may exacerbate pressure on house sparrows caused by loss of garden habitat in areas of high socioeconomic status. In addition, increasing levels of development and regeneration in many urban areas are likely to improve the socioeconomic status of human residents, but decrease the available habitat for house sparrows. The London Assembly (2005b) defines brownfield sites as “previously developed land... currently or previously occupied by a permanent dwelling”. These sites are often very rich in invertebrates and plants, and provide islands of cover in an urban landscape (Small *et al*; 2002; Eyre *et al*; 2003, Mason 2006, Muratet *et al*; 2007). The UK government is currently very keen to promote the use of these brownfield sites for development, and in 2001, 61% of new developments were built on brownfield sites (Office of the Deputy Prime Minister, 2003). Similarly, in Paris, the proportion of such land has decreased by 24% in the last 20 years (Muratet *et al*; 2007). Regeneration, whilst likely to improve socioeconomic conditions for human inhabitants, is therefore also reducing the amount of foraging habitat available to urban house sparrows. This is likely to lead to colony loss in areas where human socioeconomic conditions improve.

Green space that has not been previously developed is also under threat in many urban areas due to regeneration and 'infilling', which increases housing density due to the purchase of private gardens for development. Loss of green space in Merseyside by this means was greater in more affluent areas between 1975 and 2000 (Pauleit *et al*; 2005), when the urban house sparrow population also declined. However, the current demand for housing now suggests that areas where socioeconomic conditions are relatively poor - and therefore where sparrow numbers are potentially high - may become targets for regeneration. A recent study of randomly selected urban sites throughout the UK identified residential areas and allotments as key habitats for house sparrows, but predicted that a small increase in the proportion of flats (as opposed to houses with gardens) would be enough to cause rapid declines in house sparrow abundance (Chamberlain *et al*; 2007b). Allotments in particular are prime targets for development in London (London Assembly 2006). New housing developments are currently being built at an average density of 22 dwellings per hectare (Entec UK, 2004), and this is planned to increase to around 30 dwellings per hectare in areas such as the south east, where pressure on land is greatest (Office of the Deputy Prime Minister 2003). This is likely to reduce the foraging habitat available to sparrows in many areas undergoing redevelopment. If the trend towards increased housing density in urban areas continues, it is therefore likely that rapid declines in the house sparrow population will continue in areas where redevelopment is taking place.

Redevelopment may also reduce the number of potential nest sites available even if the number of houses increases, as modern roofs are likely to provide fewer cavities than older ones, or those in poor condition. It is also worthwhile to note however, that sparrow populations may be lost even in areas where there are nesting opportunities and

resources available. As sparrows are colonial nesters, a small decrease in the size of a colony may inhibit breeding in the remaining individuals. This could lead to the loss of the colony altogether (Summers-Smith 2003). This phenomenon is known as the Allee effect, and has been documented in a number of species (Veit & Lewis 1996). In the case of house sparrows, the increasingly patchy distribution of the remaining urban colonies in many cities has increased the probability that this effect may occur and exacerbate any decline. Once a breeding colony has been lost, the sedentary nature of house sparrows may also make it difficult for recolonisation to occur. Redevelopment of pockets of land in areas of low socioeconomic status could therefore affect adjacent house sparrow colonies, by reducing the overall number of nesting sites or foraging opportunities below the level at which the colonies can be sustained.

To date, attempts to determine the causes of decline in urban house sparrow populations have been through relatively small scale studies, as the complexity of urban habitats makes broad-scale studies difficult. We suggest that human socioeconomic status may provide a starting point for broader scale analyses of the factors affecting urban sparrow populations. In particular, the importance of urban green space and brownfield sites should be considered in relation to foraging success and predation risk, and the structure and condition of houses considered in terms of nesting opportunities. This may allow for the development of mitigation measures in areas undergoing regeneration in the future. In addition, this approach could provide invaluable information to assist the planning of new settlements in order to create an urban landscape that is more sympathetic to the needs of the house sparrow.

ACKNOWLEDGEMENTS:

Thank you to John Tully, Richard Bland and Denis Summers-Smith for information and comments on the draft. This review is the basis of work funded by the Natural Environment Research Council and The British Trust for Ornithology.

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Table 1: A summary of reported house sparrow population trends in European cities.

City	Overall trend	Reference
Berlin	Stable	Witt 2000, 2005
Bristol	Decline (pre 1994)	Bland 1998
Brussels	Stable	Weiserbs & Jacob 2005
Dublin	Decline	Prowse 2002
Edinburgh	Decline	Dott & Brown 2000
Hamburg	Decline	Mulsow 2005, 2006
Lisbon	Increase	Geraldes & Costa 2005
London	Decline	Sanderson 1996
Manchester	Stable	Prowse 2002
Moscow	Decline	Kostantinov & Kakhanov 2005
Norwich	Decline	Paston 2000
Paris	Stable	McCarthy 2006
Prague	Decline	Stastny <i>et al</i> ; 2005
Rotterdam	Decline	Heij 2006
St Petersburg	Decline	Khrabryi 2005
Warsaw	Slight decline	Luniak 2005

CHAPTER 3

ASSESSING URBAN HABITAT QUALITY IN RELATION TO SOCIOECONOMIC DEPRIVATION - IMPLICATIONS FOR SPECIES CONSERVATION

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Summary

Urban areas provide important resources for wildlife, but little is known about how the characteristics of urban households influence habitat quality, or how changes to urban residential areas will affect species occurrence. We aim to assess a number of factors that may contribute to the quality of urban habitats for the house sparrow, an urban specialist. This species currently has a very patchy distribution in many UK urban areas, and this has been linked to the incidence of socioeconomic deprivation in the human population. We find that interwar and detached housing, roof repairs, and the presence of paved areas, including driveways, are all linked to low levels of urban deprivation, and may reduce habitat quality for house sparrows. Insecticide use was higher in relatively deprived areas, but usage generally was low suggesting that this may simply be an indication of higher insect abundance in these areas overall. A direct link between these individual factors and house sparrow sightings was not detected; however a link was detected between house sparrow sightings and socioeconomic deprivation. We suggest that in combination these factors may indicate a reduction in the suitability of urban gardens as habitat for wildlife in neighbourhoods with a low level of social deprivation. This may be due to urban development in these areas, and suggests that future planning applications in urban areas should be carefully considered to mitigate the effects of habitat loss for wildlife due to urban development.

Keywords: biodiversity; conservation; *Passer domesticus*; social deprivation; urbanisation.

Introduction

Urban areas provide important resources for wildlife and have the potential to support high levels of biodiversity. Private gardens and brownfield sites support a large variety of invertebrate species (Cannon *et al*; 2005, Gaston *et al*; 2005, Smith *et al*; 2006), and can also support a wide variety of plants (Thompson *et al*; 2003, 2004). They are therefore also important as a resource for birds, particularly in the breeding season (Bland *et al*; 2004; Mason 2000). Many householders also provide a variety of supplementary food for birds, a factor that is linked to high avian species richness and abundance (Fuller *et al*; 2008). Additionally, urban buildings and artificially provided nest boxes are a valuable resource for cavity nesting birds. Maximising the provision for wildlife in both new and existing developments is therefore an important ecological problem, with high relevance to UK planning policy (Sutherland *et al*; 2006).

In spite of this, habitat quality is likely to vary substantially between different residential areas. This variation can be difficult to quantify as sight lines in urban areas are often obstructed and access to private gardens is restricted, thereby hampering efforts to survey urban habitats (Cannon, 2005). However, this variation is likely to be important as housing density in modern developments increases, and the amount of green space available to urban birds declines. It is therefore crucial to identify and address areas where habitat quality is declining, and to understand the mechanisms by which gardens may become unsuitable for urban wildlife.

The house sparrow *Passer domesticus* provides a good starting point for investigating urban habitat quality. The recent urban decline in this species throughout north-western Europe is well documented (Siriwardena, Robinson & Crick, 2002; Summers-Smith, 2003). House sparrows are relatively sedentary birds, and are strongly associated with

residential areas and gardens (Chamberlain *et al*; 2007). These areas must therefore provide suitable cavities for nesting, granivorous food for adults, and insectivorous food for chicks during the breeding season if they are to sustain colonies throughout the year. Availability of these resources will vary according to the habitat quality of individual gardens, and it is therefore important to know which features of urban gardens cause this variation. Previous work has already suggested factors that may affect habitat quality for sparrows in urban areas, including insecticide use, food availability and predation (Summers-Smith, 2003; Baker, 2005; Wilkinson 2006; Peach *et al*; 2008).

There is evidence that some of these small-scale factors may be linked to wider measures of habitat variation. Garden vegetation, particularly native plants, are linked to insect abundance and diversity (Smith *et al*; 2006, Wilkinson 2006). In addition, the incidence of supplementary feeding in gardens is known to differ according to the socioeconomic status of the area (Fuller *et al*; 2008). The availability of nesting sites in the eaves of houses; and of invertebrate food for nestlings may also vary across gardens in areas with differing socioeconomic status (Shaw *et al*; 2008). There is anecdotal evidence to suggest that house sparrow occurrence in cities may follow this pattern, potentially due to the effects of development on gardens in less deprived regions (Pauliet *et al*; 2005, Shaw *et al*; 2008). If so, socioeconomic deprivation may act as a predictor of urban habitat quality, and potentially provide a relatively easy way to obtain large-scale yet detailed information on the suitability of urban habitats for wildlife.

Here we aim to look for a link between aspects of urban habitat quality, land use, deprivation status, and house sparrow occurrence. Firstly, we aim to identify features of urban houses and gardens that may combine to affect urban habitat quality, and determine whether these are linked to the perceived occurrence of house sparrows in

Bristol, UK. We then determine whether any of these features are linked to urban deprivation, and consider reasons why deprived areas may appear to be ‘good’ habitats for wildlife.

Materials and methods:

STUDY AREA:

Bristol is a large ($\sim 110\text{km}^2$) conurbation in the south-west of England, with a population of approximately 400, 000 people (Bristol City Council, 2008). It is an area in which house sparrows were numerous prior to the 1980s, but the species showed a marked decline in the city as a whole between 1980 and 2005. The population is currently stable, but the distribution of house sparrows in the city is patchy, and recolonisation does not appear to be occurring in those areas where sparrows are no longer present (Bland; 1998). The city also shows large variations in socioeconomic status, with some parts ranking in the lowest 10% of deprived areas in England, but others in the highest 10% with regard to deprivation (Bristol City Council, 2008). Bristol also consists of a number of different phases of development, including large areas of Victorian and post-war housing, and substantial pockets of modern redevelopment. As such it provides an ideal opportunity to study house sparrow occurrence in relation to urban habitat type.

DATA SOURCES:

Data on house sparrow occurrence, and house and garden features in Bristol were obtained from questionnaire surveys distributed via local ornithological and wildlife groups in 2006 and 2007 (Appendix 3A). Householders throughout Bristol were asked to complete a number of questions and return the answers either by post, email or online via a local wildlife group website (<http://www.troopers-hill.org.uk>). The questionnaires aimed to collect information regarding house and garden features in the Bristol area, and

on the perception of house sparrow abundance in relation to these features. Information collected included the age and type of housing occupied; the presence of cats in the neighbourhood; the incidence of supplementary feeding; the presence of garden features that may help or hinder wild bird populations; and the use of insecticides. Data on house type and age was grouped into categories used by national statistics census data on housing (for details see www.statistics.gov.uk).

Each respondent was asked to provide their postcode, which was then used to identify the Lower Super Output Area (LSOA) in which they were situated. LSOA's are the smallest geographical unit at which census data are collected, and are subdivisions of electoral wards. They vary in size, but contain on average 1500 dwellings. Responses from outside the Bristol urban boundary were discounted, along with those that provided no information on location. Each questionnaire location was then assigned a value for both deprivation and land use characteristics according to their LSOA.

Deprivation status was measured by the Index of Multiple Deprivation (IMD), obtained from the English Indices of Deprivation for 2004 (Noble, 2004, Communities and Local Government 2007). This index comprises seven measures of deprivation, which are combined to produce an overall score. IMD scores range from 1–82 (including non-integers) with higher scores indicating a higher level of deprivation. For further information on the measures of deprivation used see Noble *et al*; (2004). Land use data were obtained from the Office for National Statistics (ONS, Communities and Local Government, 2007). The dataset used for this analysis was the “Land Use Statistics (Generalised Land Use Database) 2005”, and is measured in thousands of square metres, giving accuracy to the nearest 10m^2 . Further details of the land use data are given in Chapter 5.

DATA ANALYSIS:

Two measures of house sparrow occurrence were used as response variables in the models, both of which were derived from the original questionnaire responses. The first distinguished between those sites that reported sighting sparrows versus those that did not (“sighted”; “not”), and the second between those that sighted sparrows on a regular basis, versus those that sighted sparrows either infrequently, or not at all (“often”; “not often”). In order to check that these are reasonable estimates of house sparrow abundance in Bristol, the perception of house sparrow occurrence according to the questionnaire responses was compared to standardised Breeding Bird Survey (BBS, BTO/JNCC/RSPB; 2008) style counts of house sparrow presence collected by volunteers from 1995 to the present for 1km Ordnance survey grid squares throughout Bristol (Tully & Bland, unpublished data).

The Index of Multiple Deprivation for 2004 was analysed separately in relation to house and garden features obtained from the 2006 questionnaire surveys. Analyses of Variance (ANOVAs) were used to test for differences in $\log_{10}(\text{IMD})$ according firstly to factors associated with the age and type of housing, and the extent of maintenance on properties; and secondly to factors associated with the management of gardens such as the availability of feeders, the presence of driveways or paved areas, cat presence and the use of insecticides. The occurrence of house sparrows and nests were also included in the models to test for an association between these features and sightings of sparrows in gardens. Models were analysed in R 2.7.2 (R Development Core Team 2007), and simplified by stepwise deletion.

Deprivation as measured by the IMD 2004; and the area of the land use types listed above were also modelled in relation to the perception of house sparrow abundance obtained from the questionnaire surveys. Data were modelled using ordinal logistic regression and general linear modelling, with either a three level response variable (ordinal logistic regression) or a binomial response (GLMs), with a log link function. Log transformations were used where appropriate prior to analysis where land use variables were not normally distributed.

Ordination analysis of land use variables at a nationwide scale has suggested a degree of autocorrelation between the different aspects of land use, particularly domestic housing and gardens (See Chapter 5 for further details). However, each aspect of land use was analysed separately in relation to house sparrow occurrence as each is likely to reflect a separate aspect of house sparrow behaviour – for example, buildings are likely to be important for nesting birds, but gardens will be more important in relation to foraging behaviour. The total area of each LSOA was also included in the land use models as a covariate to allow for the effects of variation in total LSOA size on the area of each land use type. Data was analysed in R 2.7.2 (R Development Core Team 2007). Models were simplified using stepwise deletion, and the Akaike Information Criterion (AIC) values for each model were compared to determine the best fit; the models with the lowest AIC score being deemed the most parsimonious (Burnham and Anderson, 1998). The log-odds ratios obtained from the models were then back-transformed to give the probabilities of detecting house sparrows at GBW sites according to IMD score and land use.

Results:

QUESTIONNAIRE RESPONSES:

A total of 608 questionnaires were returned out of the 4000 that were distributed. This approximated to a return rate of 15%; however 33 responses were excluded from the study as they did not provide spatial reference information, leaving a total of 576 usable questionnaires for analysis. Questionnaire returns were received from across Bristol, although areas with high levels of deprivation were relatively under-represented. This may be because those in deprived areas are more likely to be short-term residents, and are less likely to feed garden birds (Bristol City Council, 2008, Fuller *et al*; 2008). In spite of this, deprivation scores ranged from 3 to 77, reflecting the range of scores reported at a nationwide level (Noble *et al*; 2004). The LSOAs covered in the survey had a mean deprivation score of 21 ± 0.55 , $n = 562$, substantially higher than the mean deprivation score of 15.11 ± 0.05 , $n = 40660$, recorded for 16 urban areas nationwide (Chapter 4). This is probably a reflection of the comparatively high variation in deprivation levels seen across the city as a whole.

HOUSING, GARDEN FEATURES AND THE PRESENCE OF HOUSE SPARROWS:

House sparrows were sighted often at nearly half of the properties surveyed, but a significant proportion of respondents did not report seeing sparrows in their gardens at all, and of those who did see sparrows, less than a quarter were certain that they nested on or at the property (Table 1). The age and type of housing stock occupied by survey respondents varied considerably, but pre-and interwar housing were the most common property types surveyed (Table 1).

The vast majority of houses reported either owning, or being frequently visited by cats. In addition, over 90% of respondents reported having either a lawn and/or shrubs and climbing plants, in their garden (Table 1). As a result these features were excluded from

further analysis, as it is likely that the uniformly high prevalence of these features will prevent the detection of any impacts on the house sparrow population (see Sims *et al*; 2008). Approximately half of respondents had mature trees in their gardens, but most did not use insecticides, with only 1% using them regularly. As a result, the use of insecticides was reduced to a two level factor for analysis (i.e. yes or no). Most survey respondents had a bird feeder of some kind in their gardens (Table 1).

HABITAT FEATURES IN RELATION TO SOCIOECONOMIC DEPRIVATION:

The Index of Multiple Deprivation varied significantly in relation to both housing type and age (Table 3, Figure 1). Detached houses are on average situated in significantly less deprived areas than semi-detached or terraced housing in Bristol. Interwar housing (1914-1944) is also significantly less likely to be associated with deprived areas in Bristol than other housing, particularly where householders have made repairs to their properties (Figure 1). The type of tiling used on houses was not significantly linked to deprivation or to the type or age of housing stock (Table 3). Householders who reported seeing house sparrows “often” and those who reported nesting birds lived in significantly more deprived areas on average than those who did not report sighting sparrows or nests.

Householders that reported using insecticides in their gardens on average lived in areas that were significantly more deprived than those that did not. The mean level of deprivation in areas where householders reported having either mature trees or driveways/paving in their gardens was significantly lower than in areas where only grass and shrubs were present (Table 4). Insecticide use and the presence of driveways or paving were not directly linked to either house sparrow presence or nests and deprivation (Figure 2). There was no significant difference in deprivation between

houses that provided supplementary food and those that did not, although those who did have feeders in their gardens lived in slightly less deprived areas on average.

HOUSE SPARROW OCCURRENCE IN RELATION TO HABITAT AND DEPRIVATION:

House sparrow presence as recorded by questionnaire respondents reflected that recorded via standardised BBS-style counts of sparrows in Bristol. The mean rate of house sparrow occurrence according to these counts varied significantly between those addresses that reported sighting sparrows in their gardens “never”, “sometimes” and “often” (Table 2). Mean rates of house sparrow recording on BBS-style surveys were over twice as high in areas where questionnaire respondents reported seeing sparrows “often” than in those where sparrows were “never” sighted by questionnaire respondents (Figure 3). The perception of house sparrow occurrence in gardens by Bristol questionnaire respondents increased significantly with increasing deprivation, although only for sites that reported seeing sparrows “often”, as opposed to infrequently or not at all (ordinal logistic regression, $t = 3.08$, d.f. = 1092, $p < 0.05$). This suggests that the perception of house sparrow occurrence in Bristol according to those surveyed here is a valid indicator of actual sparrow presence at these sites.

The only aspect of land use significantly related to the perception of house sparrow occurrence in Bristol was the area of gardens, which was positively linked to the probability of perceiving sparrows. The probability of sighting house sparrows also increased significantly with overall LSOA area (Table 5). However, the effect of garden size showed a significant interaction with LSOA size, decreasing as overall LSOA size increased. This suggests that the benefits of having a large garden decrease as housing density decreases. The effects of both garden size and overall LSOA size were larger when householders reported sighting sparrows on a regular basis (Table 5) Once again; the effect of garden size was significantly reduced as overall LSOA size increased.

Discussion:

Questionnaires are increasingly used in ecology to collect data on human-wildlife interactions, and it is important to consider possible sources of bias when interpreting the results they yield (White *et al*; 2005). Our results suggest that the perception of house sparrow presence in Bristol as reported by the questionnaire respondents reflects the true pattern of occurrence recorded by more standardised surveys in the city (Tully and Bland, unpublished data). The survey also shows that house sparrows appear to visit gardens in relatively deprived areas more than those in less deprived areas. This reflects anecdotal evidence from cities in the UK and mainland Europe (Shaw *et al*; 2008). Relatively deprived gardens appeared to be slightly under-represented in our survey sample, but Gaston *et al*; (2005) found no difference in the prevalence of wildlife-friendly gardens according to socioeconomic status, so this is not necessarily likely to bias our results.

It would ordinarily be expected that a survey of relatively ‘bird friendly’ gardens would result in overestimation of species occurrence if the results were extended to the area as a whole. This is not necessarily true in the case of this survey, as many urban gardens in less deprived areas still report the absence of house sparrows even when species diversity is otherwise high. However, the occurrence of house sparrows in relatively deprived areas of Bristol as a whole is likely to be slightly lower than reported here, as these gardens are from a relatively wildlife-friendly sample.

DEPRIVATION IN RELATION TO HOUSE AND GARDEN FEATURES:

The finding that terraced and semi-detached housing are more prevalent in deprived areas is not surprising, given that these houses tend to be smaller and cheaper to buy than large detached properties, which make up a relatively small proportion of Bristol

housing stock (Bristol City Council 2008). These houses are likely to present fewer nesting opportunities per house, as they have fewer roof edges available to allow access to cavities in the eaves. However, terraced and semi-detached housing is also popular for the rental market and as social housing, both of which are common types of housing in Bristol (Bristol City Council; 2008). Houses in relatively deprived areas are less likely to be owner occupied, as deprived areas generally have a high proportion of student or local authority housing (Communities and Local Government, 2008). These houses are therefore likely to be in a relatively poor state of repair, and may provide more cavities for nesting than houses that are better maintained. In areas that are less deprived, even houses that provide a large area of suitable roof space for nesting are unlikely to support nesting attempts if there is a lack of suitable foraging habitat in the surrounding area.

The lack of association between tile type and deprivation probably reflects the fact that the majority of houses in Bristol will have had their original roofs replaced at some stage. House sparrow nesting behaviour has previously been linked to curved as opposed to flat tiles (Tully and Bland, unpublished data). It is likely that these differences are on too small a scale to be detected here, but previous evidence suggests that house sparrows are more likely to nest in properties that are either built pre-1919, or in older properties that have not been repaired (Wooton *et al*; 2002), both of which are more likely to have older style, curved tiles. Mean deprivation levels did not differ according to whether or not repairs had been carried out, except where houses were built from 1914-44. In this case, areas where houses had been repaired were situated in significantly less deprived areas than those that were not. Most interwar houses are situated in relatively affluent areas of Bristol (Bristol City Council; 2008).

Householders in these areas are more likely to be owner-occupiers than those in older or

post war houses, and are therefore more likely to invest money in maintaining their houses.

We did not detect a significant difference in the incidence of supplementary feeding between deprived and more affluent areas, contrary to the findings of Fuller *et al* (2008). Our results are likely to have been taken from a relatively bird-friendly sample, which may explain this anomaly. However, although house sparrows readily visit feeders they appear to be more prevalent in areas where feeders are likely to be absent. This supports the view that a lack of food for adult birds does not appear to be influencing the urban decline in this species (Summers-Smith, 2003).

This study found that insecticide use was higher in deprived areas, where sparrows were predominantly sighted. This appears to contradict previous studies suggesting that lack of insect food for nestlings may be influencing the urban decline of the house sparrow (Vincent, 2005). However, even those houses that used insecticides used them sparingly, with only one survey respondent reporting the use of insecticide regularly, and many others using slug pellets alone. Insecticide use may in this case be an indicator of high insect abundance in local gardens, and the infrequent use of insecticides detected here may not be enough to have an impact on the foraging success of birds. Reduction in insecticide use is consequently unlikely to significantly benefit house sparrows (Wilkinson, 2006).

The prevalence of paving within gardens in less deprived areas reflects the findings of previous studies, particularly with regard to off-street car parking (Pauleit *et al*; 2005). Areas with large amounts of paving are likely to provide less suitable habitat for foraging sparrows, as they are likely to contain fewer invertebrates than gardens

predominantly full of native plant species (Wilkinson 2006). This will result in birds being unable to find enough high quality food for their nestlings and may also increase foraging risk and energy expenditure for individual birds, all of which could reduce their survival and breeding success. Such deleterious effects have already been documented in passerines including the house sparrow (Suhonen 1993, Peach *et al*; 2008). This suggests that high levels of paving in urban areas are a potential mechanism to explain the apparent link between house sparrow presence and deprivation in cities.

CONCLUSIONS:

Although most of the factors investigated here do not show direct links with house sparrow presence and/or nesting behaviour, those that differ significantly in relation to deprivation are likely to have an impact on house sparrow occurrence at a wide scale. This study shows that deprived areas have several factors in common that could benefit house sparrow populations, notably a lower prevalence of paving in gardens and housing that has the potential to provide more suitable nesting sites. Areas that currently have a low incidence of deprivation are likely to share many characteristics with new urban developments, as these typically have small gardens and a high proportion of paving in comparison with older housing (Figure 4). Preserving current house sparrow populations, and wider species diversity in urban habitats is likely to depend on how successfully modern urban developments can prevent the loss of suitable nesting and foraging sites for urban birds.

Acknowledgements:

This study was funded by the NERC, and the British Trust for Ornithology. Thank you to John Tully and Richard Bland for help and advice throughout the project, and for sharing their extensive knowledge of house sparrows in the Bristol area. Thank you also to the many organisations who helped to distribute the questionnaires, including the

Avon Wildlife Trust; Bristol Ornithological Society; the Friends of Troopers Hill; and Bristol City Council. Lastly, thank you to all of those who completed the surveys, and provided information on house sparrow populations in the Bristol area.

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Table 1: Summary of questionnaire responses from Bristol residents on the prevalence of house and garden features that may have an impact on habitat quality for urban house sparrows.

Habitat feature	Percentage of householders:
House sparrow sightings:	
never	23
sometimes	32
often	45
House sparrow nests: present	20
House type:	
detached	14
semi-detached	51
terraced	30
House age:	
pre-war (pre-1914)	38
interwar (1914-1944)	30
postwar (1945-1970)	21
modern (post 1970)	11
Tile type:	
flat	66
curved	34
Repairs to roof or guttering (in previous 5 years):	48
Visiting cats: present	83
Mature trees: present	55
Shrubs or climbing plants: present	94
Paved areas/decking: present	72
Bird table or feeder used:	69
Nestbox: present	19
Insecticide used (occasionally)	40

Table 2: Results of Analysis of Variance comparing the perception of house sparrow occurrence in Bristol according to the questionnaire surveys with the results of BBS-style surveys of house sparrow numbers. (Residual deviance = 0.22 on 2244 d.f.)

Variable	Parameter estimate	Standard error	t-value	p-value
Intercept	0.138	0.031	4.485	<0.001 (***)
house sparrow occurrence: “sometimes”	0.088	0.040	2.136	<0.05 (*)
house sparrow occurrence “often”	0.177	0.038	4.669	<0.001 (***)

Table 3: Minimum adequate model showing the relationship between house type, age, and perceived house sparrow occurrence in Bristol gardens with regard to socioeconomic deprivation, as measured by the Index of Multiple Deprivation (IMD) 2004. (ANOVA, response variable – $\log_{10}(\text{IMD}2004)$). N = 450.

Factor	Parameter estimate	F-value	d.f.	P-value
House sparrow occurrence	0.094	16.249	443	<0.005 (***)
house sparrow nests	0.114	14.889	443	<0.005 (***)
House type	0.262	30.304	443	<0.005 (***)
House age	-0.182	11.823	443	<0.005 (***)
Repairs	-0.078	1.011	443	0.315
House age*repairs	0.169	12.129	443	<0.005 (***)
Tile type	0.024	1.136	443	0.287

Table 4: Minimal adequate model showing the relationship between garden features and house sparrow occurrence in Bristol gardens in relation to socioeconomic deprivation, as measured by the Index of Multiple Deprivation (IMD) 2004. (ANOVA, response variable – $\log_{10}(\text{IMD}2004)$). N = 495.

Factor	Parameter estimates	F-value	d.f.	P-value
House sparrow occurrence	0.085	13.04	490	<0.005 (***)
house sparrow nests	0.109	3.46	487	<0.05(*)
Insecticide use	-0.063	6.89	490	<0.01 (**)
Driveways and paving	0.085	10.35	490	<0.001 (***)
Mature trees	0.056	5.62	490	<0.05(*)
Supplementary food	0.027	2.88	488	0.09
mature trees : house sparrow nests	-0.019	3.84	487	<0.01 (*)

Table 5: Parameter estimates for binomial General Linear Models relating the perception of house sparrow occurrence in Bristol to aspects of land use. Only significant effects are shown. † denotes model where house sparrows were reported as “sighted” or “not”, θ denotes model where house sparrows were either reported on a regular basis, or seen rarely/not at all.

Variable	parameter estimate	Z-value	residual deviance / d.f.	P-value
area of gardens†	0.546	2.51	117.89 / 544	<0.01 (**)
LSOA area†	0.932	2.76	117.89 / 544	<0.005 (***)
area of gardens * LSOA area†	-0.090	-2.52	117.89 / 544	<0.05 (*)
area of gardens θ	0.593	2.72	172.75 / 544	<0.01 (**)
LSOA area θ	1.128	3.33	172.75 / 544	<0.001 (***)
area of gardens * LSOA area θ	-0.104	-2.92	172.75 / 544	<0.005(***)

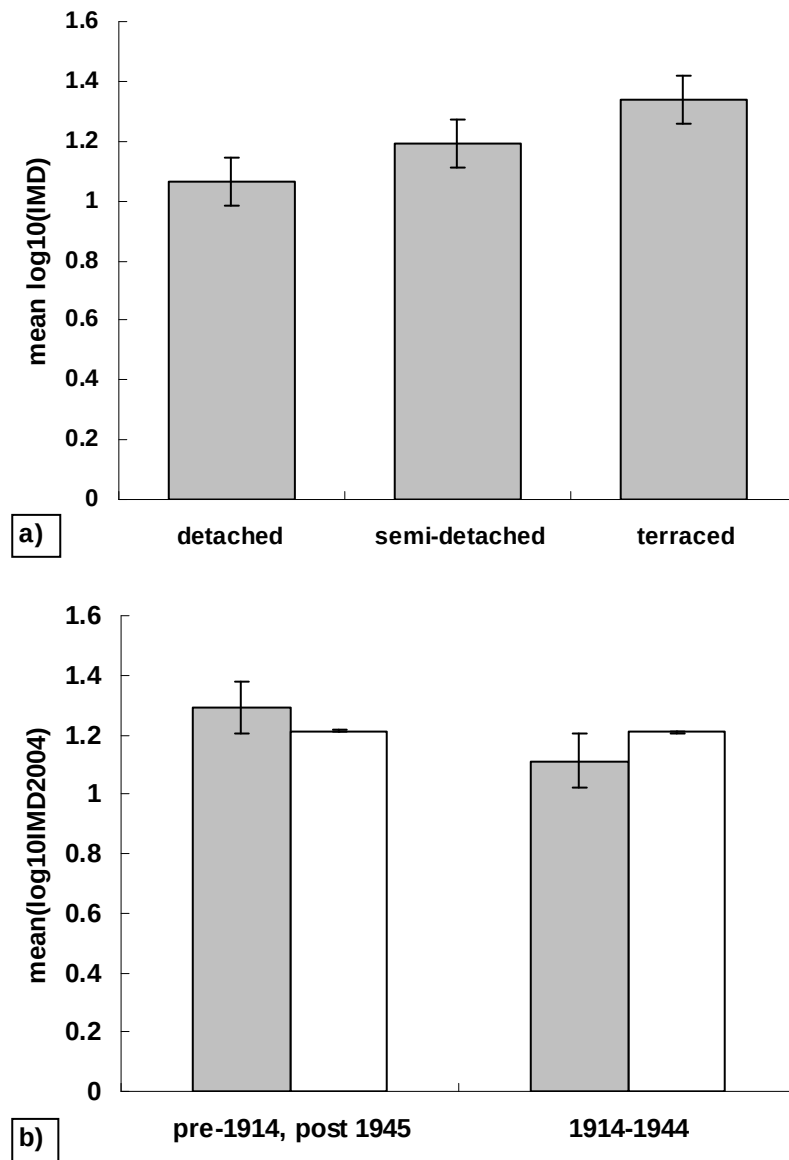


Figure 1: Variation in the type and age of housing in urban Bristol in relation to socioeconomic deprivation. a) shows mean deprivation scores and standard errors for detached, semi detached and terraced housing; b) shows mean deprivation scores for housing built from 1914-1945 versus the mean for earlier and later housing, along with the effect of repairs (grey bars) versus no reported repairs (black bars).

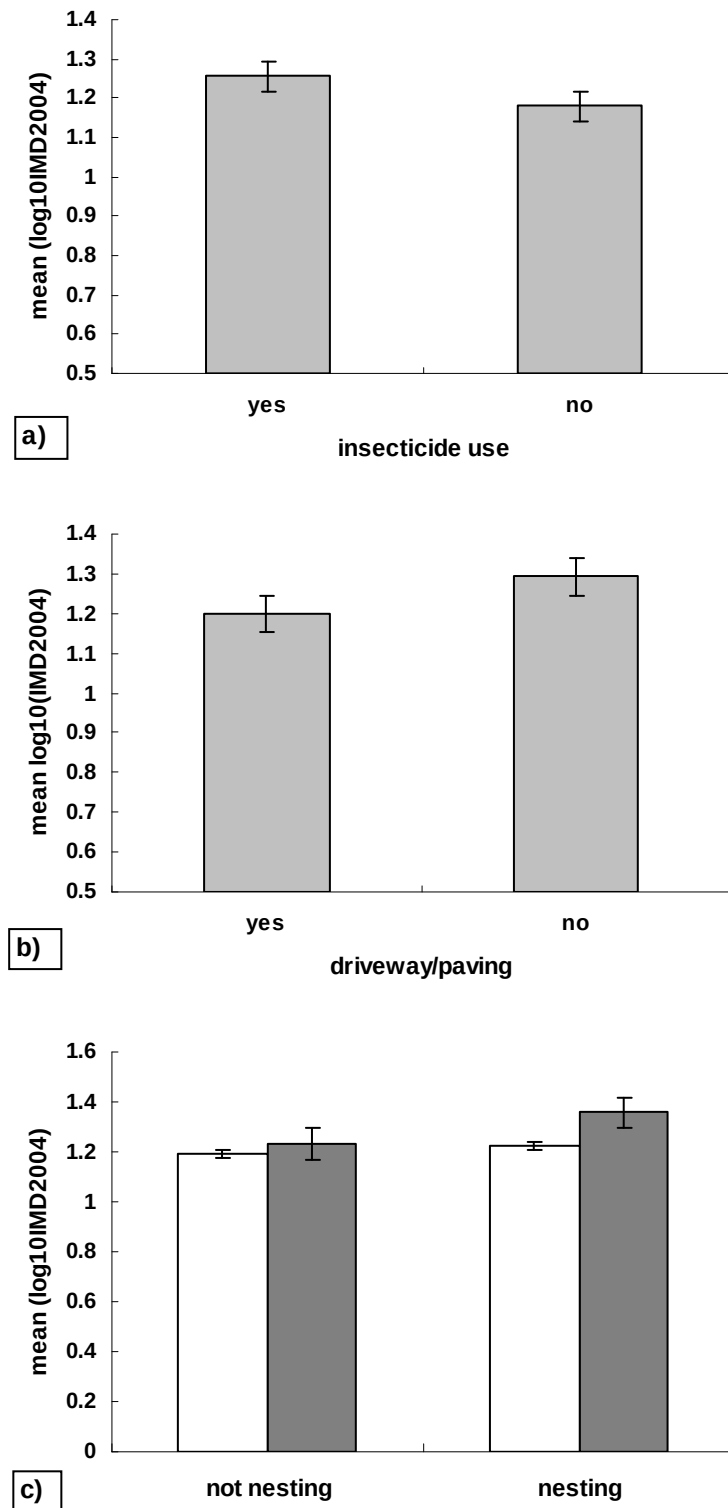


Figure 2: Differences in mean ($\pm$ S.E.) deprivation in relation to garden features in urban Bristol: (a) Insecticide use, (b) presence of driveways/paving, (c) presence of mature trees in relation to nesting house sparrows, white bars = trees present, grey bars = trees absent.

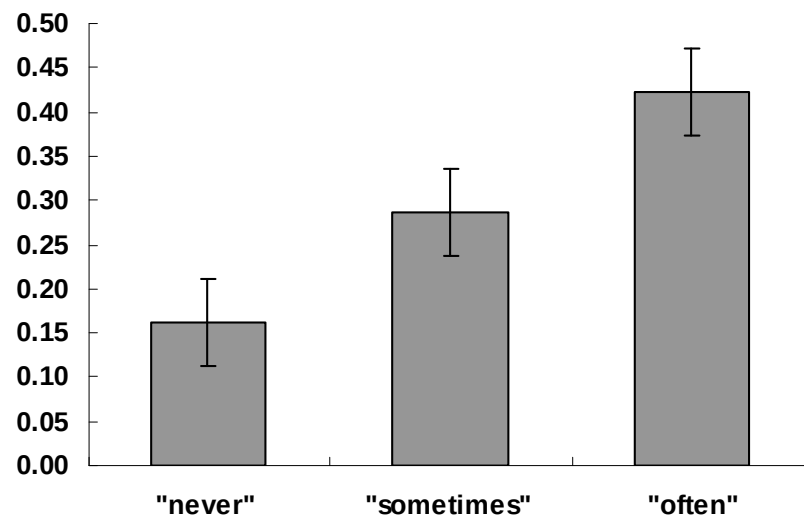


Figure 3: Mean house sparrow counts $\pm$ S.E. (birds per hour, y-axis) in relation to house sparrow occurrence as perceived by Bristol questionnaire respondents. Sparrow counts are derived from Breeding Bird Survey-style counts for 1km grid squares in Bristol.



Figure 4: Pre-war (a) and modern (b) high density terraced housing in Bristol. Pre-war housing is typically near the centre of Bristol, and is relatively deprived. Modern developments are typically in less deprived areas, or those undergoing regeneration.

APPENDIX 3A: COPY OF QUESTIONNAIRE DISTRIBUTED TO BRISTOL RESIDENTS

This questionnaire aims to record characteristics of houses and gardens in the Bristol area. I hope to compare the results obtained from this questionnaire with data on house sparrow numbers throughout Bristol, in order to identify factors that may be contributing to their decline in urban areas.

Please tick or circle the appropriate answer/s, or leave blank if not known.

- 1. Contact details:** (optional, but please give your postcode, so that your answers can be linked to data on sparrows from your area)

Name:

Email:

Address:

Contact telephone:

- 2. Postcode:**

- 3. Is your house:**

- a.** Detached
- b.** Semi-detached/ end terrace
- c.** Terraced
- d.** Flat/apartment
- e.** Bungalow

- 4. Approximately how old is your house?**

- a.** Pre-1914
- b.** 1914-1944
- c.** 1945-1970
- d.** Post 1970

5. What type of roof covering does your house have?
- a. Flat tiles/slate
 - b. Curved tiles
6. Do you have a pet:
- a. Cat
 - b. Dog
 - c. Rabbit/guinea pig
 - d. Other (please state)
7. Do any cats regularly visit your garden (other than any cats that you own)?
- a. Yes
 - b. No
8. Do you have any of the following in your garden?
- a. A shed / outbuilding
 - b. A water feature / birdbath
 - c. A bird table / feeder (if used)
 - d. A sparrow nestbox / boxes
 - e. A lawn / grassy area / flower bed
 - f. Decking / patio / driveway or parking space
 - g. Mature trees (taller than 5metres)
 - h. Shrubs or climbing plants (e.g. ivy, honeysuckle)
9. Is your garden/house surrounded by: (tick more than one if applicable)
- a. Wall
 - b. Fence
 - c. Hedge
 - d. No solid boundary

10. Do you use insecticides in your garden:

- a.** Never
- b.** Occasionally
- c.** Regularly

11. Have you ever noticed house sparrows nesting in or on your property?

- a.** Yes – please give details
- b.** No

12. On average, do you see sparrows around your property:

- a.** Never
- b.** Sometimes
- c.** Often

13. Have you carried out any repairs to your roof or guttering in the last few years?

- a.** Yes
- b.** No – do not intend to
- c.** Intend to repair roof/guttering in the near future

Thank you for taking the time to complete this questionnaire. If you have any further comments on house sparrows in your area, please feel free to add these at the end.

CHAPTER 4

SOCIOECONOMIC DEPRIVATION AS AN INDICATOR OF URBAN HOUSE SPARROW OCCURRENCE

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Summary:

Obtaining detailed information on urban biodiversity is problematic, due to the difficulty of accessing urban sites. Here nationwide datasets on socioeconomic status are used to combat this problem by comparing the distribution of house sparrows to the prevalence of urban deprivation in cities. We found that the probability of sighting house sparrows in urban gardens is significantly higher in deprived parts of English cities, suggesting that their decline may have been caused by colony loss in areas of high socioeconomic status. This effect is particularly strong in spring, suggesting that areas with high urban deprivation provide more suitable habitat for breeding birds in particular. We suggest that the availability of nest sites and of invertebrate food may be higher in areas of low socioeconomic status, and that future work should focus on comparing resources across areas of differing socioeconomic status as a starting point to increasing the understanding of urban biodiversity. In a wider context, these results show the potential for large, standardised datasets to provide indicators of urban habitat quality, highlighting the potential for using this information as a starting point for detailed habitat analysis.

Keywords: Foraging success; habitat quality; *Passer domesticus*; nesting, urbanisation.

Introduction:

Urban areas are an important resource for wildlife, but they are still relatively little studied in comparison to other habitats (Evans *et al*; 2009). Urban green spaces have the potential to support large species assemblages (Cannon *et al*; 2005, Smith *et al*; 2005), and are also important for a number of threatened species in the UK. In addition to the house sparrow *Passer domesticus*, these include song thrushes *Turdus philomelos* (Mason 2000); starlings *Sturnus vulgaris* (Robinson *et al*; 2002); and black redstarts *Phoenicurus ochruros* (www.blackredstarts.org.uk). Urban greenspace also has both physical and psychological benefits for humans, and these benefits appear to increase with increasing biodiversity (Fuller *et al*; 2007, Mitchell & Popham, 2008). As a result, maximising provision for wildlife in both existing and new urban developments has been identified as an ecological problem with high policy relevance in the UK (Sutherland *et al*; 2006).

Assessing population trends and habitat quality in urban environments is very difficult, due to the fine scale of urban habitat features, and the difficulty of accessing private land (Cannon, 2005). As a result, many studies of urban birds are small-scale, and confined to public parks rather than including private gardens, which are an important resource for wildlife (Cannon *et al*; 2005). Standardised, nationwide datasets such as those obtainable from national census statistics provide indicators of urban habitat quality at a broad scale, and may provide clues to the changes driving urban house sparrow populations. The decline of this highly commensal species is well documented (Siriwardena *et al*; 2002), but the causes are unknown, although many theories exist (Summers-Smith 2003; Shaw *et al*; 2008). The lack of data on house sparrow populations prior to their decline adds to the problem of identifying the underlying causal factors; for many years they were not included in the UK Common Bird Census

(CBC; Marchant *et al*; 1990), because they were regarded as too common. The CBC also under-represented urban habitats, but more recent surveys such as the British Trust for Ornithology's GardenBirdWatch (Toms, 2006) have helped redress the balance. The GardenBirdWatch has provided a more accurate picture of the geographic distribution of house sparrows in the UK, and has highlighted substantial variation in house sparrow numbers both within and between different urban areas (Siriwardena *et al*; 2002, Chapter 4).

This variation is as yet unexplained, in part due to the difficulties associated with collecting data on factors linked to the decline of the urban house sparrow on a national level. The number of variables involved and the complexity of the urban landscape makes this task very difficult at a large scale. However smaller scale studies suggest that these variables are very likely to be linked to the prevalence of socioeconomic deprivation within cities. Reports from many cities appear to suggest that post-decline house sparrow populations are predominantly associated with areas of low socioeconomic status (Shaw *et al*; 2008, Summers-Smith, 2003). House sparrows are heavily reliant on insects to feed their chicks during the breeding season, and are therefore likely to be affected by a reduction in invertebrate diversity in areas undergoing socioeconomic change. In addition, house sparrows commonly nest in cavities within the eaves of buildings, and if houses in poor condition are in predominantly deprived areas, nest site availability may be affected by socioeconomic status (Shaw *et al*; 2008). Areas of high socioeconomic status appear to be losing greenspace at a greater rate than more deprived areas in the UK (Pauleit *et al*; 2005). This could provide a potential mechanism for explaining the apparent pattern of association between sparrow occurrence and deprivation across Europe.

If human socioeconomic status is a predictor of nest site availability and/or foraging success, the post-decline distribution of house sparrows would be expected to mirror urban deprivation. We use nationwide survey data on house sparrows and socioeconomic status to firstly; test whether house sparrows are more likely to occur in deprived areas; and secondly to determine whether areas that become less deprived over time are more likely to lose their house sparrow populations than areas that are becoming more deprived. Our results are discussed in relation to previous work on house sparrow populations, and the specific differences that are likely to exist between habitats in deprived versus affluent areas.

Methods:

1) OCCURRENCE OF SPARROWS IN RELATION TO DEPRIVATION:

We obtained data on house sparrow occurrence from the British Trust for Ornithology's GardenBirdWatch survey (BTO GBW; Toms, 2006). The GBW survey is volunteer-based, and participants record the birds in their gardens throughout the year. This is a unique resource, as consistent surveying methodology is used on a large scale, allowing the use of large sample sizes in analysis. It is therefore extremely valuable for detecting nationwide patterns of occurrence. GardenBirdWatch trends are representative of wider-scale population changes in the house sparrow (Cannon *et al*; 2005). Data in the form of weekly presence/absence records for house sparrows were obtained for 2645 GBW sites from 16 urban areas in England (see Table 1). These cover a wide geographical area, and were defined using National Statistics census urban area boundaries. All areas had an average population density of at least 2,500 people per km² (National Statistics, 2007). Data from 1995 – the first year of the survey – through to 2006 was used for analysis, enabling changes in house sparrow occurrence over time to be identified.

The socioeconomic status of GBW sites was measured by the Index of Multiple Deprivation (IMD), obtained from the English Indices of Deprivation for 2004 and 2007 (Noble 2004, Communities and Local Government 2007). This index comprises seven measures of deprivation, which are combined to produce an overall score (Table 2). As with GBW data, the consistent scoring system allows small-scale comparisons to be applied over a wide geographical area. Each GBW location from the 16 urban areas was allocated IMD scores according to the Lower Super Output Area (LSOA) in which it was situated. LSOAs are subdivisions of electoral wards and vary in size, but contain on average 1500 dwellings. IMD scores range from 1–82 (including non-integers) with higher scores indicating a higher level of deprivation. Information on the area of each LSOA was also obtained from the “land use statistics (Generalised land use database), 2005” (ONS Communities and Local Government, 2007).

Deprivation score was modelled in relation to the presence of house sparrows using a Generalized Linear Mixed effects Model (GLMM) to allow for the inclusion of fixed and random effects. Data were analysed using R 2.5.1 (R development core team 2007) using the library “repeated” and the “rmutil” package for mixed effects models using repeated measures (Lindsay 2007). Sparrow presence/absence counts were summarised by season and year for each GBW location, with sparrows scored as present if they occurred at a site in any week during a particular season and year. The summarised data were then included in the model as the binomial response variable. The overall deprivation score (IMD) for 2004 was added as an explanatory variable in initial models. Since deprivation status is likely to be affected by housing density and therefore LSOA size, a measure of LSOA area was also included in the analysis. Season was included as a fixed effect and year effects were also considered, resulting in a maximum of four observations per site per year for use in the analysis. GBW site was

treated as a nested random factor. Non-linear terms for year and deprivation score were included; hence year was treated as a continuous variable. The model used a binomial error structure with a log-link function. The basic model structure was as follows:

House sparrow presence/absence $\sim$ (year + (year)² + deprivation + season + LSOA area + deprivation*LSOA area + deprivation*year + year*season, nested = GBW location).

City was initially included in the model as a random factor, but was excluded on the basis that some of the cities used reported sighting house sparrows at all GBW sites in all seasons where data was available. This consequently prevented the detection of a trend in house sparrow occurrence in these cities, and led to confusion when interpreting the overall model output. The cities affected were typically those with high mean deprivation scores (e.g. Middlesbrough and Liverpool). Subsequently separate analysis of each city that showed variation in both deprivation and house sparrow occurrence (e.g. Bristol) gave the same results as the overall model (Appendix 4A).

As deprivation scores are not available for each year of the GBW survey it is possible that changes in deprivation status of GBW sites in the years before and after 2004 could affect the results. Deprivation scores prior to 2004 were not available at a small scale, but are available for 2007. The analysis was therefore repeated using the deprivation scores for 2007 in place of the 2004 index. Each aspect of the deprivation index for 2004 was also tested separately by substitution into the original model in order to test whether any particular aspect of deprivation was having an effect on the presence of house sparrows. Models were simplified using stepwise deletion, and the Akaike Information Criterion (AIC) values for each model were compared to determine the best fit; the models with the lowest AIC score being deemed the most parsimonious

(Burnham & Anderson 1998). The log-odds ratios obtained from the models were then back-transformed to give the probability of detecting house sparrows at GBW sites according to IMD score.

2) VARIATION IN RELATIVE DEPRIVATION STATUS OVER TIME

Comparing the 2004 and 2007 Indices of Deprivation scores directly is not possible, as the deprivation scores for each year are obtained by comparing LSOAs within years, and therefore give only relative deprivation levels for each year. However, in each of the two years LSOAs were also ranked according to their deprivation scores for that year, and these rankings allow comparisons of relative changes in deprivation rank with the probability of house sparrow occurrence to be made between years. The rankings range from 1 – the most deprived LSOA in England, to 32 482 – the ranking of the least deprived LSOA. The indices for 2004 and 2007 are based on National Statistics data from preceding years (from 2001–2003 and 2004–2006 respectively). Therefore GBW sites that had submitted data to the BTO during both of these time periods were used in this analysis.

House sparrow presence/absence data from the GBW was summarised for each of the two time periods above. Each GBW location was allocated a deprivation ranking for each of the two time periods. A generalised linear mixed effects model (GLMM) was then used to test the effect of changes in deprivation on sparrow occurrence over time. The presence or absence of house sparrows in 2007 was used as the binomial response variable, with sparrows counted as present if they occurred in a particular season in any of the three years leading up to 2007. The deprivation rankings for 2004 and 2007 were included as continuous explanatory variables, with season included as a categorical explanatory variable. House sparrow presence\absence in 2004 was also included as a two level explanatory factor, with house sparrows recorded as present if they occurred

in a particular season in any of the three years leading up to 2004. Non-linear terms for the deprivation ranking in 2004 and 2007 were also included, as were interaction terms. As before, the models used a binomial error structure with a log-link function, and were simplified using stepwise deletion. The log-odds ratios obtained from the models were then back-transformed to give the probabilities of detecting house sparrows at GBW sites according to deprivation.

Results:

The occurrence of house sparrows at GBW sites varied substantially between different areas, but most of the cities studied showed an overall decline in the occurrence of house sparrows from 1995-2006 (Table 1). Mean deprivation scores also varied substantially, although cities in the north of the country had higher mean deprivation scores in general. The size of the LSOAs used in the analysis also showed large variation (from a minimum of 18 360 000m² to a maximum of over 680 000 000m²), reflecting the differences in housing density between inner city areas and suburbs within the urban boundaries.

The variation in deprivation and LSOA size both within and between urban areas raises the possibility that GBW sites may not be representative of the cities as a whole, leading to a potential source of bias in our results. Comparisons of the mean LSOA area within the 16 cities to the mean LSOA area of the GBW sites shows no significant differences, suggesting the size of LSOAs in our study sample are representative of the LSOAs in the 16 cities as a whole (paired t-test, $t = -1.78$, $p = 0.09$, d.f. = 15). However comparison of the mean deprivation scores in each city with the mean deprivation scores of GBW sites suggests that GBW sites are not representative of the levels of deprivation in the cities as a whole (paired t-test, $t = 11.45$, $p < 0.005$, d.f. = 15). The mean deprivation

scores for GBW sites were consistently lower than the mean deprivation scores for cities as a whole.

1) OCCURRENCE OF HOUSE SPARROWS IN RELATION TO DEPRIVATION:

The probability of GBW participants reporting the presence of house sparrows in their gardens significantly increased with increasing deprivation in the area surrounding the GBW site ($z = -6.932$, $n = 40\ 659$, $p < 0.005$). The year in which GBW data was collected also significantly influenced the probability of house sparrow sightings, which decreased overall from 1995 to 2006 ($z = -18.012$, $n = 40\ 659$, $p < 0.005$, Table 3). In addition, a significant interaction was found between the year of the GBW survey and deprivation. The probability of house sparrows being recorded at GBW sites was high overall in the initial years of the survey, regardless of the level of deprivation in the surrounding area. In later years the probability of sighting house sparrows decreased significantly at less deprived GBW sites. The more deprived areas also showed a decreased probability of house sparrow sightings from 1995 to 2000, but by 2005 the probability of house sparrow sightings had increased slightly ($z = 7.794$, $n = 40\ 659$, $p < 0.005$, Figure 1, Table 3).

As expected, the deprivation score of GBW sites was also affected by LSOA area, with deprivation decreasing significantly as LSOA area increased (and therefore as housing density decreased, Table 3). Furthermore, the effect of deprivation on house sparrow occurrence was also significantly affected by LSOA area. The overall effect of deprivation on house sparrow occurrence increased with increasing LSOA area, in all but the least deprived GBW sites (IMD scores < 15 , Table 3).

Season also has a significant effect on the probability of house sparrow sightings at GBW sites, and this effect shows a significant link to the year of the GBW survey. The

effect of deprivation on house sparrow occurrence is particularly strong in the spring, when the probability of recording sparrows at the least deprived GBW sites falls from above 0.90 overall to around 0.2 by 2005. In summer, autumn and winter, the probability of recording house sparrows still falls at the least deprived GBW sites, but even in later years is still above 0.85 overall. In addition, although the probability of sparrow sightings continues to decrease over the years at the least deprived sites, at those sites with a high deprivation score the decline in the probability of house sparrow sightings in spring has levelled off by 2003, and then begins to increase again by 2006 (Figures 1 and 2, Table 3).

The occurrence of house sparrows at GBW sites does not appear to be strongly affected by any particular aspect of social deprivation. All aspects of the combined deprivation index show a significant linear or non-linear trend in relation to house sparrow presence (Table 4). Substituting the 2007 deprivation index into the model also shows the same overall relationship between deprivation and the presence of house sparrows at GBW sites ($z = -4.048$, $n = 40\,659$, $p < 0.005$). Once again, the likelihood of seeing sparrows at the less deprived sites decreased significantly in later years of the survey ($z = -17.812$, $n = 40\,659$, $p < 0.005$). As with the index for 2004, the effect of the 2007 deprivation index on the presence of house sparrows is most pronounced during the spring, where the probability of house sparrow sightings decreases at the least deprived sites, but begins to increase again at the least deprived sites by 2006.

2) VARIATION IN RELATIVE DEPRIVATION STATUS OVER TIME

The occurrence of house sparrows at GBW sites showed significant changes over time in response to changes in the deprivation rankings of their surroundings. The deprivation rankings in 2004 and 2007 both showed a significant non-linear negative relationship with the probability of house sparrow presence at GBW sites from 2004-

2006 ($z = -6.292$, $n = 4244$, $p < 0.005$ (2004); $z = -6.776$, $n = 4244$, $p < 0.005$ (2007)).

The relative deprivation rankings (and therefore level of deprivation) for 2004 and 2007 showed a significant positive correlation ($z = 7.413$, $n = 4244$, $p < 0.005$), indicating that GBW sites covered a wide range of deprivation scores in both years. The occurrence of house sparrows at GBW sites in 2004 was significantly linked to the probability of their being sighted at the same site in 2007 ($z = -6.105$, $n = 4244$, $p < 0.005$).

Initial comparison of the mean deprivation rankings at GBW sites reveals differences between sites that record sparrows and those that do not. Sites that recorded no sparrows during 2001-2006 have much higher deprivation rankings (therefore much lower levels of deprivation) in both 2004 and 2007 than sites that recorded house sparrows throughout this time. Sites that recorded house sparrows in 2001-3 but not in 2004-6 also had a much higher mean deprivation ranking (therefore lower deprivation levels) for both years than sites that recorded sparrows throughout this time. Sites that did not record house sparrows in 2001-3 but recorded them during 2004-6 had comparatively low deprivation rankings (therefore high deprivation) in 2004 and became substantially more deprived by 2007 ($z = -6.105$, $n = 4244$, $p < 0.005$). This suggests that only those sites that started reporting sparrow sightings for the first time from 2004-6 showed significant changes to their relative deprivation status, and these sites became relatively more deprived than other LSOAs as a whole.

Discussion:

This study supports evidence from localised reports that house sparrow occurrence is linked to high levels of socioeconomic deprivation in urban areas. This is despite the finding by Fuller *et al*; (2008) that the incidence of supplementary feeding by

householders decreases with increasing social deprivation. Since house sparrows use feeders on a regular basis, it appears that food availability for adults is not a factor influencing their decline. It should be noted that the GBW is a voluntary survey, and as such presents a biased survey sample, with on average relatively low levels of deprivation in comparison to the study areas in general. However GBW gardens, irrespective of deprivation level, are more likely overall to provide supplementary food than surrounding gardens and therefore potentially to be frequented by sparrows. Whilst this may affect detectability in gardens to some extent, previous studies have suggested that supplementary feeding does not appear to be as beneficial to sparrows as to other species (Fuller *et al.*; 2008). It is therefore unlikely that supplementary feeding of adults will be useful as a conservation tool in the case of the house sparrow

The occurrence of house sparrows within cities is already known to be correlated with housing density (Tratolos *et al.*; 2007), and it is therefore possible that the observed effect of deprivation would be seen purely due to the fact that housing density tends to be high near the centre of cities. Deprivation is also likely to be high near city centres, where LSOAs tend to be smaller, as each LSOA contains the same number of households on average. Although housing density (as measured by LSOA size) was correlated to both deprivation and house sparrow presence as expected, socioeconomic deprivation alone still had a significant effect on the probability of sighting house sparrows at GBW sites.

Variation in LSOA size does however mean that within-LSOA variation in deprivation is likely to be greater on the outskirts of cities, and therefore that the Index of Multiple Deprivation may be more representative of a single GBW site in relatively deprived areas. Whilst this is likely to be true in large cities to some extent, overall variation in

LSOA size within cities is relatively small, and regeneration of city centres has lowered the levels of deprivation in many inner city areas. In addition, many areas with high deprivation scores are relatively large due to the presence of industrial buildings. It is therefore unlikely that bias due to within LSOA variation will affect the results obtained here.

The effect of deprivation on the presence of house sparrows at GBW sites is consistent across both deprivation indices, and is present when all aspects of deprivation are considered separately. The fact that no particular aspect of deprivation appears to be especially important is perhaps surprising, given the different criteria for measurement (Table 2). However, many aspects of the deprivation index (e.g. income and employment) are likely to be closely linked, and those which may have an impact on environmental conditions for house sparrows (such as air quality and housing condition) may be masked by other, less relevant aspects (i.e. traffic accidents and overcrowding).

The effect of deprivation on the probability of recording house sparrows is greatest during spring. Since house sparrows are highly detectable throughout the year, this trend is likely to reflect the reduced mobility of adult birds during the breeding season, rather than variations in detectability. Post fledging dispersal of juveniles from first broods is likely to explain why the probability of sighting sparrows in all gardens increases again from summer onwards. The foraging range of a breeding sparrow is likely to contract to 100m or less from the nest site when adult birds are feeding nestlings (Vincent 2005). This, and the apparent lack of a beneficial effect from supplementary granivorous food, suggests that less deprived areas are unsuitable for providing resources required by breeding birds in particular.

Reduction in vegetation cover, particularly of native plant species (Chace & Walsh, 2006), will lead to a loss of invertebrate abundance in areas which have lost green space due to infill development or demand for off-street parking. In addition, areas may become less deprived following regeneration of “brownfield” or derelict sites, which support large numbers of invertebrates (Small *et al*; 2002). A lack of good quality insect food for nestlings is known to affect nestling condition and lead to lower fledging success in house sparrows (Peach *et al*; 2008). There is evidence that gardens in relatively deprived areas are more likely to provide food for breeding birds than those in less deprived sites. Gardens in relatively deprived areas are likely to have a greater proportion of native plants, and are less likely to have a high proportion of paving or to be affected by infill development than those in less deprived urban areas (Pauleit *et al*; 2005, McKinney, 2008).

This supports the prediction by Chamberlain *et al*; (2007) that converting even small areas of private garden into continuous housing may lead to declines in house sparrow abundance. Loss of green space due to increased demand for off-street parking is also greater in less deprived areas, and is likely to exacerbate this effect (London Assembly 2005; Pauleit *et al*; 2005, Perry & Nawaz; 2008). In addition, new or refurbished buildings are unlikely to provide suitable cavities for nesting, meaning that areas which have undergone redevelopment may be deserted during the breeding season even if suitable foraging habitat is available (Shaw *et al*; 2008). There is no evidence at present of increased levels of pollution, competition or predation in less deprived sites over those with high levels of deprivation. However, lack of cover or food for foraging birds could increase the predation risk for individual birds in low deprivation areas, or force birds to spend increased amounts of time foraging, with consequent effects on individual fitness (Shaw *et al*; 2008).

We suggest that factors linked to urban deprivation therefore warrant further investigation as possible drivers of the urban house sparrow decline. Using this information, it may be possible to identify areas that are currently populated by house sparrows and monitor the surrounding habitat over time to record any changes which might affect sparrow numbers. At present, it is difficult to ascertain which of these potential factors are important for house sparrow populations, although nesting and foraging habitat availability are strong candidates for further investigation. It is clear that more research is needed not only into the factors affecting habitat availability for house sparrows, but also into how socioeconomic factors are linked to wider changes in the urban environment, and the potential effects these will have on urban wildlife populations as a whole.

IMPLICATIONS FOR WIDER URBAN BIODIVERSITY

Although human socioeconomic status in itself is unlikely to have a direct impact on wider species occurrence, areas which are becoming less deprived according to the English Index of Deprivation may be at risk of losing house sparrow populations. Other species such as starlings are also likely to be affected, as they share requirements for nesting cavities and insect food during breeding with the house sparrow. The densities of these two species of conservation concern are strongly positively correlated, suggesting that they are limited by a shared resource (Evans *et al*; 2009). Our results suggest that urban development in the absence of management to mitigate for its effects (e.g. provision of nest sites and invertebrate-rich foraging areas) may also have serious consequences for urban bird populations in general. Loss of these highly visible species is likely to be indicative of a reduction in overall biodiversity in the less deprived areas of cities.

This study highlights the potential for using nationwide datasets to look for broad-scale patterns of species occurrence in relation to existing data on human activity. It also provides a starting point for more detailed analysis of socioeconomic factors as an indicator of biodiversity. Identifying areas that are undergoing socioeconomic change may enable predictions to be made concerning localities where biodiversity may be under threat. For example, we would predict that areas with high UK deprivation rankings (> 21000) are likely to report a reduction in house sparrow occurrence in the near future. This may be indicative of decreased nest site availability; invertebrate abundance; or vegetation cover in these areas and a consequent reduction in overall biodiversity. We also consider potential implications of changes in human socioeconomic status on urban biodiversity. A reduction in habitat quality as deprivation status changes may affect the species composition of an area, potentially leading to the loss of native plants, and a consequent reduction in insect and avian species diversity. Environmental factors such as disease prevalence; predation rates; temperature and pollution levels may also be affected by increasing development in urban areas, and these changes will also have an effect on species diversity. (Savard, *et al*; 2000, Blair 2004, Bradley *et al*; 2008). By using the information provided by nationwide measures of urban change such as the Index of Deprivation it may be possible to monitor consequent changes in urban habitats at a large scale, and thereby identify areas at risk of losing biodiversity, in order to mitigate the effects of development more effectively.

In the UK, planning policy states that 60% of new housing should be built on previously developed land (Communities and Local Government 2006). It is therefore important to develop ways of assessing urban habitats at a broad scale in order to maintain biodiversity and evaluate the possible effects of urbanisation on species abundance

(Shochat *et al*; 2005). Regeneration of previously deprived areas is likely to lead to further habitat loss within urban areas, with house sparrow colony loss likely to be the most visible symptom of a larger problem. The use of large scale datasets such as the BTO GardenBirdWatch in relation to UK census datasets may allow areas at risk of habitat loss in the future to be identified and monitored, thereby allowing conservation effort to be targeted within cities.

Acknowledgements:

This work is funded by the NERC in association with the British Trust for Ornithology (BTO). We thank Mike Toms of the BTO for provision of GBW data on house sparrows, and for advice. Data on deprivation was obtained from:

<http://www.communities.gov.uk>. Thanks also to Steve Freeman (BTO) for advice on statistical methodology, David Carslake for advice and help with R, and Stuart Bearhop and Andy McGowan for advice and comments on the manuscript. Lastly, thank you to BTO GardenBirdWatch participants, for the collection of the house sparrow data. All experiments comply with UK law.

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Table 1: The 15 urban areas for which GBW data on house sparrows was selected. The mean deprivation score (IMD) and mean house sparrow presence/absence ratio for GBW sites within these urban area boundaries is shown, along with the total number of GBW sites in each area for which data on house sparrows was available.

Urban area	Number of GBW sites (total number of observations from each area)	Deprivation score (2004) (Mean $\pm$ S.E)	House sparrow presence/ absence ratio (Mean $\pm$ S.E)	Overall trend 1995 -2006
Birmingham	188 (2785)	22.78 $\pm$ 0.20	0.80 $\pm$ 0.01	Decline
Bournemouth	79 (1356)	15.19 $\pm$ 0.23	0.70 $\pm$ 0.01	Stable / slight decline
Brighton	35 (554)	20.73 $\pm$ 0.55	0.84 $\pm$ 0.02	Decline
Bristol	132 (1981)	15.83 $\pm$ 0.25	0.62 $\pm$ 0.01	Decline
Leeds	79 (1092)	16.59 $\pm$ 0.33	0.79 $\pm$ 0.01	Decline
Leicester	73 (868)	16.05 $\pm$ 0.33	0.89 $\pm$ 0.01	Decline
Liverpool	46 (666)	31.76 $\pm$ 0.65	0.68 $\pm$ 0.02	Decline
Manchester	40 (663)	31.30 $\pm$ 0.55	0.86 $\pm$ 0.01	Decline
Middlesbrough	28 (442)	16.78 $\pm$ 0.40	1.00 $\pm$ 0.01	Stable
Newcastle	24 (321)	16.47 $\pm$ 0.64	0.93 $\pm$ 0.02	Decline
Norwich	114 (2037)	13.78 $\pm$ 0.22	0.84 $\pm$ 0.01	Decline
Nottingham	47 (756)	22.46 $\pm$ 0.44	0.87 $\pm$ 0.01	Decline
Portsmouth	20 (279)	16.42 $\pm$ 0.79	0.99 $\pm$ 0.01	Stable / slight decline
Reading	114 (1913)	8.66 $\pm$ 0.16	0.85 $\pm$ 0.01	Decline
Sheffield	187 (2862)	11.39 $\pm$ 0.18	0.76 $\pm$ 0.01	Stable
London	1439 (22 084)	13.34 $\pm$ 0.06	0.62 $\pm$ <0.01	Decline

Table 2: Measures of deprivation used to create the overall Index of Deprivation for England 2004. The overall scores for each measure are weighted as shown and combined to form the overall index (IMD). Adapted from Noble *et al*; (2004).

Measure of deprivation	Description	Weighting (%)
Income deprivation	Households claiming income support, tax credits and jobseekers allowance	22.5
Employment deprivation	Unemployment and incapacity benefit claimants	22.5
Health deprivation and disability	Illness, disability and emergency hospital admissions	13.5
Education, skills and training deprivation	Young people: average grades, proportion entering higher education Adults: skills and qualifications	13.5
Barriers to housing and services	Wider barriers: overcrowding, access to owner-occupancy. Geographical barriers: distance to amenities	9.3
Crime	Recorded crimes: burglary, theft, criminal damage, violence	9.3
Living environment deprivation	‘Indoors’: housing in poor condition; central heating ‘Outdoors’: air quality, traffic accidents	9.3

Table 3: Generalized Linear Mixed effects Model showing the relationship between house sparrow occurrence in 16 English cities and socioeconomic deprivation, as measured by the Index of Multiple Deprivation 2004. Dispersion parameter = 1.

Model Parameter	Parameter estimate	Error	z- value	p value (***)
Intercept	8.319	0.341	24.395	<0.005 ***
Year	-0.738	0.041	-18.012	<0.005 ***
Year ²	0.020	0.002	8.506	<0.005 ***
Deprivation 2004	-0.107	0.016	-6.932	<0.005 ***
Summer	-0.085	0.187	-0.459	0.640
Autumn	-0.715	0.184	-3.881	<0.005 ***
Winter	-1.460	0.175	-8.363	<0.005 ***
LSOA area	-0.332	0.100	-3.316	<0.005 ***
Deprivation*LSOA area	0.030	0.006	5.43	<0.005 ***
Year*deprivation	0.005	0.001	7.794	<0.005 ***
Year*summer	-0.001	0.021	-0.079	0.937
Year*autumn	-0.064	0.022	-2.953	<0.005 ***
Year*winter	0.060	0.020	3.031	<0.005 ***

Table 4: Summary of the minimal adequate models for each individual aspect of the Index of Multiple Deprivation 2004 in relation to house sparrow occurrence. Also shown is the minimal adequate model for house sparrow occurrence in relation to the IMD for 2007. In all cases the presence\absence of house sparrows was treated as the binomial response variable in a binomial GLMM (Fixed effect = season, dispersion parameter = 1).

Aspect of Deprivation	Model parameter	Parameter Estimate	z-value	Residual deviance / DF	p-value (***)
Combined IMD 2007	Intercept	7.629	29.335	22 987 / 40 647	<0.001(***)
	Year	-0.702	-12.483	22 987 / 40 647	<0.001(***)
	Year ²	0.024	6.552	22 987 / 40 647	<0.001(***)
	IMD score 2007	-0.047	-4.515	22 987 / 40 647	<0.001(***)
	Season: summer	0.021	0.236	22 987 / 40 647	0.813
	Season: autumn	-1.189	-13.469	22 987 / 40 647	<0.001(***)
	Season: winter	-1.074	-12.838	22 987 / 40 647	<0.001(***)
	IMD score ²	0.001	3.491	22 987 / 40 647	<0.001(***)
	IMD score 2007*year	0.004	3.610	22 987 / 40 647	<0.001(***)
Income deprivation (2004)	Intercept	7.480	36.346	22 960 / 40 647	<0.001(***)
	Year	0.020	8.409	22 960 / 40 647	<0.001(***)
	Year ²	-0.739	-18.062	22 960 / 40 647	<0.001(***)
	Income score	-5.678	-6.376	22 960 / 40 647	<0.001(***)
	Season: summer	-0.084	-0.449	22 960 / 40 647	0.654
	Season: autumn	-0.718	-3.883	22 960 / 40 647	<0.001(***)
	Season: winter	-1.461	-8.352	22 960 / 40 647	<0.001(***)
	Income score*year	0.977	9.416	22 960 / 40 647	<0.001(***)
	Year*season: summer	-0.002	-0.084	22 960 / 40 647	0.933
	Year*season: autumn	-0.064	-2.931	22 960 / 40 647	<0 01 (**)
	Year*season: winter	0.061	3.043	22 960 / 40 647	<0 01 (**)

Aspect of Deprivation	Model parameter	Parameter Estimate	z-value	Residual deviance / DF	p-value (***)
Employment deprivation (2004)	Intercept	7.250	34.114	22 935 / 40 646	<0.001(***)
	Year	-0.766	-18.603	22 935 / 40 646	<0.001(***)
	Year ²	0.021	8.637	22 935 / 40 646	<0.001 (***)
	Employment score	0.177	0.114	22 935 / 40 646	0.909
	Season: summer	-0.083	-0.445	22 935 / 40 646	0.657
	Season: autumn	-0.712	-3.860	22 935 / 40 646	<0.001(***)
	Season: winter	-1.455	-8.326	22 935 / 40 646	<0.001(***)
	Employment score ²	-34.553	-9.428	22 935 / 40 646	<0.001(***)
	Year*employment score	1.508	9.043	22 935 / 40 646	<0.001(***)
	Year*season: summer	-0.002	-0.106	22 935 / 40 646	0.915
	Year*season: autumn	-0.064	-2.966	22 935 / 40 646	<0.01 (**)
	Year*season: winter	0.060	2.992	22 935 / 40 646	<0.01 (**)
Health deprivation and disability (2004)	Intercept	6.714	33.239	22 929 / 40 646	<0.001(***)
	Year	-0.601	-14.510	22 929 / 40 646	<0.001(***)
	Year ²	0.020	8.452	22 929 / 40 646	<0.001(***)
	Health score	-0.520	-6.617	22 929 / 40 646	<0.001(***)
	Season: summer	-0.077	-0.410	22 929 / 40 646	0.682
	Season: autumn	-0.711	-3.848	22 929 / 40 646	<0.001(***)
	Season: winter	-1.452	-8.289	22 929 / 40 646	<0.001(***)
	Health score ²	-0.048	-2.449	22 929 / 40 646	<0.05 (*)
	Health score*year	0.098	11.108	22 929 / 40 646	<0.001(***)
	Year*season: summer	-0.003	-0.138	22 929 / 40 646	0.890
	Year*season: autumn	-0.065	-2.954	22 929 / 40 646	<0.01 (**)
	Year*season: winter	0.060	2.969	22 929 / 40 646	<0.01 (**)

Aspect of Deprivation	Model parameter	Parameter Estimate	z-value	Residual deviance / DF	p-value (***)
Education, skills and training (2004)	Intercept	6.054	31.007	22 814 / 40 646	<0.001(***)
	Year	-0.676	-16.801	22 814 / 40 646	<0.001(***)
	Year ²	0.020	8.339	22 814 / 40 646	<0.001(***)
	Education score	0.099	12 416	22 814 / 40 646	<0.001(***)
	Season: summer	-0.048	-0.259	22 814 / 40 646	0.796
	Season: autumn	-0.677	-3.675	22 814 / 40 646	<0.001(***)
	Season: winter	-1.436	-8.224	22 814 / 40 646	<0.001(***)
	Education score ²	-0.001	-12.590	22 814 / 40 646	<0.001(***)
	Education score*year	0.004	5.355	22 814 / 40 646	<0.001(***)
	Year*season: summer	-0.006	-0.293	22 814 / 40 646	0.770
	Year*season: autumn	-0.067	-3.083	22 814 / 40 646	<0.01 (**)
	Year*season: winter	0.059	2.959	22 814 / 40 646	<0.01 (**)
Barriers to housing and services (2004)	Intercept	6.932	24.111	22 870 / 40 646	<0.001(***)
	Year	-0.528	-11.786	22 870 / 40 646	<0.001(***)
	Year ²	0.021	8.868	22 870 / 40 646	<0.001(***)
	Housing score	0.028	2.201	22 870 / 40 646	<0.05 (*)
	Season: summer	-0.069	-0.372	22 870 / 40 646	0.710
	Season: autumn	-0.713	-3.883	22 870 / 40 646	<0.001(***)
	Season: winter	-1.455	-8.354	22 870 / 40 646	<0.001(***)
	Housing score ²	-0.001	-4.506	22 870 / 40 646	<0.001(***)
	Housing score*year	-0.006	-6.659	22 870 / 40 646	<0.001(***)
	Year*season: summer	-0.004	-0.173	22 870 / 40 646	0.862
	Year*season: autumn	-0.064	-2.962	22 870 / 40 646	<0.01 (**)
	Year*season: winter	0.060	3.015	22 870 / 40 646	<0.01 (**)

Aspect of Deprivation	Model parameter	Parameter Estimate	z-value	Residual deviance / DF	p-value (***)
Crime (2004)	Intercept	7.329	37.309	22 996 / 40 647	<0.001(***)
	Year	-0.698	-17.259	22 996 / 40 647	<0.001(***)
	Year ²	0.023	9.449	22 996 / 40 647	<0.001(***)
	Crime score	-0.178	-6.067	22 996 / 40 647	<0.001(***)
	Season: summer	-0.084	-0.451	22 996 / 40 647	0.652
	Season: autumn	-0.741	-4.012	22 996 / 40 647	<0.001(***)
	Season: winter	-1.479	-8.450	22 996 / 40 647	<0.001(***)
	Crime score ²	-0.179	-6.118	22 996 / 40 647	<0.001(***)
	Year*season: summer	-0.002	-0.091	22 996 / 40 647	0.928
	Year*season: autumn	-0.061	-2.786	22 996 / 40 647	<0.01 (**)
	Year*season: winter	0.063	3.168	22 996 / 40 647	<0.01 (**)
	Intercept	8.571	36.873	22 937 / 40 643	<0.001(***)
Living Environment (2004)	Year	-0.751	-17.977	22 937 / 40 643	<0.001(***)
	Year ²	0.023	9.570	22 937 / 40 643	<0.001(***)
	Environment score	-0.093	-13.186	22 937 / 40 643	<0.001(***)
	Season: summer	-0.019	-0.093	22 937 / 40 643	0.926
	Season: autumn	-0.855	-4.232	22 937 / 40 643	<0.001(***)
	Season: winter	-1.693	-8.815	22 937 / 40 643	<0.001(***)
	Environment score ²	0.001	13.030	22 937 / 40 643	<0.001(***)
	Environment score*year	0.002	3.129	22 937 / 40 643	<0.01 (**)
	Year*season: summer	<-0.001	-0.002	22 937 / 40 643	0.999
	Year*season: autumn	-0.057	-2.578	22 937 / 40 643	<0.01 (**)
	Year*season: winter	0.068	3.382	22 937 / 40 643	<0.001(***)
	Environment score*season: summer	-0.004	-1.050	22 937 / 40 643	0.294
	Environment score*season: autumn	0.003	0.897	22 937 / 40 643	0.370
	Environment score*season: winter	0.008	2.164	22 937 / 40 643	<0.05 (*)

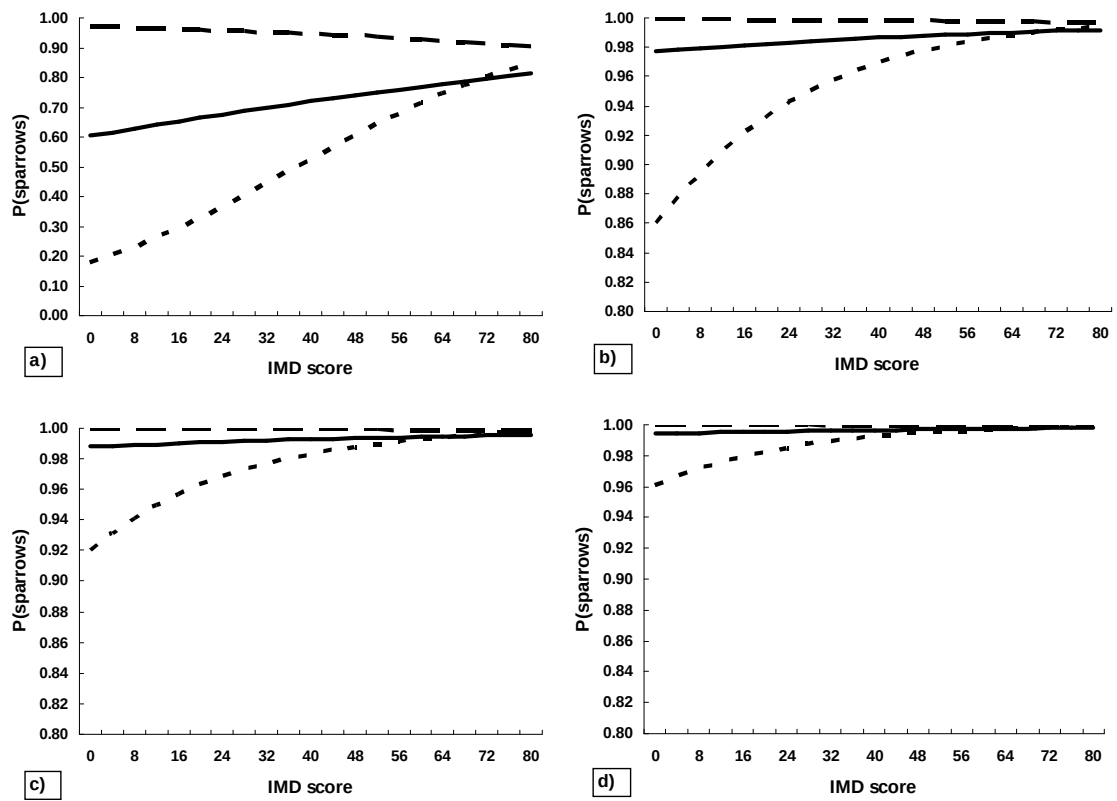


Figure 1: The probability of house sparrow occurrence at BTO GBW sites in a) spring, b) summer, c) autumn, and d) winter derived from back-transformed log-odds ratios obtained from the model. Values for three years are shown; 1995 (dashed line), 2000 (solid line) and 2005 (dotted line), according to the deprivation score (IMD), where 0 = least deprived. All values shown are for an average LSOA size of 700 000m². Note that the scale of a) differs from b), c) and d).

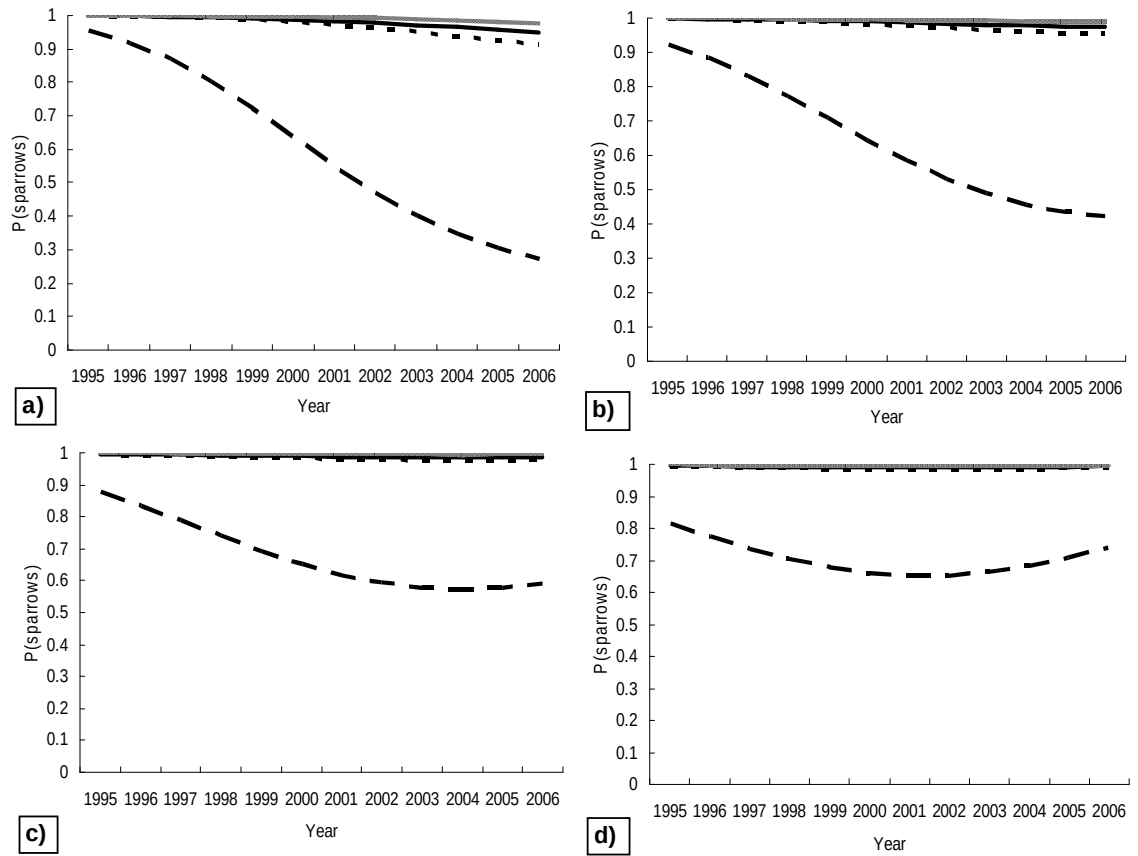


Figure 2: The probability of sighting house sparrows at GBW sites (y-axis) by year (x-axis), showing seasonal variations during the years of the GBW survey, where spring = dashed line, summer = dotted line, autumn = solid line and winter = grey line.

Variations in the level of deprivation (IMD) at GBW sites are shown, from 20 (less deprived sites) to 80 (most deprived sites). a) = IMD of 20; b) = IMD of 40; c) = IMD of 60; d) = IMD of 80.

Appendix 4A:

Parameter estimates for Generalised Linear Mixed effects Models comparing house sparrow occurrence in relation to the Index of Multiple Deprivation 2004 for individual cities in England. Only cities where models converged are listed, models shown are the minimal adequate model for that city.

City	Model parameter	Parameter estimate	z-value	Residual deviance/DF	p-value (***)
London	Intercept	14.933	27.917	13182 / 22069	<0.001 (***)
	Year	-0.815	-14.574	13182 / 22069	<0.001 (***)
	Year ²	0.024	7.369	13182 / 22069	<0.001 (***)
	IMD score	-0.216	-7.628	13182 / 22069	<0.001 (***)
	IMD score ²	0.001	-4.789	13182 / 22069	<0.001 (***)
	Season: summer	-0.233	-0.919	13182 / 22069	0.358
	Season: autumn	-0.870	-3.466	13182 / 22069	<0.001 (***)
	Season: winter	-1.718	-7.210	13182 / 22069	<0.001 (***)
	LSOA area	-2.815	-17.814	13182 / 22069	<0.001 (***)
	Year*IMD score	0.003	2.586	13182 / 22069	<0.01 (**)
	Year*season: summer	0.005	0.183	13182 / 22069	0.854
	Year*season; autumn	-0.048	-1.632	13182 / 22069	0.103
	Year*season: winter	0.100	3.707	13182 / 22069	<0.001 (***)
	IMD score*LSOA area	0.085	9.107	13182 / 22069	<0.001 (***)

City	Model parameter	Parameter estimate	z-value	Residual deviance/DF	p-value (***)
Birmingham	Intercept	16.046	10.907	1609.5 / 2774	<0.001 (***)
	Year	-1.044	-6.301	1609.5 / 2774	<0.001 (***)
	Year ²	0.026	2.696	1609.5 / 2774	<0.01 (**)
	IMD score	-0.320	-7.241	1609.5 / 2774	<0.001 (***)
	IMD score ²	0.003	4.571	1609.5 / 2774	<0.001 (***)
	Season: summer	0.050	0.233	1609.5 / 2774	0.815
	Season: autumn	-1.350	-6.372	1609.5 / 2774	<0.001 (***)
	Season: winter	-0.787	-3.898	1609.5 / 2774	<0.001 (***)
	LSOA area	-1.426	-4.337	1609.5 / 2774	<0.001 (***)
	Year*IMD score	0.018	0.329	1609.5 / 2774	<0.001 (***)
Bournemouth	Intercept	20.351	5.464	765.66 / 1344	<0.001 (***)
	Year	-0.615	-3.088	765.66 / 1344	<0.01 (**)
	Year ²	0.053	3.772	765.66 / 1344	<0.001 (***)
	IMD score	-1.244	-4.877	765.66 / 1344	<0.001 (***)
	IMD score ²	0.011	4.331	765.66 / 1344	<0.001 (***)
	Season: summer	0.007	0.022	765.66 / 1344	0.982
	Season: autumn	-1.017	-3.161	765.66 / 1344	<0.001 (***)
	Season: winter	-1.458	-4.750	765.66 / 1344	<0.001 (***)
	LSOA area	-5.439	-4.569	765.66 / 1344	<0.001 (***)
	Year*IMD score	-0.016	-2.476	765.66 / 1344	<0.05 (*)
	IMD score*LSOA area	0.383	5.388	765.66 / 1344	<0.001 (***)

City	Model parameter	Parameter estimate	z-value	Residual deviance/DF	p-value (***)
Bristol	Intercept	9.270	6.202	1303.3 / 1970	<0.001 (***)
	Year	-0.104	-1.906	1303.3 / 1970	0.057
	IMD score	-0.301	-2.696	1303.3 / 1970	<0.01 (**)
	IMD score ²	0.003	5.468	1303.3 / 1970	<0.001 (***)
	Season: summer	0.162	0.688	1303.3 / 1970	0.492
	Season: autumn	-0.990	-4.168	1303.3 / 1970	<0.001 (***)
	Season: winter	-1.179	-5.234	1303.3 / 1970	<0.001 (***)
	LSOA area	-3.064	-5.975	1303.3 / 1970	<0.001 (***)
	Year*IMD score	-0.013	-4.110	1303.3 / 1970	<0.001 (***)
	IMD score*LSOA area	0.139	3.671	1303.3 / 1970	<0.001 (***)
Sheffield	Intercept	2.312	5.011	1514.2 / 2851	<0.001 (***)
	Year	-0.051	-0.948	1514.2 / 2851	0.343
	IMD score	0.327	12.212	1514.2 / 2851	<0.001 (***)
	IMD score ²	-0.004	-6.311	1514.2 / 2851	<0.001 (***)
	Season: summer	0.641	1.054	1514.2 / 2851	0.292
	Season: autumn	0.284	0.476	1514.2 / 2851	0.634
	Season: winter	-0.457	-0.799	1514.2 / 2851	0.424
	Year*season: summer	-0.068	-0.890	1514.2 / 2851	0.373
	Year*season; autumn	-0.210	-2.701	1514.2 / 2851	<0.01 (**)
	Year*season: winter	-0.077	-1.062	1514.2 / 2851	0.288

CHAPTER 5

LAND USE IN RELATION TO NATIONWIDE HOUSE SPARROW OCCURRENCE

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Summary:

Studies of urban biodiversity are problematic, due to the difficulty of obtaining large scale, yet detailed information on urban habitat quality. This information is nevertheless important to obtain, as urban areas provide potential habitat for a wide variety of species, including many of conservation concern. Large scale datasets derived from human population censuses may provide a starting point for more detailed analysis of urban habitat quality, and enable information on urban areas to be gathered by indirect means. Here we build on existing data on urban habitat quality by examining trends in urban land use in relation to house sparrow occurrence. When individual aspects of land use were considered, the probability of house sparrow occurrence decreased as building area, paving, and garden area rose, but increased with the area of green space. The apparent anomaly whereby gardens and green space affect house sparrow occurrence in different directions may be due to the increasing trend towards paving over gardens, making them less suitable as habitats for urban birds. This is a particular problem in relatively affluent areas with low housing density, and green space may be particularly important in these areas as foraging habitat. We suggest that future research into urban habitat quality should monitor the changes in land use within urban centres due to redevelopment.

Keywords: Brownfield sites, cities, deprivation, *Passer domesticus*, urbanisation.

Introduction:

Maximising provision for wildlife in urban developments is an ecological problem with high policy relevance in the UK (Sutherland *et al*; 2006). However, urban areas are difficult to monitor effectively due to access and sight restrictions in built-up areas (Cannon 2005). As a result, monitoring urban species is relatively difficult, and studies of urban habitats are under-represented, particularly in the UK, which has a higher proportion of urban area than most other countries (Evans *et al*; 2009). It is important to address this imbalance, particularly as urban habitats are host to a number of species of conservation concern, such as the house sparrow, *Passer domesticus*, starling *Sturnus vulgaris* (Robinson, Siriwardena & Hewson, 2002), and song thrush, *Turdus philomelos* (Mason 2000).

Use of large-scale datasets may be an important resource in combating the problems associated with the study of urban habitats. For example, it has been hypothesized that the decline of the urban house sparrow is linked to changes in human socioeconomic status (Shaw *et al*; 2008), and support has been shown for this in that that house sparrows are more likely to occur in areas with relatively high levels of deprivation (see Chapter 4). This pattern may be linked to a lack of invertebrates in the breeding season, when sparrows need protein-rich food for their young; and to a lack of nest site availability (Peach *et al*; 2008, Shaw *et al*; 2008). Urban development appears to be leading to a reduction in the available habitat for house sparrows, and recent studies of urban habitats suggest that reduction in green space due to development could be occurring to a greater extent in less deprived areas (Pauleit *et al*; 2005). This may be due to changes in land use in these areas in particular, as modern development targets require new developments to be built at higher densities than existing housing (Office of the Deputy Prime Minister, 2003). It may therefore be possible to link the distribution of urban house sparrow populations with land

use measures on a similar scale. This could enable the habitat changes most likely to affect house sparrow population sizes to be identified, thereby enabling mitigation measures to be targeted effectively.

This study aims to test the potential for measures of land use from nationwide surveys to predict house sparrow occurrence in urban areas. If the availability of foraging habitat in the breeding season is a factor affecting house sparrow populations, it would be expected that house sparrow occurrence will be higher in areas with relatively few paved areas, and low building density in relation to other land use types. Here we use nationwide data to identify the aspects of land use that are related to house sparrow occurrence. We then examine these aspects in relation to existing patterns of both socioeconomic deprivation and house sparrow occurrence in order to test which features of land use, if any, might explain the current distribution of house sparrow populations within urban areas.

Methods:

DATA SOURCES:

Data on house sparrow occurrence was obtained from the British Trust for Ornithology's GardenBirdWatch survey (BTO GBW, Toms 2006). The GBW survey is volunteer-based, and participants record the birds in their gardens throughout the year. This is a unique resource, as consistent surveying methodology is used on a large scale, allowing the use of large sample sizes in analysis. It is therefore extremely valuable for detecting nationwide patterns of occurrence. GardenBirdWatch trends are representative of wider-scale population changes in the house sparrow (Cannon *et al*; 2005). Data in the form of weekly presence/absence records for house sparrows were obtained for 2645 GBW sites from 16 urban areas in England. (See Chapter 4 for further details of sites). Urban areas were

defined using National Statistics census urban area boundaries (National Statistics 2007), and covered a wide geographical range. All areas had an average population density of at least 2,500 people per km². Data from 1995 – the first year of the GBW survey – through to 2006 was used for analysis, enabling changes in house sparrow occurrence over time to be identified.

Land use data were obtained from the Office for National Statistics (ONS, Communities and Local Government, 2007). The data were collected at the scale of the Local Super Output Area (LSOA), the smallest geographical unit at which census data is collected. LSOAs are subdivisions of electoral wards and vary in size, but contain on average 1500 dwellings. The dataset used for this analysis was the “land use statistics (Generalised Land Use Database), 2005”, and is measured in thousands of square metres, giving an accuracy to the nearest 10m². Nine land use categories are identified, plus a further “unclassified” category, dealing with land that did not fit the classification criteria (this classification was very rarely used). From these categories, seven major land use types were identified: Paving (including roads, paths and other paved areas such as car parks); domestic buildings; non-domestic buildings (including industrial and public buildings); Gardens (privately owned green space); Green space (parks, publicly owned green areas and “brownfield sites”; usually defined as previously developed land); railways; and water. As many LSOAs did not contain non-domestic buildings, the areas of domestic and non-domestic buildings were combined, and the total area of buildings was used as a variable for the purposes of analysis, regardless of usage. The area of water and railways were also not included in the analysis, as each represented a relatively small proportion of urban land use, and were not present in all LSOAs. This left a total of four land use variables to be used for analysis. Due to slight overlaps when categorising the data, the true area of land in

an LSOA often differed from the sum of the estimated land use types within it. For this reason the area of each land use type was corrected according to the true LSOA area, based on the proportions of each land use type in each LSOA.

Each GBW location from the 16 urban areas was allocated scores for land use and deprivation according to the LSOA in which it is situated. Deprivation was measured using the Index of Multiple Deprivation (IMD), obtained from the English Indices of Deprivation for 2004 and 2007 (Noble 2004, Communities and Local Government 2007). IMD scores range from 1–82 (including non-integers) with higher scores indicating a higher level of deprivation. Each LSOA was also ranked according to its relative deprivation score, with a low ranking indicating a high level of deprivation. The area and proportion of each land use type at low, medium and high levels of deprivation were calculated, with scores for each land use type defined as low medium or high according to whether they were in the highest, mid, or lowest third of the Index of Multiple Deprivation rankings respectively.

ANALYSIS:

Aspects of land use were analysed individually in relation to house sparrow occurrence, using a separate model for each land use type. As these were likely to be intercorrelated, a principal components analysis was conducted in SPSS v15 (www.SPSS.com/uk) to assess the degree to which the land use types were linked (Appendix 5A). Although it would have been possible to analyse house sparrow occurrence in relation to all land use types in one model, the decision was made to analyse each aspect of land use independently. This was due to the likelihood that each aspect of land use will affect house sparrow occurrence in a different way, and impact upon different aspects of house sparrow behaviour, for example

buildings are an important nesting resource, but are not likely to be important for foraging birds.

Generalised linear mixed effects models were used to allow for the inclusion of both fixed and random effects. Data were analysed using R 2.5.1 (R development core team 2007) using the library “repeated” for repeated measures models, and the “rmutl” package for mixed effects models (Lindsay, 2007). House sparrow presence/absence counts were summarised by season and year for each GBW location, with sparrows scored as present if they occurred at a site in any week during a particular season and year. The summarised data were then included in the model as the binomial response variable, resulting in a maximum of four observations per site per year for analysis. Season was included as a fixed factor, and year effects were also included. Total area of the LSOA was included as a covariate in each analysis, enabling comparisons to be made between smaller LSOAs typically located in city centres and larger ones usually located nearer the urban boundary. The models used a binomial error structure with a log-link function, and non linear terms were also included. Log transformations of the raw land use variables were used where appropriate, and models were simplified by stepwise deletion, with the Akaike Information Criterion (AIC) values for each model compared to determine the best fit; the models with the lowest AIC score being deemed the most parsimonious (Burnham & Anderson 1998). The log-odds ratios obtained from the models were then back-transformed to give the probabilities of detecting house sparrows at GBW sites according to each aspect of land use.

Results:

LAND USE AND DEPRIVATION:

As expected, the different aspects of land use were highly intercorrelated (Table 1).

Principal components analysis showed that over a third of the variance in land use characteristics was explained by the area of domestic buildings and gardens, which were closely linked. The area of buildings in general and of paved areas explained almost two thirds of the overall variance, and this increased to over 75% when the effect of green space was also included (Appendix 5A). The areas of water and rail contributed only a very small proportion of the overall variance, reflecting their relatively minor contribution to the overall land use in urban areas.

The proportion of land use identified as “gardens” was higher for the least deprived sites than those that were more deprived. (Table 2). Furthermore, the proportion of paving in each LSOA was lowest in the least deprived third of sites, as was the proportion of residential buildings to a lesser extent. This probably reflects the fact that LSOAs each contain the same number of dwellings on average. The proportion of non-domestic buildings was highest in the most deprived areas, but this land use type represented less than 7% of the total land use in all LSOAs on average. The area of green space accounted for approximately 25% of land use on average, regardless of the level of deprivation in the surrounding area (Table 2).

LAND USE FACTORS AND HOUSE SPARROW OCCURRENCE:

PAVING:

Increased levels of paving within an LSOA had a significant negative effect on the probability of sighting house sparrows at GBW sites. In general, as the area of paving

within an LSOA increases, the chances of sighting house sparrows in a particular season and year decreased (Table 3, Figure 1). This pattern was apparent regardless of LSOA size, but the effect was slightly greater for the larger LSOAs. As expected, this trend is more pronounced in later years of the GBW survey, and in the spring.

BUILDINGS:

The area of buildings (both domestic and non-domestic) was significantly associated with a decrease in house sparrow occurrence between 1995-2005, when the effect of overall LSOA size is taken into account. This effect is particularly pronounced in the spring (Table 3, Figure 1). The probability of house sparrow occurrence decreases from above 0.8 to nearly zero in spring in larger than average LSOAs. Year effects also significantly influenced the probability of sighting house sparrows at GBW sites, regardless of the overall LSOA size or the area of built up land. Larger LSOAs showed more variation in the probability of sighting house sparrows in later seasons, particularly in more densely built up areas. In practice however, the largest LSOAs are unlikely to have a large area of land taken up by buildings, because although the buildings present may be bigger in large LSOAs, the average number of dwellings is the same for all LSOAs regardless of their size.

GARDENS:

The area of private gardens significantly influenced the occurrence of house sparrows at GBW sites overall, and this effect was also linked to overall LSOA size (Table 3, Figure 1). The probability of house sparrow sightings at GBW sites decreased significantly with increasing garden size overall from 1995 to 2005 when LSOA size was taken into account (Table 3). This effect was particularly noticeable in the spring, but the effect of garden area varied according to the overall size of the LSOA. In smaller LSOAs, the area of garden had

little or no effect on the probability of sighting house sparrows. However, in larger LSOAs, the probability of house sparrow occurrence decreased as garden area increased.

GREEN SPACE:

The probability of house sparrow occurrence increased as the area of green space within an LSOA increased, even when overall LSOA area was taken into account (Table 3, Figure 1), although this effect was relatively small in the smaller LSOAs, where the relative area of green space is likely to be smaller than other land use types. In larger LSOAs the probability of sighting house sparrows in 2005 increased from below 0.1 in areas with very little green space, to above 0.4 in spring, and from below 0.75 to nearly one in later seasons of that year.

Discussion:

The area of domestic buildings and gardens combined make up on average nearly half of the total land use in the urban areas studied here. The average proportion of urban land use given to domestic gardens in this study was over 30%, which compares favourably with a recent study of five UK urban areas where on average a quarter of land use was taken up by gardens (Loram *et al*; 2007). The high proportion of land use designated as private gardens emphasises the potential importance of both domestic gardens and the surrounding buildings as a habitat for wildlife (Cannon *et al*; 2005). Residential areas, i.e. those with a mix of both housing and gardens are already known to be a key predictor of house sparrow occurrence in the UK (Chamberlain *et al*; 2007), and as such both land use types would be expected to show a positive link to house sparrow occurrence. The negative trend seen in house sparrow sightings with increasing area of gardens and buildings is therefore surprising.

However, as all LSOAs have the same number of houses on average, a large area of houses and gardens reflects an increase in individual house and garden size, and therefore usually a reduction, rather than an increase in housing density. Low-density developments tend to have relatively low deprivation levels, a factor that has been linked to a low probability of house sparrow occurrence in cities (Chapter 4). This appears to be due to less deprived areas having fewer suitable nesting and foraging sites than areas that are relatively deprived (Shaw *et al*; 2008). As such, it appears that a difference in house and garden type between areas of high and low housing density may be the reason for the trends observed here. Furthermore, land classified as “gardens” under the land use index may not be wholly comprised of green space. Particularly in less deprived areas, lawns and flowerbeds are increasingly being paved over to create driveways, patios and decking (Pauleit *et al*; 2005). “Infill” development, where small areas of green space are converted to continuous housing, also appears to be more prevalent in these areas (London Assembly 2005; Pauleit *et al*; 2005; Perry & Nawaz, 2008). This may explain the decline in the probability of house sparrow sightings with increasing garden area seen in this study.

Previous studies show that mature, structurally diverse vegetation is preferred by the majority of urban species (Evans *et al*; 2009). Increased levels of paving in large gardens are likely to render them more homogeneous and therefore less suitable for foraging birds. Conversion of green space within gardens to paving will reduce insect abundance for foraging birds, particularly where native plants are removed, as these are associated with relatively high insect abundance (White 2005; Wilkinson 2006). Adult birds rely on insect food as a source of protein for their chicks during the breeding season, and lack of high quality food for nestlings has been linked to reduced fledging weight in juvenile birds, a

factor known to be linked to reduced subsequent survival rates (Peach *et al*; 2008). In addition, lack of suitable habitat may have other, indirect effects on adult survival. For example increasing the foraging range of individual birds may influence energy expenditure and the maintenance of fat reserves, which has been linked to predation risk in small birds, including the house sparrow (Gosler *et al*; 1995, MacLeod *et al*; 2006).

In contrast, areas with high housing density and smaller properties are commonly found near urban centres, and typically consist of older, terraced housing with small courtyard gardens. The habitat in these relatively deprived areas is therefore likely to be more structurally diverse than in larger gardens, as they contain fewer continuous open spaces such as large lawned or paved areas. Furthermore, these areas are also likely to contain mature trees and shrubs, particularly at garden boundaries, and may therefore provide more suitable foraging habitat for birds than less deprived areas. Other areas with high levels of deprivation include post-war, ex-local authority houses (Shaw *et al*; 2008). These also typically have large gardens that are structurally diverse, and contain relatively few paved areas. Anecdotal evidence from a number of urban areas suggests that this type of housing tends to have high densities of house sparrows (E.g. Bland 1998, Paston 2000), which would support the results of both this study and Chapter 4.

Many of these typically deprived urban gardens will be too small to be either converted into off-street parking, or lost due to infill development. Analysis of overall building and paving area in relation to house sparrows appears to support this, as although the probability of sighting sparrows decreases with increasing building and paving area in all cases, this effect is greatly reduced in the smaller LSOAs, where potential for future development is limited due to the high density of existing housing. It is therefore probable that gardens in smaller,

relatively deprived areas are more suitable for house sparrows than larger gardens in areas with lower housing density.

The small foraging range of a breeding house sparrow (<100m from the nest site, Vincent 2005) may also limit the suitability of large gardens in areas with low housing density. House sparrows are colonial nesters that use cavities under the eaves of houses for nest sites (Summers-Smith; 1988). High density housing may therefore be more likely than low density housing to provide both sufficient nesting holes, and foraging habitat for an adequate number of breeding pairs within the required range of the nest site. Here again, smaller, more heterogeneous gardens are likely to prove more suitable as foraging habitat than an equivalent area covered by just one large garden. Furthermore, modern houses and those in a good state of repair may not provide as many cavities for nesting birds in comparison to houses in more deprived areas where older tile types and building materials have not been replaced by impermeable plastic, and close fitting modern tiles (Palmer 2004, Shaw *et al*; 2008).

Analysis of the area of green space in relation to house sparrow occurrence suggests that this land use type may be beneficial to house sparrows. This effect is greatest in larger LSOAs, representative of suburban areas typically with low levels of deprivation. The proportion of green space varies less according to deprivation than other land use types, but it is possible that in areas that are losing suitable garden habitat for sparrows due to development, green space compensates for the loss of gardens to some extent. However the strong association of house sparrows with domestic gardens and houses on a nationwide scale (Chamberlain *et al*; 2007) suggests that green space may be a suboptimal habitat that is only utilised by non-breeding individuals, or those that cannot find sufficient food in

neighbouring gardens. This may in turn explain why green spaces, which are typically homogeneous in structure, are only utilised in areas where gardens are not as suitable for sparrows.

This study shows that large scale land use data can be a useful tool for assessing habitat quality in urban areas where detailed data is difficult to collect; particularly when combined with other large scale datasets. It also supports previous evidence suggesting that house sparrows avoid gardens in areas with low deprivation levels, as these are typically areas with low housing density and large gardens. It appears that although the proportion of land use occupied by domestic gardens is greater in less deprived areas, house sparrow occurrence is lower, signifying that gardens in deprived areas provide better quality habitat than those with lower deprivation levels. Future research into the habitat quality of urban gardens should therefore take into account the differences between gardens in deprived versus affluent areas, and the effect of urban development on housing density and habitat structure.

Even relatively deprived areas may be at risk of redevelopment, particularly in areas with disused industrial sites. UK planning policy states that 60% of new housing should be built on previously developed, or “brownfield” land (Communities and Local Government 2006). As a result, land that is currently classified as green space may be lost to high density modern developments comprising a large proportion of paving. The housing density required of new developments is above that at which the avian species diversity of urban habitats shows a sharp reduction (Tratolos *et al*; 2007). New developments often have a high proportion of paved areas, and consequent increases in the proportion of paving may also have detrimental effects on human populations, for example by increasing flood

risk (Perry & Nawaz, 2008). By continued monitoring of land use changes at a large scale, it may be possible to identify areas at risk of losing species due to development, and mitigate for the effects of land use change on both species occurrence and wider environmental concerns.

Acknowledgements:

This work is funded by the NERC in association with the British Trust for Ornithology (BTO). We thank Mike Toms of the BTO for provision of GBW data on house sparrows, and for advice. Data on land use was obtained from: <http://www.communities.gov.uk>. Thanks also to David Carslake, William Pitchers and Wiebke Schütt for advice and help with R and SPSS. Lastly, thank you to BTO GardenBirdWatch participants, for the collection of the house sparrow data.

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Table 1: Correlation matrix for land use components derived from the ONS “Generalised Land Use database” for England, following principal components analysis.

	Area of domestic buildings	Area of non- domestic buildings	Area of paving	Area of green space	Area of gardens
Area of domestic buildings	1.000	0.105	0.413	0.243	0.832
Area of non-domestic buildings	0.105	1.000	0.736	0.188	-0.113
Area of paving	0.413	0.736	1.000	0.456	0.200
Area of green space	0.243	0.188	0.456	1.000	0.226
Area of gardens	0.832	-0.113	0.200	0.226	1.000

Table 2: Summary of the mean areas and proportions of each land use type at high; medium; and low levels of deprivation.

land use type	high deprivation		medium deprivation		low deprivation	
	area	proportion	area	proportion	area	proportion
Buildings:						
Domestic	33.38	0.13	38.34	0.12	46.69	0.11
Non-domestic	21.23	0.06	13.75	0.03	9.48	0.02
Total	54.61	0.19	52.09	0.15	56.17	0.13
Gardens	91.54	0.31	131.37	0.37	207.76	0.45
Paving	83.48	0.26	83.35	0.21	80.55	0.17
Green space	114.49	0.24	161.23	0.27	176.32	0.25
Total	344.12		428.04		520.08	

Table 3: Summary of Generalised Linear Mixed effects Models for each of four major land use types in English LSOAs in relation to house sparrow occurrence. In all cases the presence\absence of house sparrows was treated as a binomial response variable, and GBW site was a nested factor. Model parameters shown are for the minimal adequate model for each land use type. n = 39,168.

Land use type	Model parameter	Parameter estimate	z-value	Residual deviance/DF	p-value (***)
Area of paving	intercept	8.366	31.879	22989 / 40647	< 0.005 ***
	year	-0.688	-17.061	22989 / 40647	< 0.005 ***
	season: summer	-0.078	-0.417	22989 / 40647	0.677
	season: autumn	-0.717	-3.897	22989 / 40647	< 0.005 ***
	season: winter	-1.457	-8.361	22989 / 40647	< 0.005 ***
	area of paving	-0.934	-13.819	22989 / 40647	< 0.005 ***
	total LSOA area	0.475	10.862	22989 / 40647	< 0.005 ***
	year ²	0.022	9.278	22989 / 40647	< 0.005 ***
	year*season: summer	-0.003	-0.139	22989 / 40647	0.889
	year*season: autumn	-0.064	-2.946	22989 / 40647	< 0.005 **
	year*season: winter	0.061	3.033	22989 / 40647	< 0.005 **
Area of buildings	intercept	-16.470	-8.643	22315 / 39154	< 0.005 ***
	year	-0.677	-16.507	22315 / 39154	< 0.005 ***
	season: summer	-0.040	-0.212	22315 / 39154	0.832
	season: autumn	-0.676	-3.624	22315 / 39154	< 0.005 ***
	season: winter	-1.425	-8.067	22315 / 39154	< 0.005 ***
	(log) area of buildings	5.980	12.230	22315 / 39154	< 0.005 ***
	(log) total LSOA area	4.953	15.189	22315 / 39154	< 0.005 ***
	area of buildings ²	0.022	8.918	22315 / 39154	< 0.005 ***
	year ²	< 0.000	19.390	22315 / 39154	< 0.005 ***
	year*season: summer	-0.007	-0.313	22315 / 39154	0.755
	year*season: autumn	-0.070	-3.151	22315 / 39154	< 0.005 **
	year*season: winter	0.057	2.824	22315 / 39154	< 0.005 **
	log(area buildings)*log(total LSOA area)	-1.264	-15.329	22315 / 39154	< 0.005 ***

Land use type	Model parameter	Parameter estimate	z-value	Residual deviance/DF	p-value (***)
Area of gardens	intercept	6.287	26.285	22217 / 39154	< 0.005 ***
	year	-0.559	-12.646	22217 / 39154	< 0.005 ***
	season: summer	-0.040	-0.213	22217 / 39154	0.831
	season: autumn	-0.674	-3.606	22217 / 39154	< 0.005 ***
	season: winter	-1.428	-8.063	22217 / 39154	< 0.005 ***
	area of gardens	0.005	6.481	22217 / 39154	< 0.005 ***
	total LSOA area	0.001	7.884	22217 / 39154	< 0.005 ***
	year ²	0.020	8.269	22217 / 39154	< 0.005 ***
	year*season: summer	-0.007	-0.303	22217 / 39154	0.762
	year*season: autumn	-0.070	-3.143	22217 / 39154	<0.01 **
	year*season: winter	0.057	2.841	22217 / 39154	<0.01 **
	year*area gardens	-0.001	-7.134	22217 / 39154	< 0.005 ***
	area gardens*total LSOA area	< 0.000	-16.116	22217 / 39154	< 0.005 ***
Area of green space	intercept	7.802	37.512	22317 / 39154	<0.005 ***
	year	-0.659	15.947	22317 / 39154	<0.005 ***
	season: summer	-0.047	-0.249	22317 / 39154	0.803
	season: autumn	-0.692	-3.683	22317 / 39154	<0.005 ***
	season: winter	-1.449	-8.153	22317 / 39154	<0.005 ***
	area of green space	0.005	20.663	22317 / 39154	<0.005 ***
	total LSOA area	-0.004	-21.476	22317 / 39154	<0.005 ***
	area of green space ²	<-0.000	-11.053	22317 / 39154	<0.005 ***
	year ²	0.021	8.565	22317 / 39154	<0.005 ***
	year*season: summer	-0.006	-0.257	22317 / 39154	0.797
	year*season: autumn	-0.067	-3.007	22317 / 39154	<0.01 **
	year*season: winter	0.060	2.959	22317 / 39154	<0.01 **
	area of green space * total LSOA area	<0.000	7.816	22317 / 39154	<0.005 ***

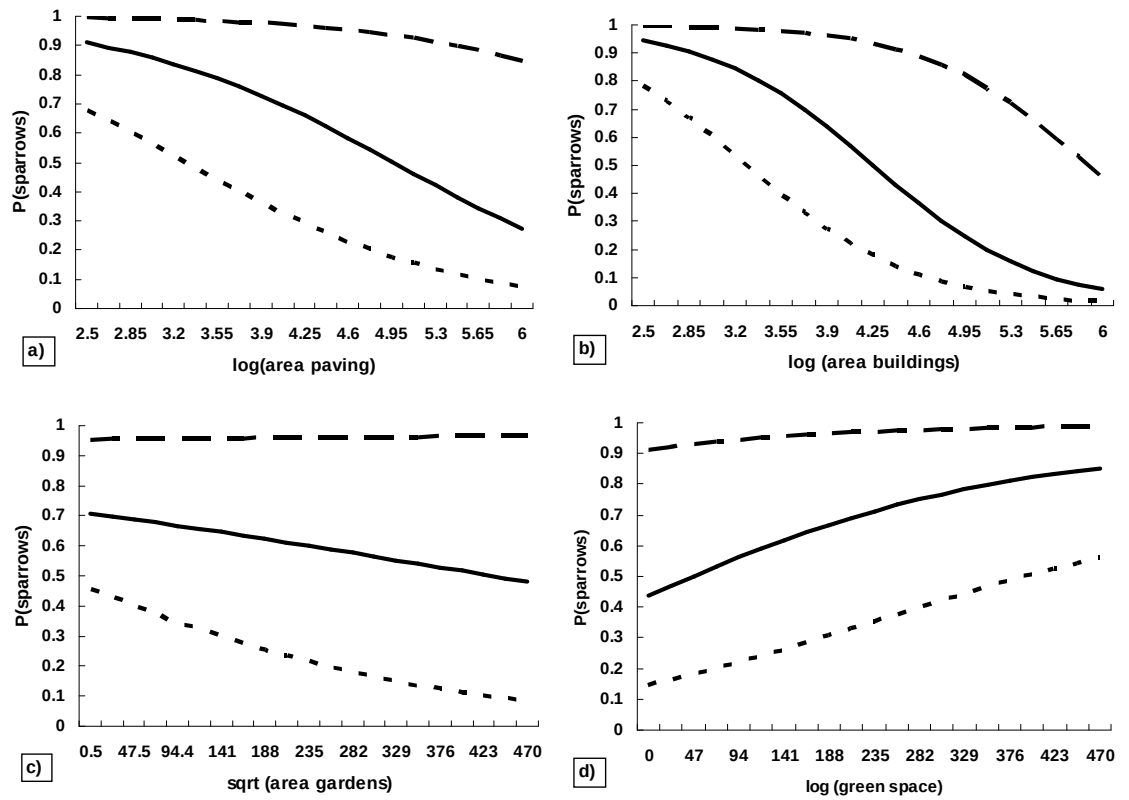


Figure 1: The probability of house sparrow occurrence at GBW sites derived from GLMM parameter estimates for the main urban land use types: a) paving, b) buildings, c) gardens and d) green space. Estimates shown are for average sized LSOAs, and three years are shown, 1995 (dashed line), 2000 (solid line) and 2005 (dotted line). Parameter estimates shown are for spring.

Appendix 5A:

Results of a principal components analysis of the seven main land use types extracted from the Land Use Statistics (Generalised Land Use Database) 2005. Analysis was completed in SPSS v15.0, using Varimax rotation with Kaiser Normalization (Field, 2000). All variables had r-squared values of between 0.1 and 0.9, and therefore none were excluded from the analysis on the basis that they were either highly correlated, or unrelated to each other (Field, 2000). The “Kaiser-Meyer-Olkin measure of sampling adequacy” (KMO) statistics for both individual variables and multiple variables were all greater than 0.5, suggesting an acceptable pattern of correlations (Field, 2000).

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Table A1: The distribution of each land use type on the first three principal components obtained by running PCA with Varimax rotation and Kaiser Normalisation. A high figure denotes that the land use type contributed a relatively large proportion of the variation on that principal component. Also shown are the initial eigenvalues for the first three principal components, and the cumulative variance explained by each. All other principal components had eigenvalues of < 0.8 . The threshold for including principal components in the analysis of intercorrelated data is generally one (Field 2000).

Land use type	Component		
	1	2	3
Domestic buildings	0.94	0.13	0.09
Non-domestic buildings	-0.04	0.98	0.06
Paving	0.24	0.67	0.28
Green space	0.14	0.11	0.97
Gardens	0.96	-0.08	0.11
Water	0.02	0.16	0.13
Rail	-0.01	0.17	0.07
Initial Eigenvalues: total	2.62	1.72	0.96
Cumulative % variance explained	37.47	62.05	75.73

CHAPTER 6

HOME RANGE AND HABITAT USE OF BREEDING

URBAN HOUSE SPARROWS

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Summary:

Recent evidence suggests that the decline of the house sparrow in urban areas may be in part due to small-scale habitat changes that reduce the availability of suitable foraging and nesting habitat for the species. The success of conservation strategies will therefore depend on gaining an accurate estimate of the habitat requirements of the species; both in terms of the area needed to sustain breeding colonies, and also of the selection preferences of birds within their range. Here we use radio telemetry to define the home range of individual birds during the breeding season, and examine the habitat composition of their home ranges in comparison to the surrounding area. As expected, we find that the home range of breeding house sparrows is very restricted, encompassing less than 800m² on average. Within this range, private gardens make up a relatively high proportion of the available habitat, and gardens with a high proportion of green space relative to paving were favoured over those without. Residential buildings were also important as a source of nest sites, but modern, high density housing developments appeared to be avoided. This reflects the importance of private gardens as a resource for house sparrows, and suggests that conservation strategies should focus on maintaining the habitat quality of gardens in close proximity to nest sites, and resisting the trend towards infill development and increased housing density in urban areas.

Keywords: Habitat selection; Kernel density; *Passer domesticus*; Radio tracking

Introduction:

The success of species-specific conservation strategies are dependent to a large extent on obtaining sufficient prior knowledge of the ecology of that species (Blaustein & Kiesecker 2002, Sutherland *et al*; 2004). This information will influence the viability of conservation efforts and mitigation measures. In order to ensure the continued presence of a population, it is important to know how large an area each individual will need to gather adequate food to survive and breed successfully. It is also important to have information about the resources available in a particular habitat, as the quality of the habitat will affect the area needed to sustain the population. In addition to this, the degree of isolation of each habitat patch will affect the viability of the populations they contain; areas that are too isolated from other suitable habitat will leave populations vulnerable to the effects of inbreeding depression (Dickman, 1987). Lack of connectivity in fragmented habitats may also reduce the probability of suitable habitat being recolonised following local extinctions, particularly in the case of species with restricted range or mobility.

The house sparrow *Passer domesticus* is listed as a species of conservation concern in both the UK and Europe, and has declined by over 50% in many urban areas (Siriwardena *et al*; 2002). There are many potential factors that could have caused this decline (Summers-Smith, 2003), but changes to urban habitats may be a contributory factor. House sparrows appear to be more prevalent in relatively deprived areas where there is less urban development, which suggests that habitat loss could be occurring in areas of relatively high socioeconomic status (Shaw *et al*; 2008). The species is highly sedentary, with adults thought to have a range of only one or two kilometres (Snow *et al*; 1998). As such they may become subject to problems associated with habitat fragmentation and population isolation, particularly in urban areas, where the

availability of suitable habitat appears to be increasingly under threat due to the effects of infill development, whereby pockets of previously undeveloped land are converted to housing (Pauleit *et al*; 2005, Chamberlain *et al*; 2007). As a colonial nester, it is potentially vulnerable to both inbreeding depression and the Allee effect, whereby an isolated colony ceases to breed when the number of individuals drops below a threshold level (Veit & Lewis, 1996). It is therefore important to gain an understanding of the habitat types most commonly used by the species in an urban environment, and those which are avoided and may prove a barrier to dispersal between populations.

Habitat selection by individuals can be considered to occur at a number of scales, divided into four broad categories (Johnson 1980). The scale at which selection occurs will influence the conclusions drawn as to the type of habitat a species favours, and it is therefore important to consider the effect of scale when comparing the usage of different habitat types. Here we concentrate on Johnson's second- and third-order habitat selection, i.e. determining the home range of an individual, and the usage of habitat components within that home range (Johnson 1980). This necessitates the definition of an animal's "home range", a task that has long been recognised as problematic.

The home range of an individual was first defined by Burt (1943) as "the area traversed by the individual in its normal activities of food gathering, mating and caring for young". However, this definition has been criticised in the past for being imprecise (White & Garrott, 1990), particularly with regard to what is considered "normal" behaviour. In addition, the area traversed by an individual is likely to vary during different stages of its life cycle: for example in many species juveniles travel further than adults in search of new territories, and there may be differences in dispersal

patterns between males and females within a species. External factors such as the time of year; resource availability between years; and the presence of competitors or predators are also likely to influence the area utilised by individuals. For this reason, Kernohan *et al*; (2001) suggest that home range should be defined as the “extent of an area with a defined probability of occurrence of an animal during a specified time period”. This definition allows some flexibility in deciding on the scope of an animal’s home range, according to the purpose for which it is measured. For the purposes of this study, I have defined the home range of a house sparrow as the area within which an individual is likely to have spent 95% of its time during the tracking period. The overall range of the birds studied here is likely to be somewhat larger than this, as outside the breeding season the range of individual birds is likely to increase. The home range estimates here are therefore likely to be an under-representation of the area an individual will cover during its lifetime.

Within an animal’s home range, habitat use is very unlikely to be uniform, reflecting differences in quality across the available space. Valuable resources are likely to be visited disproportionately more than poor quality areas, and the extent of the home range itself is likely to reflect a preference for that area over the potentially available habitat in the surrounding areas. By dividing a habitat into a number of types the usage of each type can be compared to its availability in order to determine relative habitat preferences of individuals (Aarts *et al*; 2008). Urban habitats are particularly variable at a localised scale, and as such habitat use within the home range of urban house sparrows would also be expected to vary according to an individual’s relative preferences for particular habitat types. Here we use radio telemetry to determine the range of individual house sparrows during the breeding season and determine the relative preferences of the birds for a number of available urban habitat types. We then

discuss the implications of these preferences with regard to urban habitat change and conservation strategies in urban areas.

Methods:

TAGGING BIRDS:

A total of 19 house sparrows were caught using mist nets at 6 locations within the Bristol urban boundary (Table 1). Tagging locations were private gardens, and were at least 1km apart, roughly equal to the previously documented range of adult house sparrows (Snow *et al*; 1998). Therefore overlap between individuals tagged at different sites was not expected to occur. All tagging sites were located in predominantly residential areas to the north and south east of Bristol city centre. The mean deprivation scores at the tagging sites (according to the Index of Multiple Deprivation, Noble *et al*; 2004) were very similar to the mean level of deprivation reported for Bristol as a whole; which in turn is slightly higher than mean levels of deprivation reported for other UK cities. All sites reported sighting house sparrows on a regular basis.

Birds were tagged in June 2007 and 2008. Data collection commenced approximately 24 hours after tagging to minimise the possibility of data being influenced by atypical behaviour of the birds (i.e. avoidance of the capture site) immediately after release. Individuals were then tracked for three weeks from the date of tagging using a flexible Yagi antenna on 173 MHz. The transmitters were detectable over a relatively small scale (usually 30-60m), in part due to the built up nature of the habitat, and the resulting blockage and reflection of the signal from buildings and walls. For this reason it was not possible to triangulate the position of the birds, however the small distances involved in combination with the signal strength and direction enabled the birds to be located to within two metres, either by sight or by song when individuals were located in dense cover.

Tail mounted ‘pip’ transmitters were attached to the base of the central tail feather with glue, and started either by soldering two bare wires, then covering with potting material (2007), or using integral reed switches (2008). Tags were checked to ensure they were in working order, and each bird was weighed and the wing length measured prior to release. Adult birds were tagged wherever possible, however on two occasions where insufficient adult birds were caught, juveniles were tagged and the sex of these birds is therefore unknown. Each tag weighed 0.5g. The mean weight of the tagged birds was 25.5g and each tag therefore represented approximately 2% of the body weight of the bird, comfortably under the level at which adverse effects have been reported for flying animals (Kenward, 2001).

HOME RANGE ANALYSIS:

Radio tagging data was plotted and projected in ArcGIS v9.2 using the British National Grid coordinate system (ESRI, 2009). Home range estimates and kernel densities were then calculated using the ‘Batch Fixed Kernel Density Estimator’ and ‘Percent Volume Contour’ tools in the “Hawth’s tools” ArcMap extension package (Beyer, 2004). Kernel methods were used as they are considered more accurate than other analysis methods such as Minimal Convex Polygons, and provide a means of assessing habitat selection within an individual’s potential range (Seaman *et al*; 1999). Kernel densities also give the utilisation distribution of the animal; or the probability of occurrence at each point in space. This allows the core areas of each home range to be identified, and therefore the areas in which each individual spends most of its time. Computer simulations by Swihart & Slade (1997) also found that kernel estimators of home range area were also less likely to be affected by autocorrelation than convex polygon methods.

The smoothing parameter was calculated using Least Squares Cross Validation (LSCV, Worton, 1989). Current opinion is divided over the most appropriate method to measure the smoothing parameter when calculating kernel densities, but LSCV was chosen in this case as it is a widely used and convenient method. Comparison of different smoothing methods by Seaman *et al*; (1999) has shown that the choice of smoothing parameter has no effect on the fit of kernel density estimates. It was then possible to generate area and perimeter estimates for the kernel densities of each individual using Hawth's tools (Beyer 2004).

In order to assess the accuracy of the home range estimates obtained, kernel estimates were recalculated for each individual using subsamples of 10, 15, 20, 25, 30, 35 and 40 locations wherever possible. The mean home range area estimates obtained from these subsamples were then plotted against the number of locations used in order to check the stability of the estimates (Figure 2). Analyses of Variance were used to detect differences in home range size between the sexes, and between data collected in each of the two years of the study. Juveniles were excluded from the analysis of sex differences, as it was not possible to determine their sex.

Autocorrelation, or non-independence of data points, can potentially affect the accuracy of home range estimates, and lead to the distortion of kernel density estimates. Opinion of the importance of this effect is divided, as it can only be avoided by using locations rather than individuals as the sample unit for analysis (Otis & White, 1999). This in turn causes problems due to pseudo replication. In order to combat this sampling at longer intervals has been recommended, but this can result in insufficient data being collected (Swihart & Slade, 1997). This is generally thought to cause more problems than the effects of autocorrelation itself, as low sampling effort can lead to underestimation of an

animal's home range. In this study, the effects of autocorrelation are likely to be relatively minor, as the house sparrow is a sedentary species and has a relatively small home range. As a result all data points collected were used for analysis, but the sampling interval between locations was kept at a minimum of 15 minutes. In addition, kernel densities for those individuals for which adequate data was collected were reanalysed using random subsamples of the data in order to check for the effects of autocorrelation on the size of home range estimates.

COMPOSITIONAL HABITAT ANALYSIS

Habitat usage was determined at two spatial levels; first by quantifying each habitat type within an individual's home range, then by examining habitat usage within the home range in relation to availability of habitat types in the wider area. The potential habitat available to each individual was defined as the area within a distance of 1km from each tagging site, in accordance with the estimated range of adult sparrows outside the breeding season (Snow *et al*; 1998). Habitat use was quantified by estimating the proportion of each different habitat type within the area in question, using satellite imagery from Google Earth (www.earth.google.co.uk). Habitats were divided into five broad categories: Buildings (domestic or commercial); Private Gardens (subdivided into <50% paved, and >50% paved); Green Space (parks and public areas); Paving (roads, paths, car parks etc); and Other (e.g. water; building sites). Buildings were further classified by age, using classifications defined by the Office for National Statistics (ONS, 2007).

Proportional habitat usage was then evaluated using methodology outlined by Aebischer *et al*; (1993). This uses the individual animal as the sampling unit, thereby avoiding the problems associated with data pooling across individuals. Compositional analysis also

avoids a common problem associated with the use of proportion data - namely that the proportions are non-independent - by using one habitat type as a denominator (here paving) and transforming the data using the log-ratio transformation to remove the linear dependency (Aebischer *et al*; 1993, Pendleton *et al*; 1998). All the habitat types surveyed were available to all individuals, but not all birds were detected using each habitat type. Where usage of a particular habitat type by an individual was effectively zero, a small positive number (an order of magnitude less than the smallest recorded non-zero proportion) was substituted, as recommended by Aebischer *et al*; (1993). Compositional analysis was carried out in SAS (www.SAS.com). Habitat usage within each bird's home range was analysed in relation to the log ratios that would be expected if habitat use by each bird was random. If usage by house sparrows was non-random with respect to availability, habitats were ranked in order of preference, based on the log ratios of used to available habitat types.

Results:

RADIO TRACKING EFFORT:

Of the 19 tagged birds, 12 were relocated >15 times after release (Table 1), with six of these detected on over 25 occasions. All of those that were detected within two days of release provided data throughout the three-week data collection window, with the exception of bird 16 (173.312; 2008), which appeared to lose its transmitter halfway thorough the study period. This bird was therefore excluded from the home range analysis.

The cause of transmitter failure in the remaining birds is not known, and access restrictions made it impossible to recover the transmitters. It is possible that the birds in question were not detected simply due to the fact that their home ranges were predominantly in areas that were located behind solid obstacles such as walls and

buildings and they were therefore undetectable from ground level. This is supported by the fact that weak signals were detected intermittently from other individuals in a few cases, and may potentially bias the estimates of home range obtained from some birds. However, it is unlikely that solid obstacles blocked all transmitter signals from those birds that were not detected at all after release, as virtually all gardens and green spaces within the study areas were visible either from adjacent roads or public footpaths. Furthermore, birds were detected successfully at a variety of locations around all tagging sites, making it unlikely that a bird would not be detected on any occasion. It is therefore probable either that the transmitters on these birds failed, or that they lost their transmitters by shedding the feather to which it was attached in an undetectable location.

HOME RANGE SIZE:

Figure 1 shows the plotted home range contours for four of the radio tagged individuals. The mean home range of the house sparrows in this study as defined by the 95% kernel density estimates, was approximately 760m² (Table 2). No overlap was detected between birds at different tracking sites, and no bird was detected at a distance of greater than 100 metres from the site at which it was tagged. There was no significant difference in the kernel density estimates obtained from birds tracked in 2007 and in 2008 (Analysis of Variance, $p = 0.11$, $F = 15.78$, $n = 52$). Initial analysis of home range area in relation to sex suggested that females had significantly larger home ranges than males, but this result was heavily influenced by the 95% kernel estimate for individual 173.236, which had a larger home range than all the other individuals tracked. When this individual was removed from the analysis, no significant differences between the sexes were detected (Analysis of Variance, $p = 0.24$, $F = 30.4$, $n = 44$).

Home range estimates for the 25% and 50% kernel density estimates were remarkably consistent, even with as few as 10 locations (Figure 2). The 75% kernel estimates remained stable at approximately 290m² with 30 or more locations. The 95% kernel density estimates increased slightly with an increase in the number of locations used, suggesting that overall home range estimates may be slightly underestimated. In spite of this it appears that the home range estimates obtained in this case are reliable indicators of the overall home range size for each individual, and the overall stability of the home range estimators suggests that autocorrelation did not affect the kernel density estimates obtained.

COMPOSITIONAL HABITAT ANALYSIS

The most commonly available habitat for the house sparrows tagged in this study was green space, comprising on average 34% of the 1km around each tagging site. Buildings and gardens comprised on average 21% and 23% of the available habitat respectively, and paving on average contributed 18%. Other habitat within 1km² of the tagging sites consisted mainly of building sites; allotments; rivers; and 'brownfield' sites – defined as previously developed land (London Assembly; 2005). On average these occupied only 5% of the potential habitat available to each bird. Most buildings in the study sites surveyed were residential, of which the majority were built between 1914 and 1970. Pre-war and modern buildings (built post-1970) were also present in all areas, but made up a small proportion of overall land use (typically between 2 and 3% of total land use on average).

Habitat usage within each area was non-random with respect to availability (Wilks's $\lambda = 0.054$, $P < 0.001$). House sparrows preferentially selected gardens over other habitat types, and on average over 80% of habitat within a home range was comprised of private gardens. The proportion of gardens within the home range of the sparrows that

contained over 50% paving was the same as that in the available habitat in general (typically about 5%). However, gardens with less than 50% paved surfaces comprised on average over 75% of home range habitat usage, compared to just 18% of the available habitat in general. Buildings were also heavily utilised, although non-residential and modern buildings were not used by the radio tagged birds. Both of these building types comprised a very small percentage of the overall available habitat, but were present at very close proximity to some tagging sites. House sparrows appeared to actively avoid green space, and paved areas were also relatively under-utilised (Table 3).

Discussion:

The home ranges reported for house sparrows in this study are consistent with previous estimates, and reflect the sedentary nature of the species (Snow *et al*; 1998). House sparrows are known to forage in close proximity to their nest site during the breeding season (Vincent, 2005), and as such the relatively small area covered by the birds tracked here is likely to reflect the fact that most were breeding adults. It is notable that the core of each home range (within the 25% kernel contour) often covered just one or two gardens. These are likely to be in close proximity to the nest site, and both tagged and untagged males were observed singing outside nest holes in houses adjacent to gardens within these core areas.

The gardens adjacent to these nest sites are likely to encompass the main foraging habitat for house sparrows when feeding young. Invertebrates are an important source of protein for house sparrow nestlings, and an insufficient proportion of invertebrates in the diet of nestlings has been linked to reduced fledging success in both house sparrows and other bird species (Borg & Toft, 2000; Peach *et al*; 2008). The 25% kernel

estimates generally contained a high proportion of shrubs and small native trees, which are likely to be associated with relatively high invertebrate abundance (Southwood, 1961). Locations visited on a less regular basis often contained supplementary food, and may therefore be visited by adults in order to obtain food for themselves, or to engage in other activities such as dust bathing and preening.

It is probable that the small home range of the house sparrows in this study is not representative of urban sparrows in general for a number of reasons. Firstly, house sparrows are likely to have a much larger range outside of the breeding season, particularly in the case of juvenile birds (Snow *et al*; 1993). Secondly, it is possible that radio tagging the birds affected their behaviour, for example by increasing the costs associated with flying (Kenward, 2001). Whilst we took precautions to ensure that the tags did not have a detrimental effect on the birds, this possibility cannot be ruled out entirely. However, observation of untagged birds suggests that the areas covered by tagged individuals were typical of habitat use by sparrows in general at each site. There was no evidence that habitat use by house sparrows changed following capture or tagging, and tagged individuals did not appear to subsequently avoid the sites where they were captured.

It is also possible that the estimates of home range and habitat availability are adversely affected due to the built-up nature of the habitat, in particular the potential for variation in detectability caused by blocking of transmitter signals by solid objects. This is unlikely to have affected overall home range estimates, as virtually all front and back gardens in the vicinity of the tagging sites were accessible from public land for the purposes of picking up transmitter signals. However, it is possible that the observed habitat use of some individuals was biased due to differential signal detectability in

different habitat types. Detectability is unlikely to have differed between paved areas, green space and gardens, as these habitats were all equally accessible. However, it is possible that buildings were underrepresented in the case of some birds, particularly those that were visiting nest site cavities as the signal may have been blocked by the surrounding walls.

Therefore care should be taken not to underestimate the importance of residential buildings as a habitat for house sparrows in their own right. Although non-residential buildings may also provide suitable nest sites they were not utilised in this study, probably due to their increased distance from private gardens in relation to domestic dwellings. All age classes of domestic buildings were utilised in proportion to their overall availability with the exception of modern buildings, which appeared to be avoided. This type of housing was relatively scarce within the habitat around the tagging sites, which implies that house sparrows may be actively selecting against this building type when choosing suitable sites for nesting. A substantial area of modern housing was situated adjacent to one of the tagging sites (Figure 1), but the houses in this area were the only ones in the immediate vicinity where the tagged individuals were not recorded, suggesting that this area may be unsuitable for foraging birds.

There is evidence to suggest that new buildings may provide comparatively few cavities for nesting birds such as sparrows (Shaw *et al*; 2008). This is likely to be because they are in a better state of repair than older dwellings, and the materials used in their construction present fewer opportunities for accessing the eaves of houses. It is notable that the core ranges of two of the four birds tagged at one site (Pen Park Road) were centred on an unoccupied house in a poor state of repair. Modern housing is built at a much higher density than housing constructed pre-1970, and the density of new

developments in the UK is expected to increase still further in the future (Office of the Deputy Prime Minister 2003, Entec UK, 2004). This will reduce the amount of land designated as private gardens, and the available foraging habitat for house sparrows will be further reduced in modern housing developments. A large proportion of modern developments consist of parking areas for cars, reducing the amount of space available for the plants needed to sustain invertebrate populations. Both of these factors are potentially enough to prevent house sparrows from nesting in new developments, even where nest sites are available.

Private gardens have previously been identified as an important source of food for many bird species throughout the year (Gaston *et al*; 2005). Urban specialists are reliant on the resources that gardens provide, as other suitable foraging habitats are likely to be scarce in inner city areas. The predominance of this habitat type in the home ranges of the birds studied here is therefore consistent with findings from the UK as a whole, and other European cities such as Valencia (Chamberlain *et al*; 2004; Murgui, 2009).

Allotments and brownfield sites have also been identified as important resources for urban wildlife, and allotments are key predictor of house sparrow density in the UK (Small *et al*; 2002; Chamberlain *et al*; 2007). They were not utilised by the birds in this study, probably due to their relative scarcity and the fact that suitable habitat for foraging was available closer to the tagging sites. However, both these and brownfield sites are prime targets for redevelopment in many cities (London Assembly, 2005), and their loss may exacerbate habitat loss caused by the development of private gardens.

The apparent avoidance of green space by house sparrows is perhaps surprising, given that public green spaces can be an important resource for wildlife (e.g. Mason, 2006). However, public green spaces are often intensively managed, homogeneous habitat, and

large areas of continuous green space are unlikely to provide nesting and foraging habitats in close proximity to each other. Whilst small areas of green space may prove invaluable in areas with high building density, it is possible that large continuous areas of green such as golf courses may actually prove to be a barrier to dispersal in the case of sedentary species such as the sparrow. Less intensive management of public green spaces, and the creation of habitat that is not only more suitable for wildlife, but also has beneficial effects for humans may provide a way of combating this problem (Fuller *et al*; 2007).

These results suggest that conserving habitat for breeding house sparrow populations may entail preserving a relatively small area of suitable foraging habitat, as comparatively few houses are within the range of individual birds. By the same token however, the loss of relatively small areas of private gardens is predicted to lead to a rapid decline in house sparrow abundance (Chamberlain *et al*; 2007). The small home range of the house sparrows seen here suggests that the loss of even two or three gardens around a nest site may substantially increase the foraging range of adult house sparrows. The subsequent increase in energy expenditure for adult birds feeding nestlings may be enough to render the habitat around that site inadequate for the successful rearing of young, and lead to the loss or relocation of colonies. In addition, if adult birds are forced to expend more time in order to forage effectively, they may become more vulnerable to predation. House sparrows are known to trade off the benefits of carrying extra body mass to ward off starvation against the potential costs due to mass-dependent predation risk (MacLeod *et al*; 2006). It is important therefore to consider the indirect effects of habitat change on vulnerable populations when assessing the potential impacts of urban development.

The results of this study show that habitat use by house sparrows during the breeding season occurs at a relatively localised scale. This could indicate that a relatively small area is sufficient to maintain urban house sparrow populations during the breeding season. This in turn suggests that conservation efforts may only need to be targeted at small groups of houses in order to preserve the existence of whole colonies. However, as the range of individual birds is likely to increase outside the breeding season the availability of suitable habitat in the surrounding area also needs to be addressed. In addition, the small area covered by individual birds could indicate that their range is constrained by external factors such as competition with other individuals, or unsuitable habitat in the surrounding areas. Patches of suitable habitat surrounded by extensive areas of redevelopment could also hinder the dispersal of juvenile birds, and prevent the establishment of new colonies in an area. This would inhibit gene flow between colonies and increase the likelihood of inbreeding depression and local extinction of colonies, a problem that could be exacerbated by the Allee effect (Stephens & Sutherland, 1999).

The apparently minor changes to urban habitats caused by localised developments could result in loss of colonies from those urban areas affected, as has been seen in many urban areas across Europe (Shaw *et al*; 2008). UK planning policy recommends that 60% of new housing be built on previously developed land (Communities and Local Government, 2006). This housing is likely to be built at up to twice the density seen on older developments, with consequent reductions in garden size (Figure 3). The implications of increasing housing density in modern developments and reducing the area of gardens for wildlife should therefore play an important part in planning decisions for urban habitats.

Acknowledgements:

This work forms part of an NERC case PhD studentship, in association with the British Trust for Ornithology (BTO). We thank the BTO for funding assistance, and for the loan of radio tracking equipment. The authors would also like to thank Dr Andy McGowan for help with catching and tagging house sparrows, Dr Matthew Witt for help with GIS work, and Biotrack for providing tags and advice. Thanks also to Paul Brookman and family, Pru and Brian Comben, Mitch Crossingham, Rodney Holbrook, Nicola Ramsden, David and Susan Slatter and Mrs Wilmer for allowing me access to their gardens in order to catch and tag birds, and to other householders who assisted by allowing access to their gardens.

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Table 1: Details of the 19 radio tagged birds. Birds highlighted in bold were detected on >15 occasions after release.

ID	Transmitter frequency	Sex/age	Site/ Bristol ward	weight (g)	wing length (mm)	Date tagged
1	173.246	Male	St Anne's / Brislington East	25.2	80	07/06/2007
2	173.309	Female	St Anne's / Brislington East	25.7	77	07/06/2007
3	173.275	Male	Lockleaze	26.7	79	07/06/2007
4	173.212	Female	Lockleaze	28.9	76	07/06/2007
5	173.327	Male	Henleaze	23.3	81	07/06/2007
6	173.288	Male	Henleaze	25.1	80	07/06/2007
7	173.230	Male	Bishopston	25.6	80	07/06/2007
8	173.266	Female	Bishopston	24.2	76	07/06/2007
9	173.257	Female	Southmead	29.7	78	07/06/2007
10	173.336	Male	Southmead	29.3	79	07/06/2007
11	173.278	Male	Southmead	26.5	78.0	07/06/2008
12	173.334	Male	Southmead	24.3	77.0	07/06/2008
13	173.236	Female	Southmead	24.8	76.0	07/06/2008
14	173.250	Male	St Anne's / Brislington East	25.0	80.0	21/06/2008
15	173.323	Juvenile	St Anne's / Brislington East	23.4	72.0	21/06/2008
16	173.312	Male	Bishopston	24.9	78.0	21/06/2008
17	173.212	Juvenile	Bishopston	22.6	70.0	21/06/2008
18	173.285	Male	Redland	24.3	77.0	21/06/2008
19	173.335	Female	Redland	25.4	77.0	21/06/2008

Table 2: Mean area and perimeter values for house sparrow home range kernel density estimates.

kernel %	area / m2	perimeter / m
25	97.8	41.0
50	192.9	128.2
75	387.0	256.6
95	763.3	449.8

Table 3: T-statistics and P-values comparing habitat selection by house sparrows

relative to the availability of five habitat types, as determined by compositional habitat analysis (Aebischer *et al*; 1993). Preference rankings for each habitat type are shown, with 0 = least preferred. The reference habitat type was paving.

Habitat	Buildings	Gardens	Green space	Other	Paving	Rank
Buildings	-	-4.5977	5.0163	1.69147	2.24112	3
	-	0.01	0.01	0.11	0.06	
Gardens		-	12.2341	7.02827	5.37686	4
		-	0.01	0.01	0.01	
Green space			-	-3.46527	-3.72054	0
			-	0.04	0.01	
Other				-	-0.30236	1
				-	0.72	
Paving					-	2
					-	



Figure 1: Kernel density contours for the home ranges of four of the house sparrows tagged in urban Bristol in 2007 and 2008. The tagging location is denoted by a red star. Note that the home ranges of the four birds encompassed housing in all directions except to the North-East of the tagging site – this area consisted solely of modern housing. The core home range of two of the individuals (in blue and yellow) centred on a house that had been unoccupied for eight years, and was consequently in poor condition.

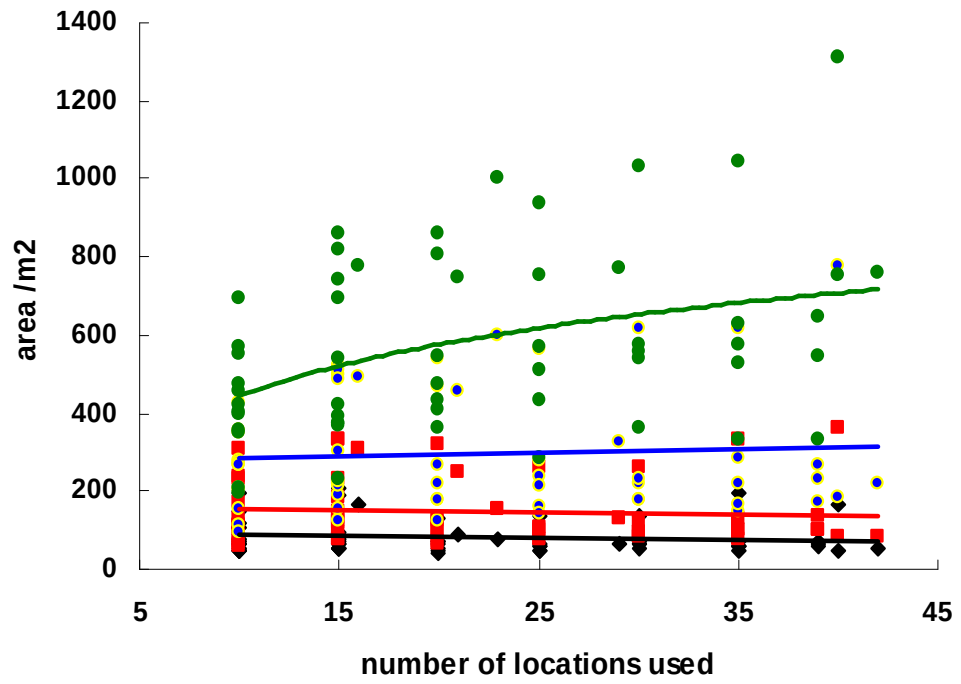


Figure 2: Kernel density area estimates for randomly selected subsamples of the radio tracking data, using varied sample sizes. Black = 25% contours; Red = 50%; Blue = 75%; Green = 95%.



Figure 3: Post war housing in Bristol (top), and modern housing (below). Modern housing in Bristol is built at up to twice the density of previous developments, leaving little room for the planting of mature trees and shrubs. Modern housing also has flatter tiles and more airtight roof spaces than older housing, limiting nesting opportunities.

Appendix 6A: Home range areas and kernel densities for house sparrows in urban Bristol.

Site	Transmitter	Kernel %	Area/m²	Perimeter/m
Hill Burn	173.288	25	66.2	29.4
Hill Burn	173.288	50	123.5	79.9
Hill Burn	173.288	75	305.1	183.9
Hill Burn	173.288	95	537.8	337.6
Hill Burn	173.327	25	164.4	64.3
Hill Burn	173.327	50	308.9	199.2
Hill Burn	173.327	75	492.5	390.9
Hill Burn	173.327	95	778.2	588.0
Shaldon Road	173.275	25	66.0	29.5
Shaldon Road	173.275	50	100.0	80.8
Shaldon Road	173.275	75	169.1	132.0
Shaldon Road	173.275	95	331.3	194.7
Shaldon Road	173.212	25	78.4	32.0
Shaldon Road	173.212	50	157.0	88.1
Shaldon Road	173.212	75	598.5	331.2
Shaldon Road	173.212	95	1002.6	702.1
Highfield Grove	173.266	25	61.6	28.3
Highfield Grove	173.266	50	98.5	74.6
Highfield Grove	173.266	75	230.1	142.9
Highfield Grove	173.266	95	544.4	289.1
Highfield Grove	173.312	25	79.2	37.3
Highfield Grove	173.312	50	147.8	112.4
Highfield Grove	173.312	75	223.2	211.7
Highfield Grove	173.312	95	366.0	310.0

Site	Transmitter	Kernel %	Area / m ²	Perimeter / m
Pen Park	173.336	25	72.6	30.5
Pen Park	173.336	50	135.7	87.7
Pen Park	173.336	75	268.7	163.2
Pen Park	173.336	95	648.3	316.4
Pen Park	173.257	25	64.9	28.6
Pen Park	173.257	50	129.8	85.4
Pen Park	173.257	75	326.9	171.6
Pen Park	173.257	95	773.7	373.1
Pen Park	173.278	25	53.3	26.2
Pen Park	173.278	50	83.9	68.6
Pen Park	173.278	75	216.7	118.1
Pen Park	173.278	95	759.6	292.1
Pen Park	173.236	25	163.9	47.6
Pen Park	173.236	50	410.6	150.3
Pen Park	173.236	75	866.6	301.5
Pen Park	173.236	95	2024.4	749.1
Brookfield Avenue	173.285	25	189.4	93.2
Brookfield Avenue	173.285	50	331.5	296.2
Brookfield Avenue	173.285	75	522.9	518.0
Brookfield Avenue	173.285	95	860.0	736.6
Wootton Crescent	173.323	25	119.8	51.5
Wootton Crescent	173.323	50	232.9	198.5
Wootton Crescent	173.323	75	357.3	360.7
Wootton Crescent	173.323	95	549.7	483.7
Wootton Crescent	173.309	25	91.9	34.1
Wootton Crescent	173.309	50	247.0	144.2
Wootton Crescent	173.309	75	453.9	309.4
Wootton Crescent	173.309	95	747.3	475.2

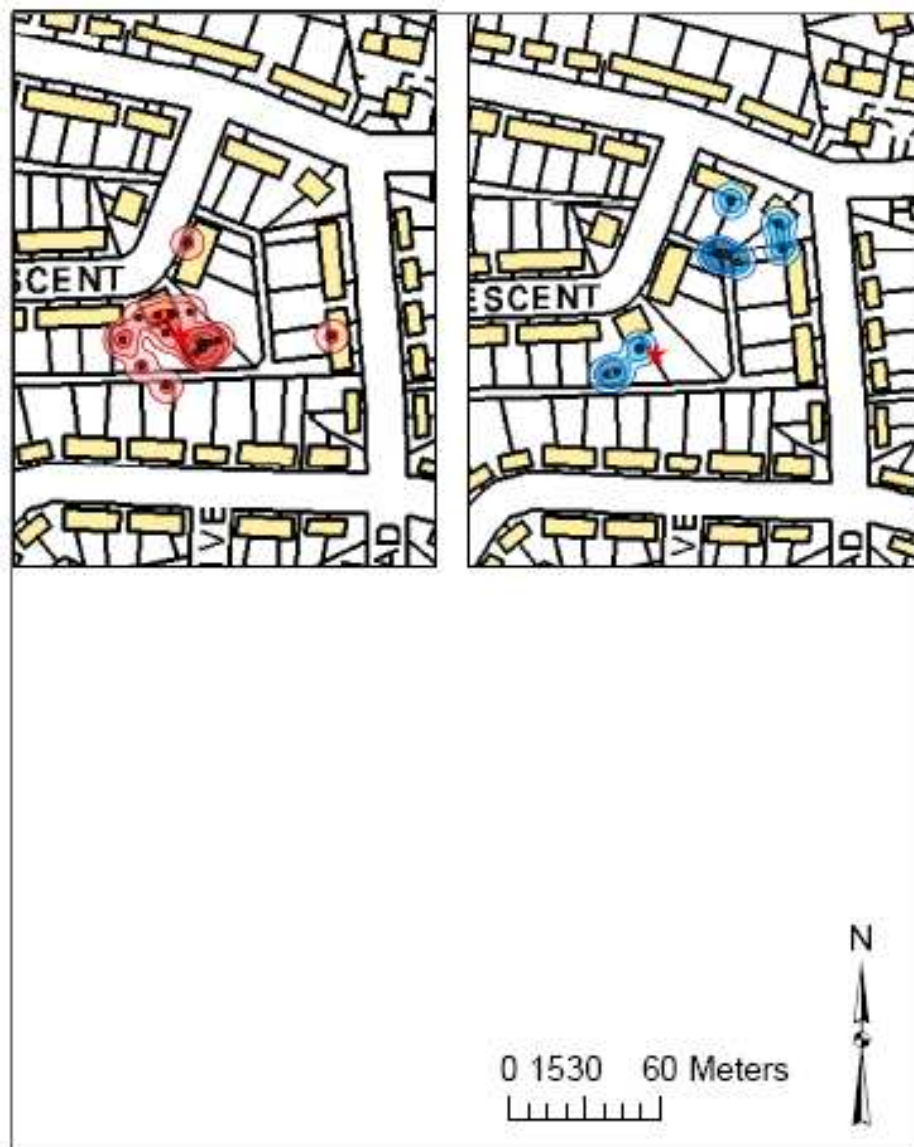
Appendix 6B:

Kernel density contours for the home ranges of all other house sparrows tagged in urban Bristol in 2007 and 2008. Bird 16 (green, fig a) lost the transmitter halfway through the tracking period, and was therefore excluded from the home range analysis. The tagging locations are denoted by red stars.





b)



c)

CHAPTER 7

DISCUSSION: IMPLICATIONS OF URBAN DEVELOPMENT FOR THE HOUSE SPARROW, AND FOR WIDER URBAN BIODIVERSITY

The environmental implications of a growing human population present many challenges for the maintenance of biodiversity. However, whilst increasing urbanisation can have a detrimental effect on species occurrence and productivity, it is also clear that in some cases urban areas can provide a valuable resource for many species, several of which are of conservation concern (Chapter 1). The aim of this thesis was to use national and local scale data to shed light on potential causes of decline in the urban house sparrow. However, the difficulties associated with surveying urban habitats can be a barrier to collecting useful data in urban environments. I have used a combination of local scale habitat data, landowner surveys and large scale indices to assess urban habitat quality for the house sparrow. In particular I have focussed on the relationship between human socioeconomic deprivation and urban land use.

Anecdotal evidence has suggested that low socioeconomic status is linked to an increased incidence of house sparrow occurrence. Reviewing the current literature showed that this pattern has been reported in cities across Europe, and that a potential explanation for the phenomenon lay in changes to house and garden features that have affected urban habitat structure in less deprived areas (Chapter 2). Surveying households in one city - Bristol, UK - showed that there are a number of land use factors that are related to low levels of socioeconomic deprivation (Chapter 3). These include the presence of detached and interwar housing; the occurrence of roof repairs, and the presence of paved areas within private gardens. These factors were not linked directly to the presence of house sparrows but all are likely to be indicative of measures of habitat quality relevant to the species; for example the availability of nest sites in the eaves of houses and of suitable sites for foraging.

As these features were linked to deprivation at a local scale they provided a potential mechanism to explain the observed pattern of house sparrow occurrence with indices of deprivation (Chapter 2). By using the BTO GardenBirdWatch survey and the England Index of Multiple Deprivation, it was possible to demonstrate a correlation between deprivation and house sparrow occurrence on a national scale (Chapter 4). House sparrows were more likely to occur in relatively deprived areas of cities, as would be expected if less deprived areas present fewer opportunities for nesting and foraging (Chapters 3 & 4). This pattern is particularly evident in spring, supporting the suggestion that the causes of decline in this species are likely to affect breeding success rather than over winter survival (Vincent 2005, Chapter 2). This would be expected if foraging habitat was lacking in less deprived areas as adult birds are granivorous, but need insect food to provide protein for their chicks during the breeding season. Lack of insect food has previously been linked to lower fledging weight, and therefore a reduced chance of survival, in juvenile house sparrows (Vincent 2005, Peach *et al*; 2008). Moreover, houses in areas that become relatively more deprived over time appear more likely to start sighting house sparrows in their gardens than those that do not (Chapter 4).

This does not imply that high levels of deprivation are a good thing *per se*, either for human or wildlife populations. Rather it is more likely that infill development and increased levels of paving in areas associated with low levels of deprivation may be negative in the sense that they render these areas less suitable as habitat for nesting and foraging birds. Comparison of house sparrow occurrence in relation to aspects of urban land use across England supports this hypothesis, as do the findings of a localised study of house sparrow home range and habitat preferences (Chapters 5 & 6). In common with previous reports (Snow *et al*; 1998, Vincent 2005), the home range of urban

breeding house sparrows was found to cover a very restricted area, and paved areas were avoided. Both urban green space and private gardens were linked to house sparrow occurrence, although the non-significant association of sparrows in relation to the area occupied by private gardens at a national scale is intriguing, as it contradicts the findings of previous studies (Chamberlain *et al*; 2007), and the data collected at a smaller scale (Chapters 3 & 6). This suggests that the benefits of private gardens do not necessarily increase with size, and may be due to the fact that many gardens contain a high proportion of paving, a factor not reflected in the nationwide land use data as this classifies all private land associated with domestic buildings as gardens, regardless of their habitat composition. This appears to be particularly the case in less deprived areas, where infill development and paved gardens are more prevalent than in areas with high levels of deprivation (Pauleit *et al*; 2005).

THE HOUSE SPARROW: FUTURE RESEARCH DIRECTIONS

The use of large scale datasets provides a convenient way of monitoring a host of interconnected factors that may influence house sparrow populations (Chapters 4 & 5). This could facilitate standardised comparisons between different urban habitats; provide a means of monitoring spatial patterns in the distribution of the species; and allow mitigation measures to be targeted to areas where they are likely to be successful at restoring or maintaining house sparrow colonies. However, it is also apparent that to really understand the mechanisms behind the population decline of the house sparrow data from smaller scale studies is also important. This is shown by the apparent anomaly between the results of Chapters 5 and 6. The nationwide study detects no significant association of house sparrows with private gardens. However, the more detailed radio tracking study shows that house sparrows spend the majority of their time in gardens, but distinguishes between those with low and high proportions of paving and finds that

sparrows avoid the latter. It is therefore important to use a combination of approaches when assessing urban habitat quality for conservation purposes (Shochat *et al*; 2006).

In the case of the house sparrow, research investigating the causes of decline has highlighted a number of factors that may contribute to a decline in urban habitat quality for the species. These include a lack of nesting opportunities in areas undergoing development; and a lack of insect food for chicks; probably due to a loss of invertebrate rich habitats to development and the demand for off street parking (Chamberlain 2007, Peach *et al*; 2008; Shaw *et al*; 2008). There is evidence that the availability of insect food is a limiting factor in some house sparrow populations, and that certain types of insect food are of higher nutritional benefit than others (Vincent, 2005, Peach *et al*; 2008). However, although the insect diversity on urban areas is becoming more widely studied, research into the key habitats and plants for wildlife is still conflicting and future studies should address this issue in more detail.

These factors are likely to be exacerbated by other issues such as the trade-off between foraging, body mass and predation risk. Small birds often carry excess fat reserves as a safeguard against unpredictable food supplies, but this can reduce their mobility and therefore make them more vulnerable to predation (Gosler *et al*; 1995). This can influence the choice of foraging habitat by individual birds (Suhonen, 1993). House sparrows may be vulnerable to habitat changes that reduce the predictability of their food supply, as they have been shown to regulate their fat reserves according to predation risk, rather than just the risk of starvation (Macleod *et al*; 2006). Loss of green space to paving could therefore affect house sparrows by increasing their risk of starvation, as well as by directly increasing their risk of predation.

Lack of cover for foraging birds could also increase their risk of predation in areas with low deprivation levels. House sparrows are preyed upon by domestic cats in large numbers, and a small scale study in Bristol, UK showed that the rate of predation of sparrows by cats was enough to have a negative impact on the population as a whole (Baker *et al*; 2005). Other predators such as sparrowhawks *Accipiter nisus* have recolonised urban areas since the banning of organochlorine pesticides, although there is no evidence to suggest that this is linked to the decline in house sparrow populations (Chamberlain *et al*; 2009b). However, sparrowhawks may have a localised impact on house sparrow numbers, particularly in less deprived areas, where large trees are available for the birds to nest.

House sparrows may also be affected by air pollution, either due to direct toxicity, or indirectly, due to detrimental effects on their insect prey. (Summers-Smith, 1999). Air pollution has been implicated in reducing overall insect diversity in urban areas, and arthropod community composition is also known to vary according to urbanisation (McIntyre, 2000). A study of urban house sparrows in Leicester, UK suggested that nestling body mass and condition were negatively correlated with local levels of nitrogen dioxide air pollution (Vincent, 2005). Pollution may therefore also have contributed to the decline of house sparrows in urban areas. However, analysis of house sparrow occurrence at a city wide scale suggests that they are more likely to occur in deprived areas, which are typically nearer the city centre and therefore presumably more polluted than suburban areas (Chapters 4 & 5). It is therefore likely that the loss of foraging habitat to development is affecting insect availability for house sparrows more than pollution, although the effect of pollution on invertebrate populations in areas where house sparrows are already under increased foraging pressure may be enough to reduce nestling survival and fledging success.

Given this information, the focus of future research should now perhaps be directed towards mitigating for the effects of these issues, and monitoring the success of these mitigation strategies. Previous studies have suggested that non breeding house sparrows could benefit from the provision of nestboxes to compensate for a lack of suitable holes elsewhere (Anderson, 1990). However, as breeding house sparrows have a very restricted range, nestboxes are unlikely to be utilised unless there is sufficient foraging habitat in the immediate vicinity (Chapter 6). This is likely to limit the effectiveness of nest box provision, especially in less deprived areas as these are more affected by habitat loss (Chapters 2, 3, 4 & 5).

Combining nest box provision with other measures such as wildlife friendly gardening may be a solution in areas with low deprivation levels. The popularity of gardening to encourage wildlife is increasing, as is demand for allotments, and this trend is encouraged by schemes such as the BTO GardenBirdWatch and the RSPB London House Sparrow Project (Toms, 2003; www.rspb.org.uk). The UK government is also planning to increase public participation in gardening for wildlife (Gaston *et al*; 2007). However, many householders may still be reluctant to alter their gardens for the benefit of wildlife in areas where demand for car parking is high and low maintenance gardens are a priority. For this reason the effectiveness of conservation strategies needs to be independently monitored and assessed (Gaston *et al*; 2005).

For these reasons conservationists need to work not only with individual landowners but also with planners and local government to implement effective strategies. The red-listed status of the house sparrow, together with the legal protection afforded to wildlife in the UK should prevent disturbance to the species, particularly during the breeding

season. In practice however, this may not be the case, particularly if people are unaware of the conservation implications. Limiting infill development and placing restrictions on the housing density of new developments are likely to be effective conservation measures, as is resisting the temptation to build on brownfield sites such as allotments, another key habitat for sparrows (Chamberlain *et al*; 2009a). However, this will inevitably conflict with the demand for more, high density housing in urban areas highlighted by the UK Government (ODPM, 2003), and this conflict may prove difficult to resolve.

CONSERVATION IMPLICATIONS FOR WIDER URBAN BIODIVERSITY:

It is important to consider the implications of urban development and place the results presented here into a wider context. There appears to be no single factor driving the decline of the house sparrow in urban areas, despite extensive research. This suggests that the many factors that appear to affect house sparrow numbers (e.g. food availability, predation, nest site availability) are acting cumulatively or interactively. This may be why broader scale measures such as the Index for Multiple Deprivation (Chapters 3 & 4) is a good reflection of house sparrow occurrence, as it can encapsulate much of this variation. This highlights the need to use multiple strategies in managing urban areas both for house sparrows, and also for wider biodiversity.

These findings have implications for other urban species, some of which are themselves of conservation concern. These include the European starling, *Sturnus vulgaris*, which is also red-listed in the UK and has declined in Europe (Robinson *et al*; 2002). Both starlings and house sparrows require cavities in buildings for nesting, and insect food for their chicks, so loss of habitat due to development and demand for car parking is likely to affect starlings in a similar way. The role of urban development in reducing habitat quality for this species therefore warrants further investigation.

Given the wide range of species that utilise urban gardens, loss of habitat to paved areas within cities is likely to have detrimental consequences for urban wildlife in general, even for species with dissimilar requirements to the house sparrow. ‘Edge species’ such as thrushes will lose both nesting and foraging sites to development, as could a number of other passerines. The distribution of many invertebrate species may also be linked to changes in deprivation status, as could the prevalence of native versus invasive plants, although as yet there is no evidence to suggest that this is the case (Gaston *et al*; 2007). Biodiversity and land use patterns within urban gardens therefore warrant further investigation and this is increasingly being recognised, as is shown by recent projects to assess urban biodiversity (e.g. Gaston *et al*; 2004).

Socioeconomic status in urban areas indicates not only the quality of habitat for sparrows and other wildlife, but potentially also acts as an indicator for detrimental impacts on the human population. The beneficial effects of green spaces on human health and wellbeing are already well known (Ulrich 1984; Fuller *et al*; 2007). National scale data on land use changes and deprivation could provide indicators of other hazards that affect urbanising landscapes, such as the risk of flooding, increased pollution, and high noise levels within cities. These are increasingly recognised as important threats both to biodiversity, and to human health (Wilby & Perry, 2006; Perry & Nawaz, 2008). This kind of information may be especially useful in developing countries, which are experiencing similar problems due to urbanisation, but where environmental awareness is relatively low (Pauchard *et al*; 2006).

Concentrating on these aspects of urbanisation may provide an avenue by which conservation strategies can win the backing of both local residents and governmental

organisations. The demand for land to house a growing population may take precedence over the conservation of an individual species or habitat, but there is growing recognition of the need for interdisciplinary research in tackling the problems posed by urbanisation, and for policy decisions to be based on sound evidence (Sutherland *et al*; 2006). United Nations projections estimate that 60% of the world's population will be residing in urban areas by 2030 (Wilby & Perry, 2006; www.un.org). This creates huge demand for development land near urban centres, and consequently a trend towards increasing the density of housing developments. The UK government wishes to increase the density of new developments from an average of 22 dwellings per hectare in 2003 to 30 dwellings per hectare in south-east England (ODPM, 2003). This demand for new housing should be viewed as an opportunity to create urban areas that are beneficial for both wildlife and people. In the case of the house sparrow specifically this could entail incorporating the needs of biodiversity into modern buildings – for example by adding nest boxes and swift bricks to new builds, and by maximising the wildlife potential of the green space available by planting native trees and shrubs and minimising the extent of paved areas; for example by incorporating green roofs into developments. Using the familiar and highly commensal house sparrow as a starting point for urban habitat monitoring may help to unite landowners, local authorities and planners to secure long term sustainability for not just the sparrow, but for human populations as well.

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APPENDIX 1

HOUSE SPARROW OCCURRENCE IN RELATION TO THE INDEX OF MULTIPLE DEPRIVATION FOR WALES (WIMD)

House sparrow populations are stable or increasing in Wales, but the Welsh Index of Deprivation (WIMD; Welsh Assembly, 2009) is not comparable to the English one, as they use different scoring systems. I therefore analysed the results for two Welsh cities (Cardiff and Swansea) separately in the same way as for the English cities in Chapter 4. The WIMD ranges from 0-80, with higher scores representing higher levels of deprivation, but the deprivation scores for GBW locations in Wales only ranged from 0-40. GLMMs were performed in R 2.5.1 using the “repeated” package for mixed effects models (Lindsay 2007, R development core team 2007). House sparrow occurrence was treated as a binomial response variable, and city and season were included as fixed factors. Non linear terms for year and WIMD were included, and models were simplified by stepwise deletion.

Results:

House sparrow occurrence in Cardiff and Swansea shows the same overall trend in relation to deprivation as is found in England (Chapter 2), although there is no significant seasonal effect in Wales. It is not possible to measure changes in deprivation status over time for Wales, as only one Index has been produced to date. However, house sparrow occurrence in Cardiff and Swansea is so high overall that back-transformed parameter estimates suggest that the probability of house sparrow sightings per season in GBW gardens is effectively one at all but the very least deprived locations (WIMD scores of <5). This could be due to the slightly lower population density of Cardiff and particularly Swansea in relation to English cities, or reduced pressure for

development in Wales in comparison to England. It is also possible that the two Welsh cities surveyed here are more deprived overall than deprived areas in England.

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Table A1: Summary of Generalised Linear Mixed Effects model linking house sparrow occurrence to the Welsh Index of Multiple Deprivation for 2005. Parameters from the minimal adequate model are shown.

Parameter	Parameter estimate	Residual deviance /d.f.	z-value	p-value (***)
Intercept	3.999	812.89 / 2032	8.926	<0.001 (***)
Wales deprivation score	0.539	812.89 / 2032	7.800	<0.001 (***)
Wales deprivation score ²	-0.014	812.89 / 2032	-7.383	<0.001 (***)
Year	-0.193	812.89 / 2032	-5.195	<0.001 (***)
City: Swansea	-3.680	812.89 / 2032	-8.993	<0.001 (***)
Wales deprivation score*city	0.181	812.89 / 2032	3.659	<0.001 (***)

APPENDIX 2

SEPARATE ANALYSIS OF HOUSE SPARROW GBW DATA FOR LONDON IN RELATION TO THE INDEX OF MULTIPLE DEPRIVATION

Aims and methods:

GBW data on house sparrows in London was initially analysed separately, with the aim of testing whether it was possible to predict house sparrow occurrence in the city by deriving probabilities of occurrence from the Generalised Linear Mixed effects models used in Chapter 2. This model was initially run excluding the London data, and the back-transformed parameter estimates from the model were used to predict house sparrow occurrence according to the deprivation score (IMD 2004, Noble *et al*; 2004) for each London GBW site.

Cohen's Kappa statistic (Petit *et al*; 2003) was then used to quantify the accuracy of the predictions. The Kappa statistic tests the agreement between the observed proportion of correctly identified sites and the proportion expected by chance, with a score above 0.4 indicating fair to good agreement (Petit *et al*; 2003). A score of less than or equal to zero indicates no agreement, other than what would be expected by chance.

Table A2: measures of prediction success for London GBW data, modified from Petit *et al*; 2003), where a = true positive, b = false positive, c = false negative, and d = true negative. The threshold for occurrence for house sparrows at GBW sites was set at 0.5, as in the original model.

Performance measure	Definition	Formula	Score
Cohen's Kappa	Agreement between the observed proportion of correct classified sites and the proportion expected by chance	$\frac{[(a+d) - ((a+c)(a+b) + (b+d)(c+d))/n]}{[n - ((a+c)(a+b) + (b+d)(c+d))/n]}$	-6035
Prediction success	Proportion of cases correctly identified	$a+d/n$	0.73
Sensitivity	Proportion of true positives correctly predicted	$a/(a+c)$	0.96
Specificity	Proportion of true negatives correctly predicted	$d/(b+d)$	0.05
Positive predictive power	Proportion of predicted presences that were real	$a/(a+b)$	0.75
Negative predictive power	Proportion of predicted absences that were real	$d/(c+d)$	0.30

As shown in Table A2, Cohen's kappa was strongly negative, indicating that there was no agreement between the observed and expected values. Inspection of the prediction success of different aspects of the model shows that it was relatively good at predicting presence, but did not predict absences. This may be because London has low house sparrow numbers on average in comparison with other cities (a mean house sparrow presence/absence ratio of 0.62 ± 0.01 ; in a study of English cities, only two other areas, Bristol and Liverpool, had a mean house sparrow presence absence ratio of less than 0.7, Chapter 4). Alternatively, the low success rate for predicting absences may be a reflection of the lower density of house sparrows in London leading to a more patchy distribution of house sparrows within the city. As house sparrows have such a small

range (< 2 km in adults, Snow *et al*; 1998), lack of connectivity between patches of suitable habitat combined with lack of movement by existing sparrow populations may mean that suitable patches may become extinct due to a combination of factors, such as inbreeding depression, low productivity and the Allee effect (Veit & Lewis, 1996) and are not recolonised. High levels of inbreeding have been reported in metapopulations of house sparrows (Jensen *et al*; 2007), suggesting that this may be a factor influencing the disappearance of isolated colonies.

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APPENDIX 3

RESULTS OF GENERALISED LINEAR MIXED EFFECTS AND LINEAR REGRESSION MODELS TO ANALYSE THE PRINCIPAL COMPONENTS DERIVED IN CHAPTER 5 IN RELATION TO HOUSE SPARROW OCCURRENCE AND DEPRIVATION

Table A3: Summary of models linking Principal Components of urban land use types to house sparrow occurrence (GLMMs, Φ) and deprivation (linear models, $\dagger$). Starred terms indicate significant interaction effects. N = 39,168.

Principal Component	Main land use types	t-statistic / z-statistic	Significance
1	Domestic buildings and gardens	-120.904 $\dagger$	<0.005 (**)
2	Non-domestic buildings, paving	41.202 $\dagger$	<0.005 (**)
3	Green space	-2.720 $\dagger$	<0.05 (*)
1	Domestic buildings and gardens	0.269 Φ	0.79
2	Non-domestic buildings, paving	-3.296 Φ	<0.005 (**)
1*2	Domestic buildings, gardens * Non-domestic buildings, paving	8.246 Φ	<0.005 (**)
1*year	Domestic buildings and gardens	-5.942 Φ	<0.005 (**)
2*year	Non-domestic buildings and paving	2.2 Φ	<0.05 (*)

Principal components (PC) one, two and three were used for analysis as they had eigenvalues greater than or equal to one (Field, 2000). When the number of variables is less than 30 (seven in this case), and the sample size is greater than 200 (here $n > 39,000$) Kaiser's threshold of one is considered appropriate (Field, 2000). Regressions and GLMMs were performed in R 2.5.1 (Lindsay 2007, R development core team 2007). Models were simplified by stepwise deletion where appropriate.

Areas with high values for both PC1 and PC2 show a higher probability of house sparrow sightings than those with high values for just one of these land use components, suggesting that a mosaic of different land use types may be more beneficial to house sparrows than larger areas at risk of development, or very high density housing. The high proportion of paving and non-domestic buildings in these areas suggests that they are likely to be relatively deprived in comparison to those areas that just score highly for PC1. Areas with high deprivation have high values for PC2; and PC3 to a lesser extent.

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